

CONTACT TOMOGRAPHY IN HANKEL INFORMATION GEOMETRY

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ABSTRACT. We study which information metrics can be recovered from the second contact jets of synchronized root observations. Under injective quadratic measurements, the conormal union of the rank-one residual cone has a sharp dimension-dependent obstruction: with two shared parameters, precisely one quadratic matrix direction is invisible, whereas with at least three shared parameters no nonzero quadratic form is invisible. In calibrated polynomial binary experiments, interpolation forces inverse efficient information into a Hankel congruence class. We give explicit three-site relations characterizing that class and prove that they remove the two-parameter ambiguity exactly under a stated transversality condition. In the smallest nonlinear case, five directional contact curvatures recover the native metric; without its Hankel constraint an entire line remains indistinguishable to second order. More generally, sharp finite dual-jet recovery transports the fixed-exposure image to a convex region of observable data with singleton boundary fibres and circular interior design fibres. We also give an intrinsic effective minimal-representation theorem whose input is only a labelled endpoint value function. Complete companion results on finite-map residuals, critical weighted jets, endpoint reconstruction, and local experiments are retained with their proofs.

1. INTRODUCTION

A common perturbation of several double roots constrains their variance vector to a quadratic image

$$q_A(\eta) = ((a_i^\top \eta)^2)_{i=1}^k.$$

After the free score directions have been profiled out, the leading contact with a target variance tb , $b \geq 0$, is the squared distance from b to this cone in the efficient information metric. For a fixed cone, a natural inverse question is whether its observed contact geometry determines the metric. Knowing a metric and evaluating a distance is immediate; recovering that metric from contact is not.

Theorem 5.1 answers this inverse question for injective quadratic measurements and native calibrated polynomial experiments. A normal vector n at $q_A(\eta)$ satisfies

$$\left(\sum_i n_i a_i a_i^\top \right) \eta = 0.$$

Thus the conormal union is the inverse image of the singular symmetric matrices. The quadratic forms vanishing on this union have dimension one for two shared parameters, and dimension zero in all higher shared dimensions. Independently, the native scalar score inverse satisfies explicit three-site linear relations. Testing the sole invisible quadratic direction against those relations gives a necessary and sufficient recovery criterion. This is the interaction between residual geometry and Hankel information used throughout the principal theorem.

Corollary 6.1 makes the criterion fully explicit for three root pairs and two shared parameters. Five scalar directional curvatures recover a binary quartic; its one-dimensional matrix ambiguity

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is removed by the native annihilator exactly when $\tau_A \neq 0$. We exhibit both a rational transverse experiment and a nondegenerate exceptional loading family. The ambiguity statement concerns second jets, not equality of all higher contact terms. For higher codimension the finite oracle is a scalar value of the dual normal quadratic form, not an unqualified count of ordinary directional derivatives.

The fixed-budget conclusion is a consequence of this recovery map. In moment coordinates the reciprocal budget value has positive definite Hessian. The recovered dual jets are an invertible linear change of those coordinates, so the entire budget image and all its design fibres become observable contact-data geometry. The reciprocal-fibre argument itself is a general convex lemma; what is forced by the experiment is the inverse-information transformation and the corank two in the endpoint-free family.

Theorem 7.2 addresses a separate companion inverse problem. Its input is the graph of a labelled scalar endpoint profile, not a reference quadratic presentation or an optimizer map. It returns every normalized representation at any prescribed dimension and, under a finite-representability promise, the full minimal fibre. Explicit first-stratum reconstructions remain in the appendices; the general effective statement is not advertised as an explicit normal-form catalogue.

The organization follows this hierarchy. The native score inverse and uniform contact hypotheses precede the principal conormal theorem. The three-root formula and intrinsic endpoint companion follow it. The complete earlier geometric and statistical machinery is placed in the appendices rather than used as an additional centre of the paper.

1.1. Relation to inverse optimization and design equivalence. Inverse parametric optimization is directly relevant to the companion endpoint problem. Hempel, Goulart, and Lygeros [18] construct optimization representations of supplied piecewise-affine maps in hybrid control. Nguyen, Oлару, Rodriguez-Ayerbe, Hovd, and Necoara [19] use convex lifting to represent continuous piecewise-affine functions on polytopic partitions as solutions of parametric linear or quadratic programs. Thus inverse synthesis of a parametric program is not a new problem introduced here. In those formulations the given object is an optimizer map (or its linear output), with its piecewise-affine data; constraints and objective are constructed. They do not start with only our scalar labelled optimal-value function. In particular one must not describe their inputs as a known hidden quadratic-program active fan.

Theorem 7.2 instead starts with the exact scalar graph, searches within positive block quadratic forms with a fixed endpoint orthant, determines the least endpoint dimension, and returns all normalized minimal representations. The earlier explicit first-stratum theorems additionally identify uniqueness modulo endpoint gauges or scalar moduli. These are different outputs from constructing one optimizer representation, not a claim that our restricted class subsumes the general inverse-optimization constructions. An optimizer alone does not determine the observed value: adding a parameter-only quadratic to the objective leaves every minimizer unchanged while changing that value. Conversely a value function need not specify a hidden optimizer or its dimension. Hence applying optimizer-map synthesis to an arbitrarily chosen map cannot by itself establish the value equality, positivity, and minimal-fibre assertions here. Quantifier elimination is the effective general tool; explicit first-stratum formulas remain separately valuable.

There is also a concrete distinction from direct information-matrix design equivalence. Harman, Filová, and Rosa [20, Lemma 1] describe fixed-information approximate designs by nonnegative weights satisfying linear moment equations, hence by a polytope. Our direct information is not being redescribed as something nonlinear. After nuisance elimination, rational interpolation forces the *inverse* efficient information to have weights $\omega_j = a_j/\lambda_j$. Unit exposure therefore becomes $\sum_j a_j/\omega_j = 1$, a reciprocal level inside an affine moment fibre, rather than the affine simplex slice of direct design equivalence. Lemma 3.1 deliberately states the convex argument in its general

form: it uses a positive covector and not polynomial observations. The native content is the forced reciprocal transformation, the corank $2 - e$, and, in Theorem 5.1, the exact transport of this geometry into observed dual contact jets. No new general equivalence theorem for optimal designs is claimed.

The principal result is different from both inverse-program synthesis and design equivalence. It classifies the quadratic forms invisible on the conormal union of a synchronized rank-one cone, then tests that invisible space against the three-site relations forced by the native Hankel congruence. For two shared parameters the test is essential: it eliminates an ambiguity present for arbitrary positive metrics. For higher shared dimension the determinantal obstruction has no quadratic part. This precise mechanism, rather than a claim of exhaustive priority over every use of conormal geometry, is the centre of the paper.

2. HANKEL GEOMETRY FORCED BY CALIBRATED BINARY OBSERVATIONS

We now return to the binary polynomial observation, rather than choosing contrasts to realize a desired metric. Fix two distinct strictly positive rank-one stochastic tables $M_a = u_a v_a^T$, $a = 1, 2$, with calibrated channels. At a clock $T > D$ observe the four-cell law

$$p(T) = \pi(T)M_1 + (1 - \pi(T))M_2, \quad \frac{\pi(T)}{1 - \pi(T)} = \frac{\alpha f_1(T)}{(1 - \alpha)f_2(T)}, \quad 0 < \alpha < 1. \quad (2.1)$$

Both component polynomials have roots in $[0, D]$ and are positive at all clocks. Fixed polynomial factors are permitted in either component. The variable factors consist of $k \geq 1$ separated interior double clusters and $e \in \{0, 1, 2\}$ simple endpoints, with no shared variable cluster between components. An endpoint is either 0 or D , and occurs at most once. Give a variable cluster the colour $\sigma_a = 1$ or -1 according as it belongs to component one or two. All other factors and the channels are held fixed. Equal prescribed component degrees can be obtained by fixed background factors; no coefficient is chosen from a target metric.

For an interior cluster at r_i use its sum displacement s_i and variance v_i through the factor $(T - r_i - s_i/2)^2 - v_i$. The variance is nonnegative. A lower endpoint has factor $T - z$, and an upper endpoint factor $T - D + z$, where $z \geq 0$. The log mixing odds is a free coordinate, as are the k interior sums. Its use only specifies a smooth local coordinate in the original rational model with mixing parameter α . The $n = k + e$ variances and inward endpoint shifts are retained.

Choose exactly $m = 2k + e + 1$ distinct clocks $D < T_1 < \dots < T_m$ and positive exposures λ_j . Exposure means that the number of observations in group j , divided by a common asymptotic scale, tends to λ_j . The total exposure is not fixed to one in the cone theorem below. This is an ordinary product of categorical experiments with its own Fisher metric. At the centre put

$$\kappa_j = \pi_j^2(1 - \pi_j)^2 \sum_{c=1}^4 \frac{(M_{1c} - M_{2c})^2}{p_{jc}} > 0. \quad (2.2)$$

Order the retained sites as $\rho_1 < \dots < \rho_n$. Set $\epsilon_a = -\sigma_a$ at an interior cluster or a lower endpoint and $\epsilon_a = \sigma_a$ at an upper endpoint. Define

$$L_j = \left(1, \left(-\frac{\sigma_i}{T_j - r_i} \right)_{i=1}^k \right), \quad D_{ja} = \frac{\epsilon_a}{(T_j - \rho_a)^{\mu_a}}, \quad (2.3)$$

where $\mu_a = 2$ at an interior site and 1 at an endpoint. The square matrix $V = [L, D]$ has m columns.

Theorem 2.1 (Native finite moment cone). *For the model (2.1), profiling the free sum and mixing coordinates gives a positive retained information matrix Q . Let $R = Q^{-1}$ and let Z be the*

last n rows of V^{-1} . Then

$$R = \sum_{j=1}^m \frac{Z_{\cdot j} Z_{\cdot j}^\top}{\lambda_j \kappa_j}. \quad (2.4)$$

Its columns are fixed by the observation model. More explicitly, set

$$h(T) = \prod_{i=1}^k (T - r_i)^2 \prod_{b \text{ endpoint}} (T - b), \quad P(T) = \prod_{j=1}^m (T - T_j), \quad p_*(T) = \prod_{a=1}^n (T - \rho_a),$$

$$h_a(\rho_a) = \left. \frac{h(T)}{(T - \rho_a)^{\mu_a}} \right|_{T=\rho_a}, \quad d_a = -\frac{P(\rho_a)}{\epsilon_a h_a(\rho_a)}, \quad \xi_j = \frac{h(T_j)}{P'(T_j)}.$$

Then

$$Z_{aj} = \frac{\xi_j d_a}{T_j - \rho_a}. \quad (2.5)$$

Let E be the coefficient matrix whose a th row is the polynomial $p_*(T)/(T - \rho_a)$ in the basis $1, T, \dots, T^{n-1}$, and put $J_* = \text{diag}(d_a)E$. It is invertible. With $w(T) = (1, T, \dots, T^{n-1})^\top$, the entire native image is

$$R = J_* H(\omega) J_*^\top,$$

$$H(\omega) = \sum_{j=1}^m \omega_j w(T_j) w(T_j)^\top, \quad \omega_j = \frac{\xi_j^2}{p_*(T_j)^2 \lambda_j \kappa_j} > 0. \quad (2.6)$$

As the positive exposures vary, H ranges over the relative interior of the finite moment cone generated by these clock tensors. This cone has linear span equal to the full Hankel subspace and dimension $2n - 1$. Consequently the native Q -image is a smooth $(2n - 1)$ -dimensional image under matrix inversion. These assertions hold for every separated clock and root configuration in the stated class, not merely generically.

Proof. Differentiating the log odds of (2.1) gives exactly (2.3). The probability derivative in a log-odds direction is $\pi_j(1 - \pi_j)(M_1 - M_2)$, so its scalar information is (2.2). The full information is $I = V^\top \text{diag}(\lambda_j \kappa_j) V$. To prove V invertible, multiply a linear combination of its rational columns by $h(T)$. The result is a polynomial of degree at most $m - 1$. If it vanishes at all m clocks it is identically zero. Its constant part at infinity and its simple and double principal parts at each site then force every coefficient to vanish. Thus $I > 0$. The lower-right block of I^{-1} is the inverse of the nuisance Schur complement, giving (2.4).

For explicit recovery, prescribe arbitrary clock log-odds increments y_j . The polynomial $g(T) = h(T)(V\theta)(T)$ is reconstructed by Lagrange interpolation as

$$g(T) = \sum_j h(T_j) y_j \frac{P(T)}{(T - T_j) P'(T_j)}.$$

At a retained site, its highest principal-part coefficient is $g(\rho_a)/(\epsilon_a h_a(\rho_a))$. This proves (2.5). The polynomials $p_*(T)/(T - \rho_a)$ are a basis of the polynomials of degree at most $n - 1$, as evaluation at the n sites immediately shows. Hence J_* is invertible and

$$Z_{\cdot j} = \frac{\xi_j}{p_*(T_j)} J_* w(T_j).$$

Substitution proves (2.6). Positive exposures give all positive ω , independently. A linear map sends the positive orthant onto the relative interior of its finitely generated cone: one direction follows by openness onto the span, and for the converse subtract a small positive multiple of the sum of all generators from a relative-interior point before representing the remainder with nonnegative coefficients.

The entries of $w(T)w(T)^\top$ have powers T^s for $0 \leq s \leq 2n - 2$. Since $m = 2k + e + 1 \geq 2n - 1$ exactly when $e \leq 2$, the Vandermonde matrix at the distinct clocks has rank $2n - 1$. The generators therefore span precisely the Hankel space. Their positive combinations are positive definite, because the vectors $w(T_j)$ span $\mathbb{R}^n$. Inversion is a diffeomorphism on the positive cone, which proves the last dimension assertion. $\square$

Corollary 2.2 (Secants, dual queries, and fixed budgets). *Let $\mathcal{V}_* = J_*\{\text{Hankel matrices}\}J_*^\top$ and $s = 2n - 1$. The normalized nonzero secants of the native inverse-information image are exactly the Frobenius unit sphere of $\mathcal{V}_*$. Recovering an unknown R from exact dual values $u^\top Ru$, with u in any specified cone with nonempty interior, requires and admits exactly s queries, both for fixed queries and for deterministic adaptive queries. The exposures are not supplied as additional numerical input to this oracle problem.*

For the information matrices themselves use normalized coordinates $\tilde{Q} = J_^\top Q J_* = H^{-1}$. A symmetric unit matrix D is a normalized secant of this image if and only if there exist H_0, H_1 in the relative-interior moment cone and $a > 0$ with*

$$H_1 D H_0 = a(H_0 - H_1). \quad (2.7)$$

This is a finite polynomial presentation in positive clock weights, a , and the entries of D . Physical-coordinate secants are obtained by $D \mapsto J_^{-\top} D J_*^{-1}$ and renormalization. The sharp dual-query count is not asserted for the different oracle $v^\top Q v$.*

If the total exposure is fixed to one, the exact image is obtained by adding, to (2.6), the condition

$$\sum_j \frac{\xi_j^2}{p_*(T_j)^2 \kappa_j \omega_j} = 1. \quad (2.8)$$

No unconstrained-cone dimension or secant assertion is imposed on that budget section.

Proof. The image of R is relatively open in $\mathcal{V}_*$. A relative ball around any image point realizes every nonzero direction as a secant; all differences lie in that span. Quadratic evaluations on a cone with interior separate $\mathcal{V}_*$, and s of them form a basis of its dual. For fewer than s adaptive evaluations at an interior point, a nonzero matrix in their common kernel gives two sufficiently small opposite perturbations in the image with identical transcripts. This proves the exact count without continuity assumptions on the algorithm. Multiplying the identity $D = a(H_1^{-1} - H_0^{-1})$ by H_1 and H_0 proves both directions of (2.7). The formula for ω_j gives (2.8) directly. $\square$

For $n = 2$, the dimension is three, consistent with the retained two-variance phenomenon. For $n \geq 3$, the exact dual image has Hankel relations rather than arbitrary symmetric entries. The coordinate congruence and the distinction between a tensor and its inverse are part of the classification, not changes in the observation metric.

Theorem 2.3 (Native endpoint visibility). *In the subclass in which all variable factors belong to one component, partition Q into interior variances and inward endpoints. For $e = 1$ and $k \geq 2$ the representation is fully visible. For $e = 2$ and $k \geq 3$ it is fully visible. Thus its minimal endpoint dimension and its full minimal fibre are reconstructed by Theorem G.3, for every positive exposure vector.*

For $e = 2, k = 2$ it has exactly three cells, with the empty cell missing; its entire minimal fibre is (H.4). For $k = 1$ and $e = 1$ or 2 , all endpoints are active on the positive retained ray; the profile is a single positive quadratic form, with minimal endpoint dimension zero. These conclusions are intrinsic to this binary family.

Proof. When the variable factors have one colour, all d_a have the same sign. Indeed, the squared interior factors in h_a are positive, and each upper-endpoint sign reversal compensates the negative factor contributed by that endpoint at an interior site. Directly checking the possible sets

$\emptyset, \{0\}, \{D\}, \{0, D\}$ gives a common sign. The matrix R is consequently a positive diagonal congruence of a Cauchy Gram matrix

$$\left(\frac{1}{T_j - \rho_a}\right)_{a,j} \text{diag}(\gamma_j) \left(\frac{1}{T_j - \rho_a}\right)_{a,j}^\top, \quad \gamma_j > 0.$$

Every minor of size h of the rectangular Cauchy matrix has the same nonzero sign $(-1)^{h(h-1)/2}$, by the Cauchy determinant formula. Cauchy–Binet therefore makes every minor of R strictly positive. Jacobi’s complementary-minor identity shows that $Q' = \Sigma Q \Sigma$, $\Sigma_{aa} = (-1)^a$, is also strictly totally positive. We use only these displayed determinant identities, not a separate hypothesis of total positivity.

For one endpoint the entries in the cross column alternate signs along the ordered interior sites. With at least two sites they have both signs, so a strictly positive vector annihilates the transpose of the cross block. Proposition G.4 proves visibility. For two endpoints at the extreme sites, after fixed column sign changes the interior rows of B have alternating signs times positive vectors (a_i, b_i) . The ratios a_i/b_i are strictly monotone, because every ordered 2×2 minor of Q' is positive. For three consecutive rows, the middle positive vector lies strictly in the cone of the two outer ones, and its sign is opposite. Hence these rows admit a strictly positive linear dependence. Additional rows can receive sufficiently small positive weights; a bounded right inverse of the three-row block corrects their residual while keeping the first three weights positive. Thus B has rank two and $B^\top h = 0$ for some $h > 0$. The same proposition proves full visibility for all $k \geq 3$.

For $e = 2, k = 2$ order the sites as endpoint, interior, interior, endpoint. Then

$$B = \begin{pmatrix} -a & b \\ c & -d \end{pmatrix}, \quad a, b, c, d > 0, \quad ad > bc.$$

The last inequality is a minor of Q' . The empty-cell conditions would require $x_2/x_1 \geq a/c$ and $x_2/x_1 \leq b/d$, which are incompatible. The endpoint block has negative off-diagonal, so its inverse is entrywise positive. Taking $b/d < x_2/x_1 < a/c$ gives $B^\top x < 0$ and a strict full-active point. At $x = (1, 0)$ the first-only inactive multiplier is $(Q_{11}Q_{24} - Q_{14}Q_{21})/Q_{11} > 0$; at $x = (0, 1)$ the second-only multiplier is $(Q_{31}Q_{44} - Q_{14}Q_{34})/Q_{44} > 0$. These are positive signed minors, and the strict inequalities persist on nearby interior rays. Exactly three cells occur, so Theorem H.3 applies.

If $k = 1$, block inversion gives $-C^{-1}B^\top = R_{E,I}R_{I,I}^{-1}$, which is entrywise positive. Every endpoint is therefore active on $x > 0$. Completing the square gives a single positive Schur-complement quadratic, proving the last assertion. $\square$

For several component colours, Theorem 2.1 and Corollary 2.2 remain unchanged. After conjugation by $\text{diag}(\text{sign } d_a)$, R is strictly totally positive; this specifies the signed native class. In particular the physical signs are not discarded when testing endpoint visibility. The all-one-colour subclass above is a naturally calibrated submodel of the original binary observations, not a realization obtained by first prescribing Q .

3. RECIPROCAL FIBRES AND UNIFORM HYPOTHESES

We separate a general convex-fibre fact from the polynomial experiment that supplies its particular matrix and corank.

Lemma 3.1 (General reciprocal-fibre geometry). *Let $T : \mathbb{R}^m \rightarrow \mathbb{R}^s$ be surjective. Assume there is a covector $u \in \mathbb{R}^s$ with $(T^\top u)_j > 0$ for all j . Fix $a_j > 0$, set $\mathcal{M} = T\mathbb{R}_{>0}^m$, and define*

$$\phi(\omega) = \sum_{j=1}^m \frac{a_j}{\omega_j}, \quad \chi(y) = \min_{\omega > 0, T\omega = y} \phi(\omega).$$

Every $y \in \mathcal{M}$ has a unique minimizer $\omega_*(y)$. The functions ω_* and χ are analytic, $\chi(ty) = \chi(y)/t$, and

$$D^2\chi(y) = (TD^{-1}T^\top)^{-1} > 0, \quad D = \text{diag}(2a_j/\omega_{*j}^3). \quad (3.1)$$

Put $d = m - s$. If $d = 0$, the image of $\{\omega > 0 : \phi(\omega) = 1\}$ is $\{\chi = 1\}$, with singleton fibres. If $d > 0$, its image is $\{\chi \leq 1\}$; fibres are singletons when $\chi = 1$, and are diffeomorphic to S^{d-1} when $\chi < 1$.

Proof. The closure of a positive affine fibre is bounded, since $\sum_j (T^\top u)_j \omega_j = u^\top y$. The function ϕ tends to infinity at any coordinate face and has positive definite Hessian. Existence and uniqueness follow by compact sublevel sets and strict convexity. At the minimizer, $\nabla\phi(\omega_*) = T^\top \nu$. The bordered linearization of these equations and $T\omega_* = y$ is nonsingular: a homogeneous solution obeys $D\delta\omega = T^\top \delta\nu$ and $T\delta\omega = 0$, hence $\delta\omega = 0$ and then $\delta\nu = 0$. The analytic implicit function theorem applies. Differentiating gives $D\omega_* = D^{-1}T^\top(TD^{-1}T^\top)^{-1}$ and $D\chi = \nu^\top$, proving the Hessian formula. Homogeneity follows by scaling the affine fibre.

When $d = 0$ each affine fibre is a point. Otherwise, within the d -dimensional kernel directions, each ray from ω_* reaches the boundary of the bounded positive affine fibre. Along it ϕ starts at its minimum, increases strictly, and tends to infinity. If $\chi < 1$ it crosses level one exactly once. The crossing radius is smooth in the unit direction by the implicit function theorem, since the radial derivative is strictly positive. This radial parametrization and its inverse are smooth and give S^{d-1} . If $\chi = 1$ strict convexity leaves only the minimizer; if $\chi > 1$ the level is empty. $\square$

The binary experiment forces $T_{rj} = T_j^r$, $s = 2n - 1$, $m = 2k + e + 1$, $a_j = \xi_j^2/(p_*(T_j)^2\kappa_j)$, and therefore $d = 2 - e$. The zeroth-moment covector supplies the positive covector hypothesis. Theorem B.3 is its native specialization, with all the explicit rational interpolation and rectangular-clock calculations retained. The function χ is a reciprocal budget value of degree minus one, not a degree-one norm gauge. No claim is made that the general strict-convexity argument is peculiar to binary observations.

At a fixed unit budget $\lambda_j > 0$, $\omega_j = a_j/\lambda_j$ and $\lambda_j \leq 1$, so $\omega_j \geq a_j$. Letting any exposure tend to zero sends ω_j , and hence the zeroth moment, to infinity. It does not approach a finite boundary point of the open moment cone. The hypersurface $\chi = 1$ is the relative boundary of the positive-exposure image inside that open cone; its exposures are all strictly positive. A boundary of the closed finite moment cone obtained by letting some ω_j vanish is a third, different boundary. These distinctions also apply after the observable linear transformation in (6.9).

3.1. Explicit uniform class for synchronized contact. In every compact-uniform assertion of Theorem 4.1, use the measurement margin

$$\mu(A) = \min_{S \in \text{Sym}_d, \|S\|_F=1} \|\mathcal{A}_A(S)\|_2 \geq \mu_0 > 0.$$

Keep the dimensions fixed, loadings bounded, separated roots and clocks in compact sets, positive probabilities and exposures bounded away from zero, and metric eigenvalues in a fixed compact subinterval of $(0, \infty)$. Targets always satisfy $b \in \mathbb{R}_+^k$ and range in a bounded set. The sufficient margin is not a sharp phase-retrieval condition.

For the nonlinear replacement of a loading, the precise perturbation class consists of functions on one fixed ball $|\eta| \leq \rho$, with $\ell_i(0) = 0$, $D\ell_i(0) = a_i^\top$ and a common bound on $\|D^2\ell_i\|$. Require C^2 regularity there, and analytic or semialgebraic regularity when that category is invoked. Taylor's theorem gives a uniform quadratic remainder. Moreover

$$\|q_A(\eta)\| \geq \mu_0|\eta|^2, \quad \|((\ell_i(\eta)^2)_i) - q_A(\eta)\| \leq C|\eta|^3.$$

After one common shrinking of ρ , the perturbed variance norm is at least $(\mu_0/2)|\eta|^2$. This proves the uniform localization used in the synchronized theorem rather than assuming it through the

word “analytic”. At $\eta = \sqrt{t}\zeta$ the remainder divided by t is uniformly $O(\sqrt{t})$ on bounded ζ . The inherited limiting residual and minimum proof therefore applies to exactly this class. The full residual includes the free score space and endpoint orthant; only its free-score quotient is the translated linear image of $\mathcal{V}_A \times \mathbb{R}_+^e$.

3.2. Conventions for the retained companion proofs. For Proposition D.4, the source is compact semialgebraic, the observation maps are polynomial or analytic and semialgebraic on the neighbourhoods used, and the marked arcs and tube data are semialgebraic with the stated positive-time analytic regularity. The determinant J is nonzero on the positive-time tube. The object being uniformized is the compact closure of its positive-time graph, not the closure of one selected inverse sheet. Every pruning retains a surjective preimage of that graph germ. These are hypotheses of that proposition; its proof uses them to compare full tube suprema. A proper weighted initial map in Theorem D.1 is a supporting sufficient normal form, not a classification of arbitrary critical germs.

In the missing-singleton proof, at $e = 3$ the two other independent columns give the required two distinct pair ranges. In arbitrary dimension, equality of a pair span with another selected pair containing a different known column would give a linear dependence among the independent columns, so the criterion cannot choose a spurious pair. In the planar exhaustion proof, a rank-one difference determines a linear form only up to sign; the sign is fixed by the labelled open regions carrying the two distinct local quadratics. Reversing the sign interchanges these regions and changes the function on an open set. Coincident quadratics are instead a visibility degeneration, which is handled separately and cannot justify the sign selection. Theorem I.5 remains a reference-based equality test; the function-input inverse statement is Theorem 7.2.

The seven-clock overidentification example in Proposition B.5 excludes extension of the specified six-clock congruence, not every possible exposure-independent congruence. In particular none of the contact-tomography statements extrapolates the exact square-clock Hankel constraint to an overidentified experiment.

4. NONLINEAR CONTACT OF SYNCHRONIZED ROOT PAIRS

Keep the calibrated binary family, score matrix $V = [L, D]$, and positive exposures of Section 2. Partition $D = [D_x, D_z]$ into its k interior and e endpoint columns. Reorder columns when necessary; every congruence below uses this same physical ordering. Fix nonzero loadings $a_i \in \mathbb{R}^d$, chosen independently of the exposures, and require the i th root pair to have the form

$$r_i + s_i/2 + a_i^\top \eta, \quad r_i + s_i/2 - a_i^\top \eta. \quad (4.1)$$

The sums s_i , mixing coordinate, and inward endpoints remain free. This is a subfamily of the polynomial observation model, specified by shared root perturbations, not by a prescribed metric. Put

$$q_A(\eta) = ((a_i^\top \eta)^2)_{i=1}^k, \quad \mathcal{A}_A(S) = (a_i^\top S a_i)_{i=1}^k, \quad \mathcal{V}_A = \mathcal{A}_A\{S \geq 0 : \text{rank } S \leq 1\}.$$

Assume the matrices $a_i a_i^\top$ span Sym_d . This convenient sufficient condition is stronger than injectivity modulo sign in general phase retrieval; it will make every identification used here elementary. It implies $k \geq d(d+1)/2$ and that the a_i span $\mathbb{R}^d$.

For $b \in \mathbb{R}_+^k$ let $p^{\text{target}}(t)$ be the law in the unrestricted calibrated family with interior variances tb , zero sums, zero inward endpoint shifts, and the central mixing coordinate. A target outside the synchronized subfamily is allowed. We use the convention

$$h^2(p, q) = \sum_c (\sqrt{p_c} - \sqrt{q_c})^2,$$

so that $4h^2(p, p + \delta) = \sum_c \delta_c^2/p_c + o(|\delta|^2)$. The infimum below is over a sufficiently small fixed neighbourhood of the central root configuration in the synchronized family.

Theorem 4.1 (Hankel-weighted nonlinear root contact). *Let the clock and root separation, probability, and exposure margins be positive. For uniform assertions, fix the dimensions, take the loadings and targets in bounded sets with $b \geq 0$, keep metric eigenvalues in a compact subinterval of $(0, \infty)$, and require $\mu(A) = \min_{\|S\|_F=1} \|\mathcal{A}_A(S)\|_2 \geq \mu_0 > 0$. The nonlinear perturbations are defined on one fixed ball, vanish at zero, have derivative $a_i^\top$ there, and have a common C^2 bound, as specified in Section 3.1. Then the following assertions hold.*

The complete normalized probability residual at scale t is

$$E_b = \{La + D_x(q_A(\eta) - b) + D_z z : a \in \mathbb{R}^{k+1}, \eta \in \mathbb{R}^d, z \in \mathbb{R}_+^e\}, \quad (4.2)$$

where the probability score blocks are those of Theorem L.1. This set is closed. Its quotient by the free score space is a translate of the injective linear image of $\mathcal{V}_A \times \mathbb{R}_+^e$. In particular, for $d \geq 2$ it is not a finite union of affine spaces.

The exact metric contact formula is

$$\lim_{t \downarrow 0} \frac{4}{t^2} \inf_j \sum_j \lambda_j h^2(p_j, p_j^{\text{target}}(t)) = \min_{\eta \in \mathbb{R}^d, z \geq 0} (q_A(\eta) - b, z)^\top Q(q_A(\eta) - b, z) = \min_{v \in \mathcal{V}_A} \hat{G}_Q(v - b). \quad (4.3)$$

Here Q is exactly the nuisance Schur complement of the unrestricted binary score system. Consequently its inverse has the model-forced Hankel form in Theorem 2.1, with the fixed-budget image and fibres in Theorem B.3 when the exposures sum to one. The right side is zero precisely when $b \in \mathcal{V}_A$; otherwise it is strictly positive and the Hellinger distance has order t .

Suppose moreover that all variable factors have one colour, and that either $e = 1, k \geq 2$ or $e = 2, k \geq 3$. The labelled function G_Q on $\mathbb{R}_+^k$ determines the right side of (4.3) for every allowed A and $b \in \mathbb{R}_+^k$. Indeed it reconstructs Q up to endpoint scales and permutations, and these ambiguities preserve $\hat{G}_Q$ on all of $\mathbb{R}^k$. Thus a fully visible endpoint profile determines the nonlinear contact of every synchronized subfamily with the same native experiment.

All residual and minimum limits are uniform for bounded b on compact parameter sets with the stated margins and a positive lower bound for the spanning measurement map. Replacing $a_i^\top \eta$ in (4.1) by analytic or semialgebraic C^2 functions $\ell_i(\eta) = a_i^\top \eta + O(|\eta|^2)$, with uniform remainder bounds, leaves the leading residual and contact formula unchanged.

Proof. The polynomial factor of pair i is exactly $(T - r_i - s_i/2)^2 - (a_i^\top \eta)^2$. The map $\mathcal{A}_A$ is injective on Sym_d by its spanning hypothesis. Its restriction to rank-one positive matrices therefore has precisely the sign ambiguity $\eta \sim -\eta$ away from zero. Since the loadings span $\mathbb{R}^d$, compactness of the unit sphere gives

$$\|q_A(\eta)\| \geq c_A |\eta|^2. \quad (4.4)$$

It follows that q_A is proper and that $\mathcal{V}_A$ is closed. This also proves the uniform version of this assertion under the stated bounds.

Write $\theta = (s, \text{mixing}, v, z)$ for the symmetric coordinates of the unrestricted calibrated family. The square score matrix V is invertible by rational interpolation. Each calibrated table determines its scalar clock mixing probability, so the observation map is locally bi-Lipschitz in θ , on an open extension of these coordinates. This statement is about the ambient symmetric model; its restriction to (4.1) is singular.

A bounded probability residual divided by t forces $\theta - \theta^{\text{target}}(t) = O(t)$. Thus sums and mixing are $O(t)$, endpoints are $O(t)$, and $q_A(\eta) = O(t)$. Equation (4.4) gives $\eta = O(\sqrt{t})$. Put sums and mixing equal to ta , endpoints equal to tz , and $\eta = \sqrt{t}\xi$. Taylor expansion in the ambient symmetric coordinates then gives

$$t^{-1}(p - p^{\text{target}}(t)) = La + D_x(q_A(\xi) - b) + D_z z + o(1)$$

uniformly on bounded sets. Bounded residual sequences have bounded rescaled variables; taking subsequences proves one inclusion in (4.2). Conversely each displayed point is realized by this

substitution, and all such roots are feasible for sufficiently small t by interior separation. The full score map is injective and has a bounded left inverse. Combined with properness of q_A , this proves that E_b is closed.

For $d \geq 2$, the cone of positive matrices of rank at most one is not a finite union of affine spaces. Any affine space contained in that cone has dimension at most one: two independent rank-one positive directions have a rank-two positive sum, while a complete nonconstant affine line cannot stay positive in both directions. More directly, its infinitely many distinct rank-one rays cannot be covered by finitely many such affine pieces. The injective quotient score map preserves this obstruction. Intersecting with $z = 0$ proves the same assertion with endpoints included. Thus the residual in (4.2) is genuinely non-affine after the free coordinates have been removed.

Taking zero free increments and $\eta = z = 0$ gives cost $O(t^2)$. The local inverse and (4.4) therefore localize every minimizing sequence as above. On the resulting bounded sets the Hellinger expansion is uniform. Its quadratic term is the score Gram matrix. Minimizing over a gives its Schur complement Q , and minimizing over $z \geq 0$ gives (4.3). Positive definiteness yields coercivity in $(q_A(\xi) - b, z)$ and hence in ξ, z ; the limiting minimum is attained. Testing a limiting minimizer gives the upper bound and localized minimizing sequences give the lower bound. The minimum is zero exactly when $q_A(\xi) = b$ and $z = 0$.

The Hankel assertion is now Theorem 2.1, applied before imposing synchronization; imposing it changes the feasible residual set, not the ambient Fisher metric. The native visibility theorem and Theorem G.3 recover the normalized endpoint representation. Endpoint congruences are changes of variables in $z \geq 0$ for every real y , not just $y \geq 0$. Hence every recovered representative has the same extension $\hat{G}_Q$ and the same nonlinear contact formula.

Finally $\ell_i(\eta)^2 = (a_i^\top \eta)^2 + O(|\eta|^3)$, uniformly. For a small enough fixed neighbourhood, the analogue of (4.4) still holds. At $\eta = \sqrt{t}\xi$ the extra normalized term is $O(\sqrt{t})$ on bounded sets. The same localization and two-sided minimum argument proves the perturbation and compact-uniform conclusions. $\square$

Example 4.2 (Two distinct observed sheets and a critical quadratic cone). Take $d = 2, k = 3$ and loadings $(1, 0), (0, 1), (1, 1)$, with any three separated interior sites and zero, one, or two endpoints. The root-variance image is exactly

$$\mathcal{V}_A = \{v \in \mathbb{R}_+^3 : (v_3 - v_1 - v_2)^2 = 4v_1v_2\}.$$

To prove the converse, choose $\xi_1 = \sqrt{v_1}$ and, when $v_1 > 0$, choose $\xi_2 = (v_3 - v_1 - v_2)/(2\sqrt{v_1})$; the identity gives $\xi_2^2 = v_2$. The case $v_1 = 0$ follows directly.

The marked fast map $F(\eta) = (\eta_1^2, \eta_2^2)$ has four real inverse points over $t(b_1, b_2)$ with $b_1, b_2 > 0$. The third variance has the two distinct observed values

$$t(\sqrt{b_1} + \sqrt{b_2})^2, \quad t(\sqrt{b_1} - \sqrt{b_2})^2.$$

They correspond to the relative sign, not the simultaneous sign which is observationally redundant. The slow graphs in fast coordinates are $g_\pm(u) = u_1 + u_2 \pm 2\sqrt{u_1u_2}$. On a tube of radius ϵt about $t(b_1, b_2)$, with fixed sufficiently small $\epsilon > 0$, their slopes are bounded and their second derivatives have supremum comparable to t^{-1} . If the target third variance is tb_3 different from at least one of the displayed values, the largest centre residual is comparable to t , so $cM \asymp 1$. Here c/r also need not vanish: the critical theorem does not assume an artificially separated inverse tube. Formula (4.2) gives the entire cone, including the loci where the fast sheets join.

With two endpoints and one colour, $k = 3$ is fully visible for every positive exposure vector. Thus in this explicit singular family the labelled endpoint profile reconstructs the native metric for the non-affine critical contact, and the fixed-budget native information lies on the hypersurface of Theorem B.3. No metric was selected to manufacture these properties.

Remark 4.3 (Exact rank constraint). The rank-one constraint in $\mathcal{V}_A$ is part of the observation geometry. Replacing it by the whole positive semidefinite cone is a convex relaxation and can change (4.3). The rank-one lifting and sign quotient are classical in phase retrieval [12, 13]; the claim here is the displayed coupling to the binary experiment's forced inverse information and reconstructed endpoint profile. Statistical tangent-cone limits in [15] likewise motivate, but do not supply, this model-specific coupling.

5. THE CONORMAL RIGIDITY THEOREM

Fix the endpoint-free native experiment of Section 2, with k interior double roots, $2k + 1$ clocks, and unknown positive exposures. Fix $d \geq 2$ and nonzero loadings $a_i \in \mathbb{R}^d$ such that

$$\mathcal{A}_A : \text{Sym}_d \longrightarrow \mathbb{R}^k, \quad S \longmapsto (a_i^\top S a_i)_i$$

is injective. Put $q_A(\eta) = \mathcal{A}_A(\eta\eta^\top)$ and $\mathcal{V}_A = q_A(\mathbb{R}^d)$. For $R > 0$ let

$$C_R(b) = \min_{\eta \in \mathbb{R}^d} (q_A(\eta) - b)^\top R^{-1} (q_A(\eta) - b), \quad b \in \mathbb{R}_+^k.$$

The full probability residual also includes the free score directions, which have been profiled out in this definition. Theorem 4.1 supplies its statistical interpretation. We call η admissible when $\eta \neq 0$ and every $a_i^\top \eta \neq 0$; then $q_A(\eta)$ is a strictly positive target.

At such a point set

$$N_\eta = \ker Dq_A(\eta)^\top, \quad H_R(\eta) = \tfrac{1}{2} D^2 C_R(q_A(\eta)).$$

Let Π_η be the Euclidean orthogonal projection onto N_η . The quadratic form $H_R(\eta)$ restricted to N_η is positive definite. For $n \in N_\eta$, its *dual second-jet value* is

$$h_R(\eta, n) = n^\top (H_R(\eta)|_{N_\eta})^{-1} n. \quad (5.1)$$

This is determined by the observed second jet and known tangent space. In codimension greater than one it is not, in general, the reciprocal of one directional second derivative. The finite query count below is for the scalar oracle (5.1); the three-root case in Section 6 reduces it to ordinary directional curvatures.

Write $\mathcal{R}_k$ for the native inverse-information cone and $\mathcal{W}_k$ for its linear span. If $d = 2$, define the symmetric matrix B_A by

$$(B_A)_{ii} = 0, \quad (B_A)_{ij} = \det(a_i, a_j)^2 \quad (i \neq j),$$

and, for $i < j < l$, put

$$\tau_{ijl}(A) = \frac{(r_j - r_i)(B_A)_{ij}}{d_i d_j} - \frac{(r_l - r_i)(B_A)_{il}}{d_i d_l} + \frac{(r_l - r_j)(B_A)_{jl}}{d_j d_l}. \quad (5.2)$$

The d_i here are the interpolation coefficients of (2.5), not the dimension d .

Theorem 5.1 (Conormal rigidity in native Hankel geometry). *All loadings, sites, clocks, and central channel parameters are fixed as above.*

(a) Complete second-jet ambiguity. *If $d = 2$, two positive definite matrices R, R' give identical second contact jets at every admissible point if and only if $R' - R \in \mathbb{R}B_A$. If $d \geq 3$, identical second contact jets imply $R' = R$. These are statements for arbitrary positive metrics, before imposing native information geometry.*

(b) Exact native criterion. *The native image has dimension $2k - 1$, and*

$$\mathcal{W}_k = \left\{ X \in \text{Sym}_k : \frac{(r_j - r_i)X_{ij}}{d_i d_j} - \frac{(r_l - r_i)X_{il}}{d_i d_l} + \frac{(r_l - r_j)X_{jl}}{d_j d_l} = 0 \quad (i < j < l) \right\}. \quad (5.3)$$

For $d = 2$, the full second-jet observation is injective on $\mathcal{R}_k$ exactly when at least one $\tau_{ijl}(A)$ is nonzero. When all vanish its fibres are exactly affine lines parallel to B_A intersected with $\mathcal{R}_k$. For $d \geq 3$ it is always injective on $\mathcal{R}_k$.

(c) Finite recovery and contact-data fibres. In each injective case, $2k - 1$ suitably chosen scalar dual second-jet values recover R linearly, and no smaller number suffices, even for deterministic adaptive choices. They therefore determine $C_R(b)$ for every $b \geq 0$. At unit total exposure the attainable vectors of these $2k - 1$ dual values are a linear image of $\{y \in \mathcal{M} : \chi(y) \leq 1\}$. Their relative boundary has a positive-Hessian analytic defining function. Each boundary datum has one positive-exposure design; each interior datum has a design fibre diffeomorphic to S^1 . The same sharp query lower bound holds on the budget interior.

Recovery is Lipschitz with the reciprocal of the smallest singular value of the selected linear measurement operator on $\mathcal{W}_k$. On compact positive-metric families it controls C_R uniformly on compact target sets. For $d = 2$, transversality in (5.2), and not positive definiteness alone, removes the otherwise unavoidable one-dimensional ambiguity.

Lemma 5.2 (Normal restriction of inverse information). *At every admissible point,*

$$H_R(\eta) = \Pi_\eta(\Pi_\eta R \Pi_\eta|_{N_\eta})^{-1} \Pi_\eta, \quad h_R(\eta, n) = n^\top R n. \quad (5.4)$$

Proof. Injectivity of $\mathcal{A}_A$ identifies the image of rank-one positive matrices away from zero with a smooth embedded manifold of dimension d . The map is proper, since $\|q_A(\eta)\| \geq c|\eta|^2$. Squared distance in the fixed metric is smooth in a local tubular neighbourhood. To verify that the local projection suffices globally, the trial at the base point bounds the minimum by the squared displacement. Positive definiteness then forces every minimizing image to approach that base point; properness rules out escape. In local coordinates, the stationary equations have the positive tangent Gram matrix as their derivative, so the implicit function theorem gives a unique smooth minimizer.

The quadratic term is obtained by minimizing over the tangent space. If u is a displacement, the minimizing residual w satisfies $R^{-1}w \in N_\eta$ and $\Pi_\eta w = \Pi_\eta u$. Writing $w = Rn$ gives $\Pi_\eta R \Pi_\eta n = \Pi_\eta u$ and the displayed quadratic form. Inverting its normal restriction proves the dual identity. $\square$

Lemma 5.3 (Quadratic forms on the conormal union). *Let $\mathcal{A}_A^*(n) = \sum_i n_i a_i a_i^\top$. A quadratic form $F(n)$ on $\mathbb{R}^k$ vanishes on N_η for every admissible η if and only if it is zero when $d \geq 3$, or is a multiple of $\det \mathcal{A}_A^*(n)$ when $d = 2$.*

Proof. The adjoint is onto Sym_d . Moreover $n \in N_\eta$ is equivalent to $\mathcal{A}_A^*(n)\eta = 0$. Consequently the union of these normal spaces is dense in the inverse image of the singular symmetric matrices. For clarity, if a singular matrix S has a null vector on an excluded loading hyperplane, choose orthogonal maps $U_j \rightarrow I$ taking that vector off all finitely many excluded hyperplanes. Then $U_j S U_j^\top$ has an admissible null vector. A fixed linear right inverse of $\mathcal{A}_A^*$ lifts these perturbations and preserves convergence to any prescribed lift of S . Thus continuity extends vanishing to all lifts of singular matrices.

Choose a linear splitting $n = L(S) + z$, $z \in \ker \mathcal{A}_A^*$. The term purely quadratic in z vanishes by taking $S = 0$. The mixed term vanishes by taking singular S and varying z : singular symmetric matrices span Sym_d for $d \geq 2$, since rank-one matrices do. It remains to classify quadratic forms $f(S)$ vanishing on singular symmetric matrices.

For $d \geq 3$, fix an orthogonal matrix U and restrict to $S = U \text{diag}(t_1, \dots, t_d) U^\top$. The resulting degree-two polynomial vanishes whenever any $t_i = 0$. Every monomial has degree at most two and hence omits some variable when $d \geq 3$; its coefficient must therefore vanish. Every symmetric matrix has such an orthogonal diagonalization, so $f = 0$.

For $d = 2$, write $S = \begin{pmatrix} x & y \\ y & z \end{pmatrix}$. Substituting $S = (u, v)(u, v)^\top$ into a general quadratic form shows that the coefficients of x^2, z^2, xy, yz vanish and that the coefficients of xz, y^2 are opposite. Thus f is a multiple of $xz - y^2$. Conversely this determinant vanishes on the entire normal union. Finally

$$2 \det \mathcal{A}_A^*(n) = 2 \sum_{i < j} \det(a_i, a_j)^2 n_i n_j = n^\top B_A n.$$

This identifies the nonzero ambiguity direction when $d = 2$. $\square$

Proof of Theorem 5.1. By Lemma 5.2, equality of second jets is equivalent to equality of the restrictions of R to all normal spaces. Polarization and Lemma 5.3 give (a).

For each clock, the Cauchy column $c_i(T) = d_i/(T - r_i)$ satisfies every three-site relation in (5.3), by multiplying through the three denominators. Positive sums therefore satisfy them. Conversely those linear relations specify a space of dimension $2k - 1$: the k diagonal entries are arbitrary, and the off-diagonal entries are specified by $k - 1$ numbers $u_2, \dots, u_k$ through

$$\frac{X_{ij}}{d_i d_j} = \frac{u_j - u_i}{r_j - r_i}, \quad u_1 = 0.$$

The native span has the same dimension by Theorem 2.1, proving (b). In dimension two, its intersection with the ambiguity line is zero precisely when at least one displayed three-site functional is nonzero on B_A . Relative openness of the native cone gives the asserted nontrivial line fibres in the exceptional case.

In every injective case the scalar functionals $X \mapsto n^\top X n$, for accessible conormals n , separate $\mathcal{W}_k$. A finite basis of $2k - 1$ such functionals therefore exists. Their linear inversion proves finite recovery. A deterministic adaptive procedure using fewer queries leaves a nonzero common kernel at any interior reference metric. Small opposite perturbations along it give the same transcript and remain native; this proves the lower bound.

The moment map has corank $(2k + 1) - (2k - 1) = 2$. Lemma 3.1 and Theorem B.3 give the budget image and its singleton or spherical fibres. The selected linear measurements are an isomorphism from moment space, so they transport that image, its positive defining Hessian, and its design fibres exactly. The budget interior is relatively open in $\mathcal{W}_k$, giving the same lower bound. Finally finite-dimensional inversion gives the singular-value estimate. Comparing minimizing residuals in two uniformly positive metrics gives uniform control of the entire minimum, without requiring unique minimizers; the explicit estimate is supplied in the proof of Corollary 6.1 below. $\square$

6. CONTACT TOMOGRAPHY AND THE HANKEL CONSTRAINT

We consider three separated interior double roots and no endpoints in the native experiment of Section 2. There are seven clocks. The roots, clocks, calibrated channels, central mixing probability, and component colours are known; the seven positive exposures are not supplied. The unknown retained inverse information belongs to the open native cone

$$\mathcal{R} = \{J_* H(\omega) J_*^\top : \omega \in \mathbb{R}_{>0}^7\}.$$

Here and below “open” for this cone means open in its five-dimensional linear span. The loading vectors $a_1, a_2, a_3 \in \mathbb{R}^2$ are pairwise linearly independent. Set

$$\ell_i(\eta) = a_i^\top \eta, \quad \Delta_{ij} = \det(a_i, a_j), \quad q(\eta) = (\ell_i(\eta)^2)_{i=1}^3.$$

The loading tensors are a basis of Sym_2 . Thus the cone $\mathcal{V}_A = q(\mathbb{R}^2)$ is a linearly embedded rank-one positive cone, with its vertex at zero. For any $R > 0$ define

$$C_R(b) = \min_{\eta \in \mathbb{R}^2} (q(\eta) - b)^\top R^{-1} (q(\eta) - b), \quad b \in \mathbb{R}_+^3. \quad (6.1)$$

For $R \in \mathcal{R}$, Theorem 4.1 identifies this function with the leading Hellinger contact of the synchronized experiment. We first take that small-time limit; the second derivatives below are derivatives of the resulting function of b . No interchange of a finite-sample derivative with the small-time limit is asserted.

The following conormal field is fixed by the loading geometry:

$$n_A(\eta) = (\Delta_{23}\ell_2\ell_3, -\Delta_{13}\ell_1\ell_3, \Delta_{12}\ell_1\ell_2)^\top. \quad (6.2)$$

Let $B_A \in \text{Sym}_3$ have zero diagonal and $(B_A)_{ij} = \Delta_{ij}^2$ for $i \neq j$. In physical root order $r_1 < r_2 < r_3$, let d_i be the nonzero interpolation coefficients in (2.5), and define $K \in \text{Sym}_3$ by

$$K_{ii} = 0, \quad K_{12} = \frac{r_2 - r_1}{2d_1d_2}, \quad K_{13} = -\frac{r_3 - r_1}{2d_1d_3}, \quad K_{23} = \frac{r_3 - r_2}{2d_2d_3}. \quad (6.3)$$

Our transversality invariant is

$$\tau_A = \text{tr}(KB_A) = \frac{(r_2 - r_1)\Delta_{12}^2}{d_1d_2} - \frac{(r_3 - r_1)\Delta_{13}^2}{d_1d_3} + \frac{(r_3 - r_2)\Delta_{23}^2}{d_2d_3}. \quad (6.4)$$

Its vanishing is unchanged by an invertible change of the two shared parameters: each squared determinant is multiplied by the same positive factor. This is a relation between the conormal of the synchronized cone and the native information hyperplane, not a positivity condition on a freely selected metric.

Choose five nonzero vectors $\eta_1, \dots, \eta_5$ in distinct real projective directions, avoiding the three lines $\ell_i = 0$. Then $b_j = q(\eta_j) > 0$. Put $n_j = n_A(\eta_j)$ and $v_j = n_j / (n_j^\top n_j)$. The measured scalar contact jet and its reciprocal are

$$c_j(R) = \lim_{\epsilon \rightarrow 0} \epsilon^{-2} C_R(b_j + \epsilon v_j), \quad h_j(R) = c_j(R)^{-1}. \quad (6.5)$$

Both signs of sufficiently small ϵ give admissible targets. These are exact limiting scalar data; their number is not a sample-complexity assertion.

Corollary 6.1 (Five directional contact curvatures). *In the experiment and with the five projective directions just specified, the following statements hold.*

(i) Sharp transversality. *The five contact jets determine the native inverse information $R \in \mathcal{R}$ if and only if $\tau_A \neq 0$. In that case they determine $C_R(b)$ for every $b \geq 0$. More precisely,*

$$h_j(R) = n_j^\top R n_j, \quad \text{tr}(KR) = 0. \quad (6.6)$$

These six equations have one solution in Sym_3 . If R_0 is any solution of the first five equations, that solution is

$$R = R_0 - \frac{\text{tr}(KR_0)}{\tau_A} B_A. \quad (6.7)$$

Five is the least number of such scalar contact-jet queries sufficient on the open native image, even for deterministic adaptive queries. It is also least on the interior of the fixed-total-exposure image.

(ii) Exact ambiguity without the native constraint. *For arbitrary positive definite R, R' , all second contact jets at the smooth positive part of $\mathcal{V}_A$ coincide if and only if $R' - R \in \mathbb{R}B_A$. Thus arbitrary positive metrics have a one-dimensional ambiguity which the native constraint removes precisely when $\tau_A \neq 0$. When $\tau_A = 0$, the native fibre of any attainable jet data is exactly*

$$(R_0 + \mathbb{R}B_A) \cap \mathcal{R}. \quad (6.8)$$

It contains a nontrivial open interval about every one of its points. At fixed total exposure one intersects this fibre with the budget image; a nontrivial interval persists at every budget-interior point, but need not persist at its boundary. Equality of second jets is not asserted to imply equality of the entire contact functions for the ambiguous metrics.

(iii) Geometry of the observable budget image. Suppose $\tau_A \neq 0$ and $\sum_j \lambda_j = 1$. Write $y = (y_0, \dots, y_4)$ for the moment coordinates of H , and let χ be the reciprocal budget function of Theorem B.3. The linear map

$$\mathcal{L} : y \mapsto (n_j^\top J_* H(y) J_*^\top n_j)_{j=1}^5$$

is invertible. The attainable reciprocal contact-jet vectors form exactly

$$\{h \in \mathcal{L}(\mathcal{M}) : \chi(\mathcal{L}^{-1}h) \leq 1\}. \quad (6.9)$$

This is a convex region. Its relative boundary inside $\mathcal{L}(\mathcal{M})$ is the analytic hypersurface $\chi(\mathcal{L}^{-1}h) = 1$, with positive definite Hessian of its defining function. A boundary datum comes from one positive exposure vector. An interior datum comes from a fibre diffeomorphic to S^1 . The circle is a differential-topological assertion, not an affine circle in exposure coordinates.

(iv) Stability. Let $\mathcal{W} = \{X \in \text{Sym}_3 : \text{tr}(KX) = 0\}$ and

$$\sigma_{\text{tom}} = \min_{X \in \mathcal{W}, \|X\|_F=1} \left(\sum_{j=1}^5 |n_j^\top X n_j|^2 \right)^{1/2}. \quad (6.10)$$

Then $\sigma_{\text{tom}} > 0$ if and only if $\tau_A \neq 0$, and

$$\|R - R'\|_F \leq \sigma_{\text{tom}}^{-1} \|h(R) - h(R')\|_2 \quad (R, R' \in \mathcal{R}).$$

On compact positive-metric families, this estimate controls the entire contact function uniformly on compact target sets, including targets with nonunique minimizing root configurations. All constants are uniform on compact configurations with σ_{tom} bounded below.

6.1. Conormal measurements and their unique missing direction.

Lemma 6.2 (Second contact jet). *For $\eta \neq 0$ with all $\ell_i(\eta) \neq 0$, the function C_R is smooth in a neighbourhood of $q(\eta)$ and*

$$D^2 C_R(q(\eta)) = \frac{2n_A(\eta)n_A(\eta)^\top}{n_A(\eta)^\top R n_A(\eta)}. \quad (6.11)$$

Proof. The map $S \mapsto (a_i^\top S a_i)_i$ is a linear isomorphism on Sym_2 : a quadratic form vanishing on three distinct projective lines is zero. The rank-one positive matrices away from zero form a smooth embedded surface. Therefore $\mathcal{V}_A$ is smooth there. The elementary three-vector determinant identity gives $Dq(\eta)^\top n_A(\eta) = 0$, and $n_A(\eta) \neq 0$.

Nearest-point projection in a fixed positive metric is smooth in a sufficiently small tubular neighbourhood of any point of this surface. Here this can be proved directly by solving the stationary equations in two local surface coordinates: at a point on the surface the coordinate Hessian is twice the positive definite tangent Gram matrix, so the implicit function theorem applies. This local minimizer is the global one near the point. Indeed the trial at $q(\eta)$ makes the minimum tend to zero, hence positive definiteness forces every minimizing image point to tend to $q(\eta)$. Closedness and properness of the quadratic cone rule out escape; the local surface chart then contains every minimizer.

For a displacement δb , minimizing its quadratic cost modulo the tangent plane $n^\top u = 0$ gives residual $Rn(n^\top \delta b)/(n^\top Rn)$ and cost $(n^\top \delta b)^2/(n^\top Rn)$. This is the quadratic term of the squared distance and proves the formula, including the factor two. $\square$

Lemma 6.3 (Conormal kernel). *The map*

$$\mathbf{p} : \text{Sym}_3 \longrightarrow \mathbb{R}[\eta_1, \eta_2]_4, \quad X \longmapsto n_A(\eta)^\top X n_A(\eta)$$

has rank five and kernel $\mathbb{R}B_A$. Evaluation at any five distinct real projective directions has the same kernel.

Proof. The three coordinates of n_A are independent quadratic forms. To see independence, take a nonzero vector on $\ell_i = 0$; exactly the i th coordinate of n_A is nonzero. They are therefore a basis of binary quadratics. Products of binary quadratics span all binary quartics, a five-dimensional space. Hence $\mathbf{p}$ is onto and has a one-dimensional kernel.

The matrix $S(\eta) = \sum_i n_i(\eta) a_i a_i^\top$ kills η , since $Dq(\eta)^\top n_A(\eta) = 0$. Thus

$$0 = 2 \det S(\eta) = 2 \sum_{i < j} \Delta_{ij}^2 n_i(\eta) n_j(\eta) = n_A(\eta)^\top B_A n_A(\eta).$$

As $B_A \neq 0$, this identifies the kernel. A nonzero homogeneous binary quartic has at most four distinct real projective zeros. Five evaluations are therefore injective on its coefficient space and have precisely the same kernel on Sym_3 . $\square$

Lemma 6.4 (Native annihilator). *The linear span of $\mathcal{R}$ is $\mathcal{W} = K^\perp$.*

Proof. For $c(T) = (d_i/(T - r_i))_{i=1}^3$, multiplication by $(T - r_1)(T - r_2)(T - r_3)$ gives

$$c(T)^\top K c(T) \prod_i (T - r_i) = (r_2 - r_1)(T - r_3) - (r_3 - r_1)(T - r_2) + (r_3 - r_2)(T - r_1) = 0.$$

Every native inverse information is a positive sum of these Cauchy tensors, so it lies in $K^\perp$. Theorem 2.1 gives span dimension five. Since $K \neq 0$, its orthogonal hyperplane also has dimension five, proving equality. $\square$

Proof of Corollary 6.1. Lemma 6.2 gives $c_j(R) = 1/(n_j^\top R n_j) > 0$. By Lemma 6.3, two symmetric solutions of these data differ by a multiple of B_A . Lemma 6.4 shows that this line is transverse to the native hyperplane exactly when $\text{tr}(K B_A) \neq 0$. This proves the formula and both directions of the injectivity assertion, as well as the general-metric ambiguity. When the line is contained in the hyperplane, its intersection with the relatively open convex cone is a relatively open interval. A sufficiently small segment about an interior point remains in that cone. The same reasoning applies at a budget-interior point.

For the lower bound on the number of scalar queries, run a proposed deterministic adaptive procedure at an interior native metric. Each scalar answer, after reciprocation, imposes one linear equation on $\mathcal{W}$. Fewer than five queries leave a nonzero common kernel direction. Small opposite perturbations in this direction remain feasible and give the same transcript, including the adaptively chosen directions. The fixed-budget interior is also relatively open in $\mathcal{W}$, by Theorem B.3, so the argument is unchanged there. This lower bound concerns exactly the scalar oracle in (6.5), not an oracle returning an arbitrary function.

The linear moment parametrization and the now injective five measurements give the isomorphism $\mathcal{L}$. At seven clocks the moment map has kernel dimension $7 - 5 = 2$. Theorem B.3, or the abstract reciprocal-fibre Lemma 3.1 below, identifies the budget image by $\chi \leq 1$, its boundary by $\chi = 1$, and its fibres by a point or S^1 . Under the linear change $\mathcal{L}$ the defining Hessian remains positive definite. Euler's identity $y^\top \nabla \chi(y) = -\chi(y)$ makes the gradient nonzero on the boundary. This proves (iii).

The smallest-singular-value estimate gives (iv). For completeness, suppose the eigenvalues of R, R' lie in $[a, b] \subset (0, \infty)$ and $|b_0| \leq M$. A minimizer for $C_R(b_0)$ obeys

$$C_R(b_0) \leq |b_0|^2/a, \quad |q(\eta) - b_0|^2 \leq (b/a)M^2.$$

Testing this minimizer in the other metric, and conversely, bounds the contact difference by $(b/a)M^2\|R^{-1} - R'^{-1}\|$, which is at most $(b/a^3)M^2\|R - R'\|$. This argument does not require uniqueness of the minimizer. Conversion between c and $h = 1/c$ is uniformly Lipschitz when the metric and the chosen nonzero normals range over compact nondegenerate sets. Continuity and compactness give the last uniform assertion. $\square$

6.2. The exceptional locus and an exact native example. For one-colour roots the three d_i have the same sign. With $a_1 = (1, 0)$, $a_2 = (0, 1)$, and $a_3 = (x, y)$, $xy \neq 0$, the condition is

$$\tau_A = \frac{r_2 - r_1}{d_1 d_2} - \frac{(r_3 - r_1)y^2}{d_1 d_3} + \frac{(r_3 - r_2)x^2}{d_2 d_3}.$$

It is not identically zero, and for every nonzero x there is a nonzero y making it zero. Thus the exceptional algebraic hypersurface is nonempty, and the transverse class is open and dense in this loading chart. An assertion of recovery for all spanning loadings would be false.

For a rational example, take roots $(1, 2, 3)$, $D = 7/2$, clocks $(4, 5, 6, 7, 8, 9, 10)$, and all variable factors in component one. Use any distinct positive calibrated product tables, an interior mixing probability, and positive exposures. The interpolation factors are

$$(d_1, d_2, d_3) = (-45360, -40320, -1260).$$

For loadings $(1, 0)$, $(0, 1)$, $(1, 1)$ one obtains

$$n_A(u, v) = (-v(u + v), -u(u + v), uv), \quad \tau_A = -\frac{1}{67737600} \neq 0.$$

The five directions $(1, j)$, $j = 1, \dots, 5$, are admissible. In the variable order

$$(R_{11}, R_{22}, R_{33}, R_{12}, R_{13}, R_{23}),$$

the six linear equations (6.6) have determinant $-1/58800$. Changing only the third loading to $(1, \sqrt{37}/8)$ gives $\tau_A = 0$ and a genuine one-dimensional native ambiguity of second jets. Both assertions follow by substitution in the displayed formulas. The accompanying exact checks are finite algebraic checks of these examples, not certification of the universal theorem.

The coupling is now explicit: the conormal has one quadratic ambiguity, while interpolation gives one independent linear restriction on inverse information. Their transversality determines whether observable contact curvature recovers the native metric. Neither positivity alone nor the rank-one lifting identity alone yields this conclusion.

7. INTRINSIC RECOVERY FROM A LABELLED VALUE FUNCTION

The contact theorem concerns a metric reconstructed from conormal observations. Endpoint profiling gives a different inverse question, for which it is important not to supply the answer as an unobserved reference matrix. In this section the input is only a labelled function on $X = \mathbb{R}_+^k$. Labels mean that the value at each x is known; they do not name hidden active sets.

Definition 7.1 (Exact function input). An exact function input is a quantifier-free semialgebraic formula $\Gamma(x, t)$ over the real algebraic numbers whose realization is the graph of a continuous, degree-two homogeneous function $G : X \rightarrow \mathbb{R}$. An equivalent permissible input is a finite rational or real-algebraic polyhedral conic cover C_α of X , together with symmetric matrices S_α over that field, specifying $G(x) = x^\top S_\alpha x$ on C_α , with agreement on overlaps. Hidden endpoint dimensions, active-set names, and any reference representation are absent. Exact real coefficients may instead be used in a real-closed-field oracle model; the effective assertion below is for the stated algebraic input.

For $e \geq 0$, write $Q = \begin{pmatrix} A & B \\ B^\top & C \end{pmatrix} > 0$ and

$$G_Q(x) = \min_{z \in \mathbb{R}_+^e} (x, z)^\top Q(x, z).$$

Normalize $C_{ii} = 1$. For $e = 0$, this condition is void. Positive diagonal endpoint rescaling puts every positive representation in this normalization. Only endpoint permutations remain as the elementary finite gauge; other points in a fibre need not be gauge equivalent.

Theorem 7.2 (Function-input minimal representation fibre). *From the exact input Γ and any nonnegative integer e , one can construct a finite quantifier-free semialgebraic description of*

$$\mathcal{F}_e(G) = \{Q > 0 : C_{ii} = 1, G_Q(x) = G(x) \text{ for every } x \in X\}.$$

This construction requires no reference representation of G , and is valid at every visibility and column-rank stratum. It decides emptiness of $\mathcal{F}_e(G)$ and, when nonempty, returns an algebraic sample representation and a description of all normalized representations of that dimension.

If the input is promised to have a finite positive orthant-quadratic representation, enumeration of $e = 0, 1, \dots$ terminates at its intrinsic minimum $e_{\min}$ and returns the entire fibre $\mathcal{F}_{e_{\min}}(G)$. Endpoint permutations and positive diagonal rescalings recover all unnormalized minimal representations. Without the representability promise, the same procedure decides the question for every supplied finite upper bound on e ; termination of unbounded search in the nonrepresentable case is not asserted.

Proof. Positive definiteness is expressed by strict positivity of the leading principal minors. For a fixed e , the unknown coefficients of A, B, C are finitely many real variables. Introduce $w = B^\top x + Cz$ and the formula

$$z_i \geq 0, \quad w_i \geq 0, \quad z_i w_i = 0 \quad (1 \leq i \leq e), \quad t = x^\top A x + 2x^\top B z + z^\top C z. \quad (7.1)$$

All expressions are polynomial, including products of unknown coefficients and quantified variables. Since $C > 0$, the objective is strictly convex and coercive on the orthant. Its unique minimizer satisfies these conditions, and these conditions are sufficient: for another feasible z' , the objective difference is

$$2w^\top(z' - z) + (z' - z)^\top C(z' - z) \geq 0.$$

Complementarity gives $w^\top z = 0$.

Conjoin positivity and normalization with

$$\forall x \forall t [(x \in X \wedge \Gamma(x, t)) \implies \exists z (7.1)]. \quad (7.2)$$

By the definition of the input graph, there is exactly one such t for every $x \in X$. The KKT equivalence therefore proves that this formula defines exactly $\mathcal{F}_e(G)$, in both directions. Quantifier elimination over real closed fields gives a quantifier-free formula in the coefficients of Q ; sign determination and sample-point algorithms decide emptiness and return a sample over the real algebraic numbers [5]. No coefficients of a hidden reference Q appear in this operation.

If a finite positive representation exists, diagonal normalization preserves it, and the enumeration eventually reaches its dimension. The first nonempty fibre is therefore precisely the minimal one. Conversely every point of that fibre is a minimal representation, by emptiness at smaller dimensions. Applying positive diagonal congruences in endpoint coordinates and endpoint permutations gives exactly the unnormalized fibre. This proves the claims, including the termination qualification. $\square$

Proposition 7.3 (A finite atlas implementation). *For the polyhedral-quadratic input in Definition 7.1, the equality test for a fixed candidate Q can be made by intersections of observed cones with candidate KKT cones. It never requires the active fan of a reference representation. If $K_I(Q)$ is a candidate KKT cone and $S_I(Q)$ its value matrix, equality holds if and only if, for every α, I and every pair of generators u, v of $C_\alpha \cap K_I(Q)$,*

$$u^\top (S_\alpha - S_I(Q))v = 0. \quad (7.3)$$

The graph formula (7.2) gives a uniform semialgebraic description while the candidate cones and their generators vary.

Proof. The observed cones and the candidate KKT cones cover X . On an intersection the difference is one homogeneous quadratic polynomial. It vanishes identically on the cone if and

only if the corresponding symmetric bilinear form vanishes on its linear span. Testing generators in pairs is sufficient. It is also necessary: vanishing on generators gives the diagonal terms, and vanishing on their sums gives the mixed terms by polarization. This includes lower-dimensional intersections. The resulting identities give equality everywhere on the cover. For variable candidates it is unnecessary to fix a combinatorial type of their generators: the polynomial quantified formula above already handles the varying fan exactly. $\square$

These results distinguish three levels of inversion. Theorems G.3, H.1, and H.3 give explicit formulas and low-dimensional moduli on the first strata. Theorem I.5 is retained as a reference-presentation equality test. Theorem 7.2 supplies the missing function-only input and the complete effective minimal fibre at arbitrary strata. Quantifier elimination is not a polynomial-time claim, and a finite semialgebraic description is not an explicit catalogue of normal forms. Hardt triviality gives qualitative finite partitions for finite-dimensional families; it does not change this distinction.

For instance, if $G(x) = x^\top Sx$ with $S > 0$, the algorithm stops at $e = 0$, even if the function was originally produced by a much larger representation with nonnegative cross block and identically zero optimal endpoints. Recovering a hidden presentation's endpoint count would give the wrong invariant. The same distinction applies at all visibility degenerations.

APPENDIX A. MARKED OBJECTS AND THEIR EQUIVALENCES

Time is a marked coordinate throughout. A residual germ is the feasible image of $R(x, t) = O(x) - o(t)$, together with $t > 0$, its limiting observation, and a positive quadratic metric on observation increments. A source equivalence is a germ of a homeomorphism of feasible sets over time which preserves O . It preserves the residual image, whether or not the source coordinates are regular. An observation coordinate change with derivative $T \in \text{GL}_m(\mathbb{R})$ transports

$$(E, H, \mathcal{C}(E)) \longmapsto (TE, T^{-\top}HT^{-1}, TC(E)T^\top).$$

Thus the numerical metric cost is unchanged. Replacing t by s^a is a change of marking, not an equivalence preserving the numerical contact order.

For a finite-map presentation the fast map F is additionally marked. Fast coordinate differentiation means differentiation in F , not in an arbitrary source chart. The derivative certificates below are invariants of this marked presentation and transported tubes. The residual set that they construct is an invariant of the observation germ without this additional presentation.

An endpoint object consists of a labelled retained cone $X = \mathbb{R}_+^k$ and the function $G : X \rightarrow \mathbb{R}$. A representation is a positive matrix

$$Q = \begin{pmatrix} A & B \\ B^\top & C \end{pmatrix} > 0, \quad G_Q(x) = \min_{z \in \mathbb{R}_+^e} (x, z)^\top Q(x, z). \quad (\text{A.1})$$

Its fibre means *all* positive representations inducing the same labelled function. Positive diagonal changes and permutations of z always preserve that function. They need not exhaust its fibre on a nonvisible stratum. Retained coordinate changes are stated separately and are not silently divided out. A minimal distance representation uses $L \in \text{GL}_e(\mathbb{R})$, $T \in \mathbb{R}^{e \times k}$ and $K = L\mathbb{R}_+^e$ in the ambient space $\mathbb{R}^e$. The equivalence $(T, K) \sim (OT, OK)$ uses $O \in \text{O}(e)$. Embeddings in larger ambient spaces are not included in this minimal category.

Finally a labelled statistical quotient is the Gaussian experiment $\{N(Qv, Q) : v \in \mathcal{K}\}$. Statistical equivalence means Markov kernels in both directions preserving each parameter label. Theorem K.1 identifies its equivalence class with Q in the fixed retained coordinates. The dual tensor $R = Q^{-1}$ determines the same matrix object, but a value query $u^\top Ru$ is not a value query $v^\top Qv$. All query statements below specify which of these two oracles is supplied. None is a sample-count statement. These distinctions fix the meaning of “intrinsic” in each result.

APPENDIX B. INTERPOLATION AND FIXED-TOTAL-EXPOSURE GEOMETRY

The finite moment cone in Theorem 2.1 has fixed support. It is therefore polyhedral in its moment coordinates, although its linear span is the Hankel space. We first isolate the coefficient inverse used to derive it, and then impose the exposure constraint.

Lemma B.1 (The rational coefficient inverse). *Let $p = 2k + e + 1$, let h be the degree- $p - 1$ polynomial in Theorem 2.1, and let $T_1, \dots, T_p$ be distinct clocks outside its zeros. The rational functions*

$$1, \quad -\frac{\sigma_i}{T - r_i} \quad (1 \leq i \leq k), \quad \frac{\epsilon_a}{(T - \rho_a)^{\mu_a}} \quad (1 \leq a \leq n)$$

form a basis of the p -dimensional space of rational functions with denominator h and numerator degree at most $p - 1$. Their evaluation matrix $V \in \mathbb{R}^{p \times p}$ is invertible. If $y \in \mathbb{R}^p$ is the vector of evaluations, the retained coefficient vector is Zy , where $Z \in \mathbb{R}^{n \times p}$ and

$$Z_{aj} = \frac{h(T_j)}{P'(T_j)} \frac{-P(\rho_a)}{\epsilon_a h_a(\rho_a)(T_j - \rho_a)}. \quad (\text{B.1})$$

The polynomial coefficient matrix $E \in \mathbb{R}^{n \times n}$, whose rows are $p_(T)/(T - \rho_a)$, is invertible, and*

$$Z_{\cdot j} = \frac{\xi_j}{p_*(T_j)} \text{diag}(d_a) E w(T_j).$$

Proof. A linear relation among these rational functions has zero constant part at infinity. At an interior site its highest principal part forces the retained double-pole coefficient to vanish; its remaining simple-pole coefficient then vanishes. At an endpoint the simple principal part forces the endpoint coefficient to vanish. This proves independence, and the numerator space has the same dimension p .

For a coefficient vector θ , put $g = h(V\theta)$, interpreting $V\theta$ here as the rational function whose evaluations are prescribed. Polynomial interpolation gives

$$g(T) = \sum_{j=1}^p h(T_j) y_j \frac{P(T)}{(T - T_j) P'(T_j)}.$$

At a retained site, the coefficient of its highest pole is $g(\rho_a)/(\epsilon_a h_a(\rho_a))$. In evaluating the displayed interpolation formula, $\rho_a - T_j = -(T_j - \rho_a)$ supplies the minus sign in (B.1). This proves every entry of the $n \times p$ inverse block. Evaluation at $T = \rho_b$ makes the a th polynomial row of E vanish unless $a = b$, when its value is $\prod_{c \neq a} (\rho_a - \rho_c) \neq 0$. Thus these rows form a basis. Multiplying their evaluation vector by $\text{diag}(d_a)$ proves the final identity. $\square$

Example B.2 (An exact four-clock inverse). Take one interior site $r = 1$, a lower endpoint at 0, one colour, and clocks 2, 3, 4, 5. In column order mixing, interior sum, endpoint, variance,

$$V = \begin{pmatrix} 1 & -1 & -1/2 & -1 \\ 1 & -1/2 & -1/3 & -1/4 \\ 1 & -1/3 & -1/4 & -1/9 \\ 1 & -1/4 & -1/5 & -1/16 \end{pmatrix}, \quad Z = \begin{pmatrix} -20 & 240 & -540 & 320 \\ -8 & 72 & -144 & 80 \end{pmatrix}.$$

Direct multiplication gives $ZV = [0_{2 \times 2}, I_2]$. This illustrates the dimensions and signs; Lemma B.1, rather than this finite instance, proves the general formula.

Fix the roots, clocks, channels, and central mixing. Set

$$s = 2n - 1, \quad m = 2k + e + 1, \quad d_0 = m - s = 2 - e, \quad a_j = \frac{\xi_j^2}{p_*(T_j)^2 \kappa_j} > 0.$$

Use moment coordinates $y = (y_0, \dots, y_{s-1})$ for a Hankel matrix, so that $H_{ab} = y_{a+b}$ with indices $0 \leq a, b < n$. Let

$$T_{rj} = T_j^r \quad (0 \leq r < s), \quad \mathcal{M} = T(0, \infty)^m, \quad \phi(\omega) = \sum_{j=1}^m \frac{a_j}{\omega_j}.$$

The letter T in this paragraph denotes the moment matrix, not an individual clock. Its rank is s . For $y \in \mathcal{M}$ define

$$\chi(y) = \min\{\phi(\omega) : \omega > 0, T\omega = y\}. \quad (\text{B.2})$$

Theorem B.3 (Fixed-budget image and all exposure fibres). *The minimum in (B.2) exists and has a unique minimizer $\omega^*(y)$. The function χ is real analytic and strictly convex on the open moment cone $\mathcal{M} \subset \mathbb{R}^s$, and*

$$\chi(ty) = t^{-1}\chi(y), \quad \nabla^2\chi(y) = (TD_y^{-1}T^\top)^{-1} > 0, \quad D_y = \text{diag}\left(\frac{2a_j}{\omega_j^*(y)^3}\right). \quad (\text{B.3})$$

Under the fixed budget $\sum_j \lambda_j = 1$, the native inverse-information image is obtained from the following exact moment image by $y \mapsto J_*H(y)J_*^\top$.

If $e = 2$, the image is the smooth hypersurface $\chi(y) = 1$, of dimension $s - 1$. Each of its points has exactly one positive exposure vector.

If $e = 0$ or 1 , the image is the convex region

$$\{y \in \mathcal{M} : \chi(y) \leq 1\}. \quad (\text{B.4})$$

Its relative boundary in $\mathcal{M}$ is the smooth hypersurface $\chi = 1$. A boundary point has exactly one exposure vector. At an interior point $\chi < 1$, the entire positive-exposure fibre is diffeomorphic to S^{d_0-1} . Thus it consists of two points for $e = 1$, and is a circle for $e = 0$. These statements also describe the native information image after matrix inversion.

Proof. The zeroth moment gives $\sum_j \omega_j = y_0$. Hence the closure of the positive weight fibre is compact. The function ϕ tends to infinity at its boundary because every a_j is positive. It attains a minimum in the positive fibre. Its Hessian is the positive diagonal matrix in (B.3); strict convexity proves uniqueness.

The Lagrange equations have the form

$$\nabla\phi(\omega^*) + T^\top\nu = 0, \quad T\omega^* = y.$$

Their bordered derivative matrix is invertible: if $D_y u + T^\top v = 0$ and $Tu = 0$, then $u^\top D_y u = 0$, so $u = 0$, and full row rank of T gives $v = 0$. The analytic implicit function theorem makes ω^* and ν analytic. Differentiating these equations yields

$$d\omega^* = D_y^{-1}T^\top(TD_y^{-1}T^\top)^{-1}dy, \quad d\nu = -(TD_y^{-1}T^\top)^{-1}dy.$$

Since $\nabla\chi = -\nu$, this proves the Hessian formula and strict convexity. Rescaling the weight fibre proves the homogeneity formula. Euler's identity $\nabla\chi(y) \cdot y = -\chi(y)$ makes every positive level regular.

The exposure-to-weight change is a diffeomorphism of positive orthants: $\omega_j = a_j/\lambda_j$. Its budget equation is exactly $\phi(\omega) = 1$. If $d_0 = 0$, the moment map is invertible and the fibre is a point, proving the $e = 2$ assertion.

Suppose $d_0 > 0$. Identify $\ker T$ with $\mathbb{R}^{d_0}$ by an orthonormal basis. The positive fibre is a bounded open convex set in this affine space, containing its unique minimizer ω^* . On every ray $\omega^* + ru$, $|u| = 1$, the first derivative of ϕ vanishes at $r = 0$ and its second derivative is strictly positive. Its value therefore increases strictly from $\chi(y)$ to infinity at the first boundary point of the ray. Consequently the level $\phi = 1$ is empty when $\chi > 1$, is the singleton minimizer when $\chi = 1$, and otherwise meets every ray in exactly one point. The derivative at that intersection

is positive, so the implicit function theorem makes its radius a smooth function of u . Radial parametrization gives the asserted diffeomorphism with S^{d_0-1} . The coordinatewise inversion from weights to exposures preserves it. Convexity and the boundary description of (B.4) follow from the strict convexity and homogeneity of χ . Congruence and inversion are diffeomorphisms on the positive definite matrix cone, proving the last statement. $\square$

The dimension loss at two endpoints is therefore structural: the moment map has no weight kernel. With fewer endpoints, changing the exposure distribution inside an unobserved moment fibre allows the reciprocal budget to increase from its minimum. The region in (B.4) is a statement about fixed-clock moment coordinates; its nonlinear image under inversion need not be convex.

Corollary B.4 (A scalar budget optimum). *For a fixed retained vector u , put $b_j = (Z_j^\top u)^2 / \kappa_j$. Then*

$$\inf_{\lambda > 0, \sum \lambda_j = 1} u^\top Q^{-1} u = \left(\sum_j \sqrt{b_j} \right)^2.$$

If every $b_j > 0$, the unique optimum is $\lambda_j = \sqrt{b_j} / \sum_i \sqrt{b_i}$. If some but not all b_j vanish, the infimum is approached on the boundary and is not attained with all exposures positive. If all vanish, the objective is identically zero.

Proof. The inverse-information identity gives the objective $\sum_j b_j / \lambda_j$. Cauchy–Schwarz gives the lower bound and its equality conditions. Approximate the indicated boundary design when necessary. $\square$

This scalar allocation identity is a classical design calculation, not a separate novelty claim. The image and complete design fibres in Theorem B.3 are the additional geometric statements.

Proposition B.5 (More than the minimal number of clocks). *Let the same root pattern be observed at $m > p = 2k + e + 1$ distinct clocks, with positive weights $W = \text{diag}(\lambda_j \kappa_j)$. Then the rectangular score matrix $V \in \mathbb{R}^{m \times p}$ has full column rank and the retained Schur complement is positive. If $E_0 \in \mathbb{R}^{p \times n}$ selects the retained coordinates, its inverse is*

$$R = E_0^\top (V^\top W V)^{-1} E_0, \quad u^\top R u = \min_{V^\top a = E_0 u} a^\top W^{-1} a.$$

The minimizer is $a = W V (V^\top W V)^{-1} E_0 u$. At fixed scores the native image is a rational semialgebraic image of the positive weights. Sending the weights at all but a selected p -clock subdesign to zero recovers its square-design inverse information.

The fixed congruence to a Hankel space supplied by a selected square subdesign does not, in general, persist after additional clocks receive positive weight.

Proof. Any p rows form an invertible matrix by Lemma B.1, proving full column rank. Complete the square in the constrained quadratic minimization, or use its multiplier equations, to obtain the displayed optimizer and value. Positivity, rationality, and the boundary limit follow from the Gram formula and continuity of inversion at the positive definite subdesign matrix.

For the final assertion take two interior sites 1, 2, a lower endpoint 0, one colour, and clocks 4, 5, 6, 7, 8, 9, 10. Choose positive exposures so that $W = I$. In column order mixing, the two sums, endpoint, and the two variances, the row at clock T is

$$(1, -(T-1)^{-1}, -(T-2)^{-1}, -T^{-1}, -(T-1)^{-2}, -(T-2)^{-2}).$$

Let J_* be the square-design congruence from the first six clocks. Its diagonal factors are (15120, 20160, 2520) and its polynomial matrix is

$$E = \begin{pmatrix} 2 & -3 & 1 \\ 0 & -2 & 1 \\ 0 & -1 & 1 \end{pmatrix}.$$

Exact rational inversion of the displayed seven-row Gram matrix gives, for $H = J_*^{-1} R J_*^{-\top}$ with indices 0, 1, 2,

$$H_{02} - H_{11} = -\frac{473241649}{108060048} \neq 0.$$

These formulas constitute a rational counterexample which can also be checked by clearing denominators. A common normalization of the exposures only multiplies R and this nonzero difference by the same positive scalar. Thus it also supplies a fixed-budget counterexample. No assertion that every possible alternative congruence is excluded is needed here. $\square$

APPENDIX C. REDUCTION FROM A POLYNOMIAL FINITE MAP

Let $N \subset \mathbb{R}^r$ be compact semialgebraic, the closure of its interior. Polynomials are defined on a neighbourhood of N . Suppose $F^{-1}(a_0) \cap N$ is finite, nonempty, and contained in the interior of N , and $G = b_0$ on that fibre. Let $a(t), b(t)$ be continuous semialgebraic arcs tending to a_0, b_0 . Put

$$\Delta = F(\{x \in N : \det DF(x) = 0\}) \cup F(\partial N). \quad (\text{C.1})$$

Assume $a(t) \in F(N) \setminus \Delta$ for all sufficiently small positive t . Choose a positive continuous semialgebraic function $r(t) \rightarrow 0$ with

$$4r(t) < \text{dist}(a(t), \Delta).$$

When Δ is empty the distance restriction is omitted. The set $\Omega_t = \{x \in N : |F(x) - a(t)| \leq 2r(t)\}$ contains no critical or boundary point. Define the rational matrix and its fast derivative by

$$A_F(x) = DG(x)DF(x)^{-1}, \quad M(t) = \sup_{x \in \Omega_t} \|D_x A_F(x) DF(x)^{-1}\|. \quad (\text{C.2})$$

The last norm is the operator norm of the derivative of a matrix-valued function. Also put

$$c(t) = \max\{|G(x) - b(t)| : x \in N, F(x) = a(t)\}.$$

All suprema are finite and semialgebraic. This follows from compactness, nonvanishing of $\det DF$ on Ω_t , and quantifier elimination.

Definition C.1 (Polar-gap conditions). We say the marked residual satisfies the polar-gap conditions at radius r if

$$\sup_{\Omega_t} \|A_F\| \leq K_0, \quad c(t)/r(t) \rightarrow 0, \quad \varepsilon(t) := r(t)M(t) \rightarrow 0. \quad (\text{C.3})$$

These are inequalities on the original polynomial map and the marked arc, not assumptions about the existence of a normal form.

Theorem C.2 (Intrinsic finite-sheet reduction). *Under the preceding hypotheses and (C.3), the solutions of $F(x) = a(t)$ can be labelled by finitely many semialgebraic arcs $x_b(t)$, $1 \leq b \leq B$. There are inverse maps $\phi_b(U, t)$ on $|U| \leq r(t)$ with $F(\phi_b(U, t)) = a(t) + U$, and these maps exhaust the indicated fast tube. Set*

$$C_b(t) = G(x_b(t)) - b(t), \quad D_b(t) = A_F(x_b(t)).$$

Then

$$R(\phi_b(U, t), t) = (U, C_b(t) + D_b(t)U + E_b(U, t)), \quad |E_b(U, t)| \leq \frac{1}{2}M(t)|U|^2. \quad (\text{C.4})$$

For H in a compact subset of Sym_{r+q}^{++} , let

$$q_b(H, t) = \min_{U \in \mathbb{R}^r} \|(U, C_b(t) + D_b(t)U)\|_H. \quad (\text{C.5})$$

There is a uniform K such that, for all sufficiently small t ,

$$(1 - K\varepsilon(t)) \min_b q_b(H, t) \leq \min_{x \in N} \|R(x, t)\|_H \leq (1 + K\varepsilon(t)) \min_b q_b(H, t). \quad (\text{C.6})$$

The distance is exactly zero if and only if some $C_b(t) = 0$.

If no C_b is eventually zero, write $C_b(t) = c_b t^{\rho_b} + o(t^{\rho_b})$, $c_b \neq 0$, and set $\rho = \max_b \rho_b$. The matrices $D_b(t)$ have limits D_b^0 , and the complete leading residual set is

$$\mathcal{E}_\rho = \bigcup_{\rho_b = \rho} \{(u, D_b^0 u + c_b) : u \in \mathbb{R}^r\}. \quad (\text{C.7})$$

Here $\mathcal{E}_\rho$ is defined directly as the zero-time section of the closure of $\{(t, t^{-\rho} R(x, t)) : x \in N, t > 0\}$. In particular it does not depend on the inverse-sheet construction.

Proof. The restriction of F over $B(a(t), 2r(t))$ is a proper local diffeomorphism with finite nonempty fibres. Properness follows by taking inverse images inside the compact set N ; the absence of $F(\partial N)$ keeps them in its interior. A proper local diffeomorphism is a covering: disjoint inverse neighbourhoods of the finite fibre cover a sufficiently small target neighbourhood, since otherwise a sequence of additional inverse points has a compact limit in the fibre. The ball is simply connected, so every covering component maps diffeomorphically onto it.

Apply the same argument to the map $(t, x) \mapsto (t, F(x) - a(t))$ over $\{0 < t < t_0, |U| < 2r(t)\}$. After shortening t_0 , the arcs are analytic and the base is homeomorphic to an interval times a ball. The covering is therefore trivial. Its components and graphs are semialgebraic; alternatively their graphs are the finitely many definable continuations of the inverse fibre. This supplies the asserted labels and charts, including the closed smaller ball.

Differentiating on a sheet gives $D_U(G \circ \phi_b) = A_F \circ \phi_b$. Its derivative has norm at most $M(t)$ by (C.2). Taylor's integral formula on the convex U -ball proves (C.4).

For a direct relative comparison set $V = C_b + D_b U$. The first coordinate implies $|U| \leq |(U, V)|$, and compact metric bounds imply

$$| |(U, V + E_b)\|_H - \|(U, V)\|_H | \leq K\varepsilon(t) \|(U, V)\|_H.$$

The constant is independent of C_b , including $C_b = 0$. The unrestricted minimizer in (C.5) is

$$U_b^* = -(L_b^\top H L_b)^{-1} L_b^\top H(0, C_b), \quad L_b = \begin{pmatrix} I_r \\ D_b \end{pmatrix}.$$

Since L_b contains I_r and D_b is uniformly bounded, $|U_b^*| \leq K'|C_b| = o(r)$. Thus this minimizer is a feasible trial. The trials $U = 0$ have cost $O(c(t)) = o(r(t))$. Outside the fast tube the norm of the first residual coordinate exceeds $r(t)$, so norm equivalence puts every global minimizer in a chart. Apply the preceding comparison there for the lower bound and at U_b^* for the upper bound. This proves (C.6). A zero forces $U = 0$, and then forces $C_b = 0$; the converse is immediate.

Every bounded semialgebraic matrix arc has a limit, and every nonzero semialgebraic vector arc has a nonzero rational-power leading vector. Uniform bounds on D_b show $q_b(H, t)$ is bounded above and below by positive constant multiples of $|C_b(t)|$: use the inverse of the bounded linear shear $(U, V) \mapsto (U, V + D_b U)$. Hence (C.6) selects the largest order ρ .

For completeness we identify the entire residual set, not only its minimum. A bounded normalized residual has $U = O(t^\rho) = o(r(t))$ and hence belongs to a chart. In that chart $t^{-\rho} E_b = o(1)$. A branch with $\rho_b < \rho$ cannot have a bounded normalized second coordinate because $D_b U = O(t^\rho)$. On a branch of order ρ every limit is $(u, D_b^0 u + c_b)$. Conversely choosing $U = t^\rho u$ realizes each such vector. These are the two inclusions in (C.7). $\square$

Corollary C.3 (Coordinate invariance and uniformity). *Feasible source reparametrizations leave ρ and $\mathcal{E}_\rho$ unchanged. An invertible linear change T of residual coordinates sends $\mathcal{E}_\rho$ to $T\mathcal{E}_\rho$ and its tensor envelope to $TCT^\top$. The same conclusion holds for a smooth observation-coordinate change at the common limiting observation with invertible derivative T . Time is marked; replacing t by s^m multiplies the order by m .*

In a semialgebraic parameter family the estimate (C.6) is uniform whenever the metric bounds, K_0 , and the two limits in (C.3) are uniform. Formula (C.7) is uniform on compact strata on which the stated leading-vector and slope limits are uniform. No uniformity across a vanishing gap is implicit.

Proof. The first statements follow from the definition by the feasible residual image. For a smooth observation change, the difference of its values at two observations is $TR + o(|R|)$ uniformly as both observations approach the common limit, by the integral derivative formula. Every estimate in the proof of Theorem C.2 uses only the displayed uniform constants and limits. This proves the final assertions. $\square$

The distinct affine spaces in (C.7), after duplicates are removed, are uniquely determined by the residual set: they have the same dimension r , and a real affine space cannot be covered by finitely many proper affine subspaces. This assertion concerns $\mathcal{E}_\rho$, not recovery of every sheet from the smaller tensor envelope.

Proposition C.4 (Finite order tests). *For fixed semialgebraic data, the nonzero quantities in (C.3) and (C.2) admit Puiseux leading orders. Thus the two strict limits are equivalent to*

$$\text{ord } c > \text{ord } r, \quad \text{ord } r + \text{ord } M > 0,$$

with the convention $\text{ord } 0 = +\infty$. Boundedness of A_F is tested entrywise on the tube supremum. The branch residues c_b and limiting slopes D_b^0 , together with these order tests, extract the sheets. Scalar order data alone do not specify them.

Proof. Graphs of the suprema and of the finite inverse fibres are semialgebraic by quantifier elimination. One-variable semialgebraic preparation gives each nonzero leading power with nonzero coefficient. Taking the quotient or product gives the displayed equivalences. Apply the same preparation to C_b and D_b . Appendix M explains how proper real charts recover these orders without introducing inaccessible divisor faces. $\square$

Corollary C.5 (Coupled weighted initial maps). *Give x_i positive rational weight w_i . Suppose*

$$F(x) = f(x) + F_{>1}(x), \quad f(t^{w_1}x_1, \dots, t^{w_r}x_r) = tf(x),$$

where every monomial of $F_{>1}$ has weighted degree greater than one and $f^{-1}(0) = \{0\}$. Let η be a regular value of f with a nonempty real fibre. Suppose $a(t) = t\eta + O(t^{1+\delta})$ for some $\delta > 0$, every monomial of G has weighted degree at least $\nu > 1$, and $b(t) = O(t^\nu)$. On a sufficiently small fixed neighbourhood of zero, Theorem C.2 applies with radius $r(t) = c_0t$. In particular

$$c(t) = O(t^\nu), \quad \sup_{\Omega_t} \|A_F\| = O(t^{\nu-1}), \quad M(t) = O(t^{\nu-2}),$$

and the relative error is $O(t^{\nu-1})$. The fast initial equations may be coupled.

Proof. A weighted sphere is compact, and f is nonzero there. Weighted homogeneity therefore implies properness of f : its norm is bounded below by a positive constant times the weighted radius. Consequently the regular fibre at η is finite. A sufficiently small closed target ball about η avoids all critical values and is covered by finitely many smooth inverse charts.

Put $x_i = t^{w_i}\xi_i$. After division by t , the rescaled fast map converges in C^2 on bounded ξ sets to f . No inverse point over that target ball can escape to infinity in ξ . Indeed, on a small fixed

weighted neighbourhood the higher-weight part of F is at most a small multiple of the lower bound for f , so $|F(x)| = O(t)$ implies bounded rescaled weighted radius. The inverse function theorem and compactness now give all inverse charts and uniform first and second derivatives in ξ . They also give a target ball of radius a fixed multiple of t avoiding critical and boundary values. Choose c_0 small enough to meet the factor four in the gap condition.

In a rescaled target coordinate $u = U/t$, the function G on these charts and its first two u derivatives are $O(t^\nu)$. One derivative with respect to U divides by t , and two divide by t^2 . These are exactly the estimates for A_F and M . The centre residual has the asserted bound as well. Thus $c(t)/r(t) = O(t^{\nu-1})$ and $r(t)M(t) = O(t^{\nu-1})$, proving the claim. Uniform versions follow on compact sets of the stated regular values, coefficients, and positive lower bounds. $\square$

Example C.6 (Approaching a ramified discriminant). Take

$$F(x, y) = (x^2 + y^2, xy), \quad a(t) = (t, t/2 - t^{1+\lambda}), \quad \lambda > 0 \text{ rational.}$$

The real critical-value set is the union of the rays $a_1 = 2a_2 \geq 0$ and $a_1 = -2a_2 \geq 0$. For small positive t the four inverse points are given by

$$x + y = \sigma \sqrt{2t - 2t^{1+\lambda}}, \quad x - y = \tau \sqrt{2t^{1+\lambda}}, \quad \sigma, \tau \in \{-1, 1\}.$$

Thus the two inverse separations have different orders and the marked arc approaches a genuine critical-value wall. Choose $r(t) = c_0 t^{1+\lambda}$ with a sufficiently small positive c_0 . If G is any vector-valued homogeneous polynomial of ordinary degree $n > 2 + 2\lambda$ and $b(t) = O(t^{n/2})$, then the polar-gap conditions hold. In fact, uniformly on the fast tube,

$$\begin{aligned} \|(DF)^{-1}\| &= O(t^{-(1+\lambda)/2}), & \|A_F\| &= O(t^{(n-2-\lambda)/2}), \\ M(t) &= O(t^{(n-4-3\lambda)/2}), & c(t)/r(t) &= O(t^{n/2-1-\lambda}). \end{aligned}$$

To check the second-derivative bound, differentiate $DG(DF)^{-1}$ once and multiply by $(DF)^{-1}$. The terms are bounded by $\|D^2G\| \|(DF)^{-1}\|^2$ and $\|DG\| \|D^2F\| \|(DF)^{-1}\|^3$; the latter gives the displayed, weaker bound. Hence $r(t)M(t) = O(t^{(n-2-\lambda)/2}) \rightarrow 0$. Theorem C.2 applies to all four competing sheets and every exact cancellation of their vector residuals. This example is not covered by requiring the target to remain a fixed relative distance from the rescaled discriminant.

APPENDIX D. CRITICAL INITIAL MAPS AND MARKED DERIVATIVE CERTIFICATES

The residual-scale certificate proves affine asymptotics when its strict inequalities hold. At a critical scale, the full weighted leading map can instead survive. The next theorem supplies a normal form in that regime without postulating an affine residual or a small derivative certificate.

For positive rational weights $w = (w_1, \dots, w_r)$ write $\delta_t x = (t^{w_1} x_1, \dots, t^{w_r} x_r)$ and $\nu_w(x) = \max_i |x_i|^{1/w_i}$. A closed cone K below is invariant under all δ_t , and its weighted unit section is compact. Examples include $\mathbb{R}^r$ and coordinate orthants.

Theorem D.1 (Weighted critical residual and coefficient stability). *Let $O(x) = P(x) + H(x)$ be a polynomial observation near zero on K . Suppose that $P(\delta_t x) = tP(x)$, that $P^{-1}(0) \cap K = \{0\}$, and that every monomial of H has weighted degree greater than one. Let the marked target be $o(t) = tb + \epsilon(t)$, where $|\epsilon(t)|/t \rightarrow 0$. On a sufficiently small fixed weighted neighbourhood in K , the complete normalized residual set is*

$$E_b = P(K) - b. \tag{D.1}$$

It is closed, and for every positive definite observation metric G ,

$$\lim_{t \downarrow 0} t^{-2} \min_x \|O(x) - o(t)\|_G^2 = \min_{\xi \in K} \|P(\xi) - b\|_G^2. \tag{D.2}$$

The right side is positive precisely when $b \notin P(K)$. In that case the minimum residual norm has exact order one. With time reparametrized by $t = \tau^q$, its order is q .

These conclusions, including convergence of the minima, are uniform on compact coefficient families with fixed monomial support and weights, uniformly bounded targets, metric eigenvalues in a fixed compact subinterval of $(0, \infty)$, a uniform target remainder, and

$$\inf_{\nu_w(\xi)=1, \xi \in K} |P(\xi)| \geq c_0 > 0. \quad (\text{D.3})$$

In particular the leading polynomials themselves may vary. The limiting minimum is continuous in these coefficients, targets, and metrics. Its positivity is an open condition; on compact subfamilies with a positive lower bound for that minimum, the exact order-one conclusion and its positive leading coefficient are uniform. No simultaneous resolution or constancy of divisorial orders is required for these stability assertions.

Proof. Compactness of the weighted unit section and the absence of a nonzero zero give $|P(x)| \geq c_0 \nu_w(x)$ on K , by homogeneity. Let $\delta > 0$ be the least excess over one of a weighted degree occurring in H ; if $H = 0$, omit the corresponding bounds. On $\nu_w(x) \leq \epsilon$,

$$|H(x)| \leq C \nu_w(x)^{1+\delta}.$$

Choose ϵ so that $C\epsilon^\delta \leq c_0/2$. Then $|O(x)| \geq (c_0/2)\nu_w(x)$. If $(O(x) - o(t))/t$ stays bounded, so does $O(x)/t$, and consequently $\nu_w(x) = O(t)$. Thus $x = \delta_t \xi$ with bounded $\xi \in K$. On such sets,

$$t^{-1}(O(\delta_t \xi) - o(t)) = P(\xi) - b + o(1)$$

uniformly. Subsequences give one inclusion in (D.1), and the substitution $x = \delta_t \xi$ gives the other. The same lower bound for P proves its properness on K , hence the closedness of its image.

The trial $x = 0$ bounds the minimum by $O(t^2)$. Norm equivalence and the preceding coercive estimate therefore bound every rescaled minimizing sequence. Uniform expansion on this bounded set proves the lower bound in (D.2); testing a minimizer of the proper limiting problem proves the upper bound. Its minimum is zero exactly on $P(K)$.

For a compact coefficient family, the degree gap δ , the higher-order constant C , the radius ϵ , and all metric constants can be chosen uniformly. Bounded targets and the trial at zero give a common bound for the rescaled minimizers. The same expansion proves uniform convergence. The limiting minimizers also stay in a common compact subset of K . Uniform continuity of the polynomial objective on that compact set proves continuity of its minimum under coefficient, target, and metric changes. Positivity is consequently open, and a positive uniform lower bound allows taking square roots in the uniform minimum limit. The margin (D.3) is itself open in the leading coefficients, and higher-weight coefficients can be perturbed independently inside bounded sets. This proves genuine coefficient stability, including variations that need not preserve a resolution type. $\square$

The theorem allows coupled leading polynomials of the same weighted order. It is not a claim that every critical germ is weighted homogeneous. On a resolved chart with a proper weighted leading map, it applies to that chart's leading jets; feasibility and coverage of the charts must still be checked.

For synchronized roots, use weight $1/2$ on the shared splitting variables and weight one on sums, mixing, and endpoints. The leading probability map is $\text{La} + \text{D}_x q_A(\eta) + \text{D}_z z$. Its injective ambient score and the coercivity of q_A verify the hypotheses directly on $\mathbb{R}^{k+1} \times \mathbb{R}^d \times \mathbb{R}_+^e$. The model is analytic rather than polynomial in these coordinates, but its Taylor remainder has the same uniform higher-weight estimate used in the proof. Thus the proof applies verbatim. This explains why its critical contact depends on the quadratic splitting map, rather than merely on a polar order. For the three-cluster example, $b = (1, 1, 1)$ does not satisfy the cone equation. It

therefore has a strictly positive contact coefficient, persisting under sufficiently small loading, target, and native-metric perturbations within the stated class.

D.1. Changing a fast presentation. The following chain rule states precisely a useful invariance category for derivative certificates. It also makes clear why an arbitrary replacement of the fast observables is not harmless.

Proposition D.2 (Transverse fast-coordinate transition). *Let an observation sheet be $(U, g(U))$, with $A = Dg$ and $M = \sup \|D^2g\|$ on a specified source region. Let $\mathcal{T} = (\mathcal{T}_F, \mathcal{T}_G)$ be a C^2 observation-coordinate diffeomorphism, and set*

$$H_F = D\mathcal{T}_F \begin{pmatrix} I \\ A \end{pmatrix}, \quad H_G = D\mathcal{T}_G \begin{pmatrix} I \\ A \end{pmatrix}.$$

Assume that A , the first two derivatives of $\mathcal{T}$ and $\mathcal{T}^{-1}$, and H_F^{-1} are uniformly bounded there. Wherever the transformed sheet is expressed in its new fast coordinates, its slope and second fast derivative satisfy

$$\widehat{A} = H_G H_F^{-1}, \quad \|\widehat{A}\| \leq C, \quad \widehat{M} \leq C(1 + M). \quad (\text{D.4})$$

If the corresponding inverse transition obeys the same bounds, these estimates hold in both directions on the same transported source region. Consequently residual-scale second-derivative smallness is equivalent when the compared residual scales are uniformly comparable and tend to zero. Localization and tube-covering hypotheses must also hold for each marked presentation; they are not consequences of the displayed chain rule alone.

Proof. Differentiate $\widehat{U} = \mathcal{T}_F(U, g(U))$ and $\widehat{g} = \mathcal{T}_G(U, g(U))$. Their derivatives are H_F and H_G , giving the first identity. The derivative of either H in U is bounded by a constant times $1 + \|D^2g\|$: its terms contain a second derivative of $\mathcal{T}$ applied to two bounded graph tangents, or a first derivative of $\mathcal{T}$ applied to $(0, D^2g)$. Differentiate $H_G H_F^{-1}$, use $D(H_F^{-1}) = -H_F^{-1}(DH_F)H_F^{-1}$, and multiply by H_F^{-1} to change the last derivative to $\widehat{U}$. This proves (D.4). Apply the same computation to the inverse transition. If $\widehat{c} \asymp c \rightarrow 0$, the two inequalities imply $cM \rightarrow 0$ if and only if $\widehat{c}\widehat{M} \rightarrow 0$. $\square$

Example D.3 (One residual germ, two unequal certificates). Consider $O(x) = (x, x^3)$ with target $(t, t^3 + t^4)$. Marking $F = x$ gives $A_F = 3x^2$, $M = 6|x|$ on a tube of radius t^2 , and $c = t^4$. The residual-scale certificate holds. Instead mark $\widehat{F} = x^3$ and use x as slow observable. A tube of radius t^4 around $t^3 + t^4$ avoids the critical value for small time, but the slow slope is $1/(3x^2) \asymp t^{-2}$ and is unbounded. The coordinate swap preserves the full residual germ and transports its metric, while this derivative certificate fails. The first transition has no uniformly bounded inverse fast Jacobian. This example proves that certificate invariance without a transversality condition would be false; it does not affect invariance of the normalized residual image itself.

D.2. A simultaneous proper-real presentation. The next statement makes the resolution dependency of the divisorial criterion explicit. All suprema are over the original tube; no selected inverse branch replaces that tube.

Proposition D.4 (Simultaneous resolution of the tube certificate). *Assume a compact semialgebraic source, polynomial or analytic-semialgebraic observation maps, and semialgebraic marked arcs and tube data with the stated positive-time analytic regularity. Assume $J \neq 0$ on the positive-time tube. Uniformization and every pruning below retain a surjective preimage of the positive-time graph closure. Let Γ be that compact closure, namely the closure of*

$$\{(t, x, a(t), b(t), r(t), c(t)) : 0 < t \leq t_0, x \in \Omega_t\},$$

where $c(t) > 0$ and the tube satisfies the geometric hypotheses preceding Definition C.1. The graph is considered only near its zero-time fibre. Let $J = \det DF$, $P = DG \operatorname{adj}(DF)$, and $N_2 = (JD_x P - PD_x J) \operatorname{adj}(DF)$. There is a proper real analytic uniformization followed by a proper normal-crossings modification and a finite collection of retained closed corner boxes covering the germ. In each such box the pullbacks have the forms

$$t = u_t z^a, \quad r = u_r z^s, \quad c = u_c z^h, \quad J = u_J z^j, \quad P = z^p P_0, \quad N_2 = z^q N_0,$$

with all scalar units bounded away from zero, and all displayed nonzero vector units having norms bounded above and away from zero. Coordinates z_i are nonnegative; nondivisorial coordinates range in compact boxes. A numerator which vanishes identically on an analytic component is retained as the zero block, not assigned a fictitious order.

The images of the boxes cover the entire tube germ. Every coordinate face used in a necessary bound is reached by holding the other divisor coordinates positive and approaching that face inside a positive-time orthant. For every nonnegative function of the recorded coordinates, its supremum on the tube equals its supremum on the full retained preimage. Consequently boundedness of P/J , convergence of c/r to zero, and convergence of cN_2/J^3 to zero are equivalent to the finite coordinatewise tests in Theorem E.3.

Proof. The graph coordinates are analytic functions on the ambient space of the compact semialgebraic graph. Apply real uniformization [3, Theorem 0.1] and pull back all these coordinates. Prune to the closure of the positive-time part. Lemma M.2 proves that this remains surjective onto Γ . On every retained analytic component, t, r, c, J are not identically zero: they are nonzero at positive time, including J by the tube condition. A numerator block may vanish identically; in that case its norm is identically zero and its associated boundedness or smallness condition is automatic on that component.

On each component resolve the product of the nonzero factors among

$$t r c J^2 \left(\sum_{a,b} P_{ab}^2 \right) \left(\sum_{a,b,c} (N_2)_{abc}^2 \right).$$

Normal crossings follows from [3, Corollary 4.9], applied near the compact inverse image of the zero-time fibre. Prune again. The scalar factorization and sum-of-squares statements of Lemma M.1 give precisely the asserted scalar monomials and nonvanishing vector units. In particular no separate product of norms discards cancellation inside a numerator.

Choose finitely many coordinate neighbourhoods of the compact inverse zero-time fibre and smaller closed boxes whose relative interiors cover it. Retain all orthants with positive-time interior and replace signed divisor coordinates by nonnegative ones. The units have the required bounds after shrinking. Properness proves that their images cover a full sufficiently small time germ: a sequence of omitted tube points with time tending to zero would have lifts accumulating at the covered inverse zero-time fibre. The transverse arcs in the statement lie in the retained boxes and map to feasible tube points. This also handles faces with zero time exponent: their interior points have positive time when all time-divisor coordinates are held positive.

Surjectivity gives equality of suprema before any estimates. For a nonzero monomial ratio uz^d with bounded invertible unit, boundedness holds exactly when every $d_i \geq 0$. Uniform convergence to zero at $t = 0$ holds exactly when $d_i > 0$ on time-divisor coordinates $a_i > 0$ and $d_i \geq 0$ on the others. Negative exponents or a zero exponent at a time face are excluded by the transverse arcs. Conversely, with $\delta = \min_{a_i > 0} d_i/a_i > 0$, boxes rescaled to $z_i \leq 1$ give $|uz^d| \leq Ct^\delta$. Apply this to P/J , c/r , and cN_2/J^3 , omitting identically zero numerator blocks. Their exponents are respectively $p - j$, $h - s$, and $h + q - 3j$. The last tube supremum is $c(t)M(t)$, proving the assertion. $\square$

Remark D.5 (Order faces and the meaning of persistence). The admissible exponents of the divisorial criterion satisfy both weak and strict linear inequalities. The whole admissible set need not be relatively open in the full exponent space. Partition the weak-inequality cone into faces according to which weak inequalities are equalities. On each such face, the additional strict tests cut out a relatively open rational polyhedral subset, possibly empty. Whenever a fixed-order region is called relatively open below, this facewise meaning is intended. Refinement invariance is equivalence to bounds on the same marked tube. Persistence of strict margins with fixed orders and uniform units is not a simultaneous-resolution theorem for arbitrary parameter changes. The coefficient stability of Theorem D.1 is a distinct positive stability result which does allow the leading map to vary.

APPENDIX E. RESIDUAL-SCALE REDUCTION AND RAMIFICATION CERTIFICATES

The radius in Definition C.1 describes a region on which inverse branches exist. It need not describe the region explored by a minimizing residual. This distinction permits a stronger reduction theorem.

Theorem E.1 (Reduction at the residual scale). *Keep the geometric hypotheses preceding Definition C.1, and suppose*

$$\sup_{\Omega_t} \|A_F\| \leq K_0, \quad c(t)/r(t) \longrightarrow 0, \quad c(t)M(t) \longrightarrow 0. \quad (\text{E.1})$$

All conclusions of Theorem C.2 hold, with its relative error $K\varepsilon(t)$ replaced by $Kc(t)M(t)$. This includes its exact-zero assertion and its description of the entire leading residual set. The constants are uniform under uniform metric bounds, the displayed bound on A_F , and uniform limits in (E.1).

Proof. Lemma N.1 below supplies the inverse charts without any derivative-gap assumption. Taylor's formula still gives $|E_b(U, t)| \leq M(t)|U|^2/2$. The trial $U = 0$ on any branch has norm at most $Kc(t)$. Consequently every actual global minimizer has $|U| \leq K'c(t)$, by the first residual coordinate and the lower metric bound. Since $c = o(r)$, such a minimizer is in the smaller inverse tube. The unrestricted affine minimizer also has $|U_b^*| \leq K'|C_b| \leq K'c$, by the formula in the proof of Theorem C.2. On either set of points, writing $V = C_b + D_bU$, we have

$$|\|(U, V + E_b)\|_H - \|(U, V)\|_H| \leq K''M(t)|U|^2 \leq K'''c(t)M(t)\|(U, V)\|_H.$$

Use actual minimizers for the lower bound and affine minimizers for the upper bound. The argument makes no division by C_b or by a possibly zero distance. A zero again forces $U = 0$ and then $C_b = 0$.

If the branch residues are not eventually zero, bounded slopes make their affine costs comparable with $|C_b|$. The selected order is therefore $\rho = \max_b \rho_b$. A bounded normalized residual has $U = O(t^\rho)$. Since $t^\rho = O(c(t))$, its normalized remainder is $O(M(t)t^\rho) = o(1)$. The two inclusions identifying (C.7) now follow exactly by taking limits and by testing $U = t^\rho u$, respectively. Every constant used above has the asserted uniform dependence. $\square$

Corollary E.2 (Continuation across the radius boundary). *If $c/r \rightarrow 0$, A_F is bounded, and $rM = O(1)$, the affine reduction remains asymptotically exact. In particular, $rM \asymp 1$ is not, by itself, a boundary at which the metric quotient becomes nonlinear.*

Proof. $cM = (c/r)(rM) \rightarrow 0$, so Theorem E.1 applies. $\square$

We next give finite geometric certificates for the two versions of the gap. They use the derivative numerators of the original map rather than a one-variable graph of a previously computed supremum. Write

$$J = \det DF, \quad P = DG \operatorname{adj}(DF), \quad N_2 = (JD_x P - PD_x J) \operatorname{adj}(DF). \quad (\text{E.2})$$

In the last expression the source derivative index is contracted with the adjugate. Entrywise differentiation gives

$$A_F = P/J, \quad D_x A_F D F^{-1} = N_2/J^3.$$

This identity retains cancellations in the numerator. Replacing it by a product of separate norm bounds would generally lose the exact criterion.

Theorem E.3 (Divisorial derivative criterion). *Suppose c is not identically zero, and shorten the interval so that $c > 0$. Take the compact closure of the positive-time tube graph, recording a, b, r, c as coordinate functions. There is a finite proper real corner presentation of its germ at $t = 0$ on which, on every retained component with nonzero indicated numerators,*

$$\begin{aligned} t &= u_t z^a, & r &= u_r z^s, & c &= u_c z^h, \\ J &= u_J z^j, & P &= z^p P_0, & N_2 &= z^q N_0. \end{aligned} \tag{E.3}$$

The scalar units and the norms of P_0, N_0 have positive lower and finite upper bounds; $z_i \geq 0$. Every face used below is accessible from positive time. Put $\beta_i = p_i - j_i$ and $\gamma_i = q_i - 3j_i$. Then (E.1) is equivalent to the following finite tests on all retained charts:

$$\begin{aligned} \beta_i &\geq 0 && \text{for all } i, \\ h_i - s_i &> 0, & h_i + \gamma_i > 0 & \text{if } a_i > 0, \\ h_i - s_i &\geq 0, & h_i + \gamma_i \geq 0 & \text{if } a_i = 0. \end{aligned} \tag{E.4}$$

The original condition $rM \rightarrow 0$ is obtained by replacing $h_i + \gamma_i$ by $s_i + \gamma_i$. An identically zero numerator imposes no test for its corresponding derivative. If $c \equiv 0$, the minimum is identically zero.

These tests are unchanged in truth value by further proper real refinement and by feasible source reparametrization transporting the marked fast map. On a family with a fixed such presentation and uniform unit bounds, strict inequalities imply uniform power bounds. They persist under perturbations preserving the displayed orders and those unit bounds.

Proof. All coordinate graphs are semialgebraic. Apply proper real uniformization to their compact closure, then resolve the product of $t, r, c, J^2, \sum P_{ab}^2$ and $\sum (N_2)_{abc}^2$ on each component on which those factors are not identically zero. Prune to the closure of positive time after each proper map. Lemmas M.1 and M.2 give (E.3), nonvanishing vector units, and accessible corner boxes covering the germ. A component where a numerator vanishes identically has zero derivative there and can be treated separately. No component where J vanishes identically can meet positive time.

For a monomial ratio $f = uz^d$, with u bounded above and away from zero, boundedness on the retained boxes is equivalent to $d_i \geq 0$ for every coordinate face. Necessity follows by fixing all other divisor coordinates strictly positive and approaching that face. Moreover, uniform convergence $f \rightarrow 0$ as $t \rightarrow 0$ is equivalent to $d_i > 0$ on every face with $a_i > 0$ and $d_i \geq 0$ on the remaining faces. Necessity follows from the same transverse arcs. For sufficiency choose $\delta = \min_{a_i > 0} d_i/a_i > 0$. After rescaling the boxes to $z_i \leq 1$, $|f| \leq Ct^\delta$. Faces with $a_i = 0$ and a negative exponent would also contradict boundedness at positive time. Applying these facts to P/J , c/r , and cN_2/J^3 proves exactly (E.4). The supremum of the last ratio is cM , so this is a criterion on the whole tube, not merely along a selected branch.

Every refinement presents the same boundedness and uniform-limit statements; hence it has the same answer. Fast derivatives are unchanged by a source coordinate change, by the chain rule, when the tube and F are transported. Finally the minimum of the finitely many positive ratios and the uniform unit bounds give uniform powers in a fixed presentation. The perturbation assertion follows with these same constants. $\square$

The criterion classifies the derivative certificate, not every possible mechanism that could yield an affine limit. Its refinement invariance does not assert that arbitrary perturbations admit

one simultaneous resolution. On a fixed resolution type the admissible order vectors form the explicitly given relatively open rational polyhedral region. Its faces distinguish loss of residual localization, slope boundedness, and second-derivative smallness. Corollary E.2 identifies which apparent radius faces can be crossed while retaining the affine quotient.

Proposition E.4 (A nonlinear residual at the true scale). *Let $\lambda > 0$ and, near $(y, x) = (0, 0)$, set*

$$F(y, x) = (y, x^3 + y^2x), \quad a(t) = (t, 0), \quad G(y, x) = yx^2, \quad b(t) = \lambda t^3.$$

One may take $r(t) = t^2$ on a sufficiently small time interval. The fast fibre has a unique real point $(t, 0)$, the slopes are bounded on the tube, $c/r \rightarrow 0$, and $M(t) \asymp t^{-3}$. The complete normalized residual set at order three is the closed nonlinear set

$$E_\lambda = \{(u, \xi^3 + \xi, \xi^2 - \lambda) : u, \xi \in \mathbb{R}\}. \quad (\text{E.5})$$

For every positive metric H , the squared leading cost is $\min_{E_\lambda} y^\top H y$. In the Euclidean metric it equals

$$\min_{z \geq 0} \{z^3 + 3z^2 + (1 - 2\lambda)z + \lambda^2\}. \quad (\text{E.6})$$

For $\lambda > 1/2$ this is strictly less than the affine-centre prediction λ^2 .

Proof. $J = 3x^2 + y^2$ vanishes only at the origin. The critical value is therefore zero, and boundary values stay away from the central target after choosing a compact neighbourhood whose central fibre is interior. The equation $x^3 + y^2x = 0$ at $y = t > 0$ has only $x = 0$. On the tube $y = t + O(t^2)$. Direct differentiation gives

$$A_F = \left(\frac{3x^2(x^2 - y^2)}{3x^2 + y^2}, \frac{2xy}{3x^2 + y^2} \right).$$

Its entries are bounded. In coordinates $y = t\eta$, $x = t\xi$, with η bounded above and away from zero, the three distinct second fast derivatives are

$$\begin{aligned} & -\frac{12t\xi^2\eta(9\xi^4 + 4\xi^2\eta^2 - \eta^4)}{(3\xi^2 + \eta^2)^3}, \\ & \frac{6\xi(\xi^2 + \eta^2)(3\xi^2 - \eta^2)}{t(3\xi^2 + \eta^2)^3}, \quad -\frac{2\eta(3\xi^2 - \eta^2)}{t^3(3\xi^2 + \eta^2)^3}. \end{aligned}$$

The rational factors are bounded for all real ξ . At $\xi = 0, \eta = 1$ the last derivative is $2t^{-3}$. Thus $M \asymp t^{-3}$ and $cM \asymp 1$, whereas $c/r = \lambda t$.

A bounded residual divided by t^3 forces $y = t + t^3u$ with bounded u . The second fast coordinate then forces $x = t\xi$ with bounded ξ , since $\xi^3 + (y/t)^2\xi$ is proper and strictly increasing. Substitution gives (E.5), and conversely every bounded (u, ξ) is realized by this substitution. These same coercive bounds localize every bounded-cost sequence, proving convergence of minima for each $H > 0$. For $H = I$, minimize first over u and put $z = \xi^2$. Expansion gives (E.6); its derivative at zero is $1 - 2\lambda$. The minimum is positive, because $\xi^3 + \xi = 0$ forces $\xi = 0$, where the last coordinate is $-\lambda$. Thus the order really is three. $\square$

The nonlinear set in (E.5) has the same intrinsic meaning as the affine sets in (C.7): it is the zero-time section of the normalized feasible residual image. Its metric tensor envelope is still $\mathcal{C}(E_\lambda)$. No finite affine or semidefinite description of that nonlinear envelope is presumed.

APPENDIX F. THE TENSOR QUOTIENT EXTRACTED BY REDUCTION

For a closed residual set $E \subset \mathbb{R}^m$ define

$$\mathcal{C}(E) = \text{cl}(\text{conv}\{yy^\top : y \in E\} + \text{Sym}_m^+).$$

The order and the residual set in Theorem C.2 give, for every $H > 0$,

$$\min_N \|R(x, t)\|_H = a_H t^\rho + o(t^\rho), \quad a_H^2 = \min_{y \in \mathcal{E}_\rho} y^\top H y. \quad (\text{F.1})$$

The second equality follows either directly from (C.6) or by the two normalized-residual inclusions in its proof. We record the finite representation to make the reduction's output explicit.

Theorem F.1 (Finite moment lift and metric faces). *Let $E = \bigcup_{b=1}^B (c_b + L_b \mathbb{R}^{r_b})$, where L_b has full column rank and $c_b \notin \text{ran } L_b$. Put $A_b = [L_b, c_b]$. Then*

$$\mathcal{C}(E) = \left\{ \sum_b A_b M_b A_b^\top + P : M_b \geq 0, P \geq 0, \sum_b (M_b)_{r_b+1, r_b+1} = 1 \right\}. \quad (\text{F.2})$$

This set is closed without taking an additional closure. For $H > 0$, let $y_b(H)$ be the unique H -nearest point to zero on the b th sheet, and put $q_b(H) = y_b(H)^\top H y_b(H)$. The exposed face at the minimum of $S \mapsto \text{tr}(HS)$ is

$$\text{conv}\{y_b(H)y_b(H)^\top : q_b(H) = \min_c q_c(H)\}. \quad (\text{F.3})$$

The envelope is recovered from its positive-metric costs by

$$\mathcal{C}(E) = \bigcap_{H>0} \{S \in \text{Sym}_m : \text{tr}(HS) \geq \min_b q_b(H)\}.$$

The active-sheet sets form a finite semialgebraic partition of the positive metric cone. Applied to (C.7), all these objects are invariants of the marked feasible observation germ.

Proof. The second moments of finitely supported affine random vectors give the identity

$$\text{conv}\{(u, 1)(u, 1)^\top : u \in \mathbb{R}^r\} = \{M \geq 0 : M_{r+1, r+1} = 1\}.$$

For clarity, if $M = \begin{pmatrix} K & m \\ m^\top & 1 \end{pmatrix}$, its covariance is $K - mm^\top \geq 0$. Diagonalize the covariance and realize each eigendirection by symmetric pairs about m ; the resulting finite distribution has second moment M . Conversely every second moment is positive semidefinite.

For each matrix in (F.2), let λ_b be its last diagonal entry. When $\lambda_b > 0$, divide by λ_b and use the identity just proved. When $\lambda_b = 0$, positivity makes its last row and column zero, so $A_b M_b A_b^\top$ may be absorbed into P . This proves equality before closure. Full column rank of A_b gives

$$\text{tr}(A_b M_b A_b^\top) \geq \sigma_{\min}(A_b)^2 \text{tr } M_b.$$

For a convergent sequence of represented matrices all positive summands therefore have bounded traces. Taking subsequences of their moment matrices and of P proves closedness.

Complete the square on each affine sheet. Equality in the minimum over (F.2) forces $P = 0$, zero covariance on every positive-weight sheet, and weight only on minimizing sheets, since H is positive definite. This gives (F.3). To recover the envelope, strictly separate an exterior matrix by a symmetric G . The positive-upper-set property forces $G \geq 0$. Replacing G by $G + \epsilon I$ preserves the strict separation for sufficiently small $\epsilon > 0$, since the envelope consists of positive semidefinite matrices. The asserted intersection follows. Finally

$$y_b(H) = c_b - L_b(L_b^\top H L_b)^{-1} L_b^\top H c_b$$

shows that all comparisons of q_b are semialgebraic. The final invariance is Corollary C.3. $\square$

The lift is a finite representation, not a claim of minimal semidefinite extension size. Nonzero ties of the leading costs do not determine which branch minimizes at finite time. The exact-zero assertion of Theorem C.2, by contrast, is a finite-time equivalence.

APPENDIX G. RECONSTRUCTION AFTER INVISIBLE ENDPOINT ELIMINATION

Use (A.1). For $I \subseteq [e] = \{1, \dots, e\}$ put

$$\begin{aligned} z_I(x) &= -C_{II}^{-1}B_I^\top x, \quad z_{I^c}(x) = 0, \\ K_I &= \{x \geq 0 : z_I(x) \geq 0, B_j^\top x + C_{jI}z_I(x) \geq 0 \ (j \notin I)\}, \\ S_I &= A - B_IC_{II}^{-1}B_I^\top. \end{aligned} \tag{G.1}$$

Empty blocks have their natural meanings, in particular $S_\emptyset = A$. Strict positivity of C makes the endpoint minimizer unique. The stationarity and complementary inequalities show that the cones K_I cover the positive orthant and $G_Q(x) = x^\top S_I x$ on K_I . The formulas agree on overlaps. Thus G_Q is continuous and is quadratic on finitely many polyhedral regions. We use the quadratic coefficient S_I , namely one half the Hessian, as the label of a region.

Definition G.1 (Full visibility). A representation is fully visible if B has column rank e and for every $I \subseteq [e]$ there is an $x \in (0, \infty)^k$ with

$$z_I(x) > 0, \quad B_j^\top x + C_{jI}z_I(x) > 0 \quad (j \notin I).$$

The first inequality has no entries when I is empty. These strict inequalities give an open region for each active set.

Let $\mathcal{H}(G)$ be the set of distinct symmetric matrices S such that $G(x) = x^\top S x$ on some nonempty open subset of $(0, \infty)^k$. The set, and the open regions carrying its labels, are determined by the function itself.

Lemma G.2 (The visible Hessian lattice). *For a fully visible representation, $\mathcal{H}(G_Q) = \{S_I : I \subseteq [e]\}$ consists of exactly 2^e matrices, and*

$$S_I \geq S_J \iff I \subseteq J. \tag{G.2}$$

For $I \subseteq J$, $\text{rank}(S_I - S_J) = |J \setminus I|$. In particular A is the unique largest local matrix and $S_{[e]}$ the unique smallest one.

Proof. Set $P_I = A - S_I$. A block Schur complement, with $J = I \cup H$, gives

$$P_J - P_I = (B_H - B_IC_{II}^{-1}C_{IH})(C_{HH} - C_{HI}C_{II}^{-1}C_{IH})^{-1}(B_H - B_IC_{II}^{-1}C_{IH})^\top.$$

The middle matrix is positive definite and the outer matrix has column rank $|H|$, because the columns of B are independent. This proves the forward implication and its rank assertion. Conversely $P_I \leq P_J$ implies $\text{ran } P_I \subseteq \text{ran } P_J$: every vector annihilated by P_J has zero quadratic value for P_I and is annihilated by it as well. Since $\text{ran } P_I = \text{ran } B_I$, independence of the columns forces $I \subseteq J$. This proves (G.2) and distinctness. Full visibility makes all these polynomials occur on open sets. No additional polynomial can occur: any open set meets an interior cell of the finite polyhedral cover, and equality of quadratic polynomials on an open set implies equality of their matrices. $\square$

Theorem G.3 (All-dimensional endpoint reconstruction). *Suppose $G = G_Q$ has a fully visible representation with $e \geq 1$ endpoints. The function G determines its minimal endpoint dimension, which is e , and every representation with exactly e endpoints as follows.*

Read $A = \max \mathcal{H}(G)$ and $S = \min \mathcal{H}(G)$. The e matrices covered by A in the ordered set (G.2) have rank-one deficits

$$A - S_{\{j\}} = u_j u_j^\top.$$

Choose x_0 in the open region with local matrix A , avoiding all cell boundaries, and orient each u_j by $u_j^\top x_0 > 0$. Put $U = (u_1, \dots, u_e)$ and $U^+ = (U^\top U)^{-1} U^\top$. Then

$$\bar{C} = [U^+(A - S)(U^+)^\top]^{-1}, \quad Q_{\text{norm}} = \begin{pmatrix} A & U \\ U^\top & \bar{C} \end{pmatrix}, \quad \text{diag } \bar{C} = \mathbf{1}. \quad (\text{G.3})$$

These data are unique up to simultaneous permutation of the u_j and endpoint coordinates. A lexicographic ordering of the distinct oriented columns fixes a representative. All e -endpoint representations are its positive diagonal endpoint congruences and permutations. Their continuous fibre dimension is e .

The equivalent distance representations

$$G(x) = x^\top S x + \text{dist}(Tx, K)^2, \quad K = L\mathbb{R}_+^e, \quad (\text{G.4})$$

of minimal dimension have the same S and the same pair (T, K) up to an orthogonal change of ambient coordinates. The normalized Gram matrix of the generating rays is $\bar{C}$.

Proof. In the original representation set $D_0 = \text{diag}(\sqrt{C_{11}}, \dots, \sqrt{C_{ee}})$. The rank-one deficits give $u_j = \pm B_j / \sqrt{C_{jj}}$. On the strict empty-active region all $B_j^\top x$ are positive, so the prescribed orientation selects the positive sign. This orientation does not depend on x_0 in that region. Thus $U = BD_0^{-1}$, up to permutation. The normalized coupling is $\bar{C} = D_0^{-1}CD_0^{-1}$ and

$$A - S = U\bar{C}^{-1}U^\top.$$

Multiplication by U^+ and its transpose gives (G.3). The normalized matrix is positive definite by congruence from Q . Its diagonal condition follows directly from the construction.

We next prove minimality without presuming visibility of an alternative representation. In any representation with e' endpoints, the range of every difference of two local quadratic matrices lies in the range of its cross block B' , by (G.1). The intrinsic difference $A - S$ has rank e . Hence $e \leq \text{rank } B' \leq e'$. No smaller representation is possible.

If $e' = e$, then B' has full column rank. The alternative representation has at most 2^e active-set polynomials, while G has exactly 2^e distinct local polynomials. Therefore every active set supplies a distinct full-dimensional cell. Each such cell has a strict point: a zero-coordinate or zero-multiplier equality is a nonzero linear equation because the columns of B' are independent, so finitely many such hyperplanes cannot cover an open cell. The alternative representation is fully visible. The preceding reconstruction applies to it and gives exactly the same normalized matrix, up to permutation. Thus every representation is obtained from (G.3) by

$$Q_D = \begin{pmatrix} A & UD \\ DU^\top & D\bar{C}D \end{pmatrix}, \quad D = \text{diag}(d_1, \dots, d_e), \quad d_j > 0,$$

and a permutation. Conversely changing z by Dz preserves the nonnegative orthant and the infimum, so all these matrices represent G . The parameters d_j are freely and injectively specified by the endpoint diagonal. Permutations are discrete. This proves the fibre assertion.

For the distance statement, completing the square gives $S = A - BC^{-1}B^\top$, $C = L^\top L$ and $B = -T^\top L$. Conversely every such square expansion gives the positive matrix with $A = S + T^\top T$. Under the preceding congruence the generators become LD , up to permutation, which leave the cone unchanged. Any two invertible factors of a fixed Gram matrix differ by a left orthogonal map, and $B = -T^\top L$ forces the same orthogonal map on T . Thus (T, K) has exactly the stated ambiguity. Normalizing the lengths of the rays gives $\bar{C}$. $\square$

Proposition G.4 (An open visible class in every dimension). *Suppose B has column rank e and its kernel on the transpose contains a strictly positive vector:*

$$\text{there is } h \in (0, \infty)^k \text{ with } B^\top h = 0.$$

Then the representation is fully visible, for every $C > 0$ and every A with $A - BC^{-1}B^\top > 0$. These conditions describe an open nonempty class when $k \geq e + 1$. In particular one may take $k = e + 1$, let the first e rows of B form I_e and its last row be $-\mathbf{1}^\top$, choose any $C > 0$, and put $A = BC^{-1}B^\top + I_k$.

Proof. Surjectivity of $B^\top$ gives, for any $y \in \mathbb{R}^e$, a solution of $B^\top x = y$. Adding a sufficiently large multiple of h makes x strictly positive. Thus $B^\top(0, \infty)^k = \mathbb{R}^e$. For an active set I , choose $z_I > 0$, $z_{I^c} = 0$, $w_I = 0$, and $w_{I^c} > 0$. There exists an $x > 0$ with $B^\top x = -Cz + w$. These are precisely the strict stationarity and complementary inequalities, so I is visible.

The property persists under small perturbations of B . Indeed, solve the small residual in $B'^\top h$ using a bounded right inverse of $B'^\top$; the correcting vector tends to zero, so the corrected kernel vector remains positive. Rank, positive definiteness, and the Schur inequality are open conditions. The displayed matrix has positive kernel vector $\mathbf{1}$, and proves nonemptiness. Additional rows for $k > e + 1$ may initially be zero and the same positive-kernel argument applies. Thus this is an open family of genuinely coupled endpoint problems, not merely the diagonal case. $\square$

Proposition G.5 (Finite redundancy criterion). *For an arbitrary positive definite Q , deleting endpoint coordinate j preserves G_Q if and only if, for every active cone $K_I^{(-j)}$ of the reduced endpoint problem,*

$$(B_j^\top - C_{jI}C_{II}^{-1}B_I^\top)x \geq 0 \quad \text{for every } x \in K_I^{(-j)}. \quad (\text{G.5})$$

Each condition is membership of a specified linear functional in the dual of a polyhedral cone and can be tested by a finite linear feasibility problem. Independently of visibility, the invariant

$$\delta(G) = \dim \text{span}\{\text{ran}(S - S') : S, S' \in \mathcal{H}(G)\}$$

is a lower bound for every endpoint dimension. In the fully visible class it equals the minimal dimension e .

Proof. The reduced minimizer is feasible for the full problem with its j th coordinate zero. Its gradient in that coordinate, divided by two, is the left side of (G.5). The reduced stationarity conditions already hold in all other coordinates. The additional nonnegative gradient is therefore necessary and sufficient for full optimality. Strict convexity shows that equality of the two value functions for every x is equivalent to this condition at the unique reduced minimizer for every x . Substituting the finitely many active formulas gives (G.5), including boundaries. The range bound follows because every local matrix is $A - B_IC_{II}^{-1}B_I^\top$; every difference has range in $\text{ran } B$. In the visible class $A - S_{[e]}$ already has range of dimension e , giving equality. $\square$

The coordinate-deletion criterion does not assert that repeated coordinate deletion finds a minimal representation in every nonvisible problem. Theorem G.3 proves global minimality and the complete minimal-dimensional fibre on its stated open class. Outside that class the finite equality test by intersections of active cones remains valid, but a full classification of all nonminimal representations is a separate question.

APPENDIX H. THE FIRST STRATA OF VISIBILITY LOSS

The loss of an open active cell and the loss of cross-block rank are different phenomena. In the former, all algebraic active-set Hessians remain distinct. The following theorem shows that a missing cell need not destroy reconstruction.

Theorem H.1 (One missing cell in dimension at least three). *Let $e \geq 3$, let B have column rank e , and suppose exactly $2^e - 1$ active sets have a strict feasible point in X° . Then G_Q has minimal endpoint dimension e . Its entire fibre among e -endpoint positive representations consists of positive diagonal endpoint congruences and permutations. The normalized representation is*

reconstructed from the labelled open-cell quadratic matrices even when the missing active set is empty, full, or a singleton. Its continuous fibre dimension is e . No coordinate is deletable, and no two algebraic active-set Hessians coalesce.

Proof. The algebraic rank and order statements in Lemma G.2 use only column independence, not visibility. Consequently every S_I is distinct, and strict feasibility fails on an open cell only if that cell is absent. The observed local polynomials are precisely the $2^e - 1$ remaining matrices. The span of the ranges of their differences has dimension e . If both extreme vertices are present, use $A - S_{[e]}$. If one extreme is missing, differences between the other extreme and the e vertices next to the missing one have ranges spanning $\text{ran } B$. Thus every representation has at least e endpoints. Any competing e -endpoint representation has independent cross columns and exactly one absent active set, by the same polynomial count and the strict-point argument in Lemma N.2.

We give explicit reconstruction in each possible intrinsic configuration. If both extremes and all e rank-one covers of the largest matrix are observed, the formula (G.3) applies without using the missing cell. This covers every missing nonextreme nonsingleton vertex. If the largest matrix A is present and has only $e - 1$ rank-one covers, the absent set is a singleton, say $\{j\}$. The known columns are $u_i = B_i / \sqrt{C_{ii}}$, $i \neq j$, oriented on the empty-active region. Among the observed rank-two deficits $A - S_I$, those whose ranges contain just one known column line are precisely the pairs $I = \{i, j\}$. The intersection of two such ranges is $\text{ran } B_j$, by independence. Choose a unit vector v on this line, oriented positively on the empty region. For one such pair express its deficit in the independent basis (u_i, v) :

$$A - S_{\{i,j\}} = [u_i, v]K[u_i, v]^\top.$$

For some $d > 0$ and $|a| < 1$ the matrix K has diagonal entries $1/(1 - a^2)$ and $d^2/(1 - a^2)$. Hence $d = \sqrt{K_{22}/K_{11}}$ and $u_j = dv$. The full smallest matrix is observed, so (G.3) again recovers the coupling.

If the full set is missing, the local order has e minimal vertices and one largest vertex A . All singleton and pair deficits are observed, since $e \geq 3$. Recover the oriented u_i from the singleton deficits. For each pair the identity

$$A - S_{\{i,j\}} = [u_i, u_j] \bar{C}_{\{i,j\}, \{i,j\}}^{-1} [u_i, u_j]^\top$$

recovers that 2×2 principal submatrix by a left inverse. These submatrices determine every entry of $\bar{C}$.

If the empty set is missing, there are e maximal vertices and one smallest matrix $S = S_{[e]}$. Use the dual data

$$D = C^{-1}, \quad H = BC^{-1}, \quad S_I - S = H_J D_{JJ}^{-1} H_J^\top, \quad J = I^c.$$

All dual singleton and pair matrices are observed. Orient $h_j = H_j / \sqrt{D_{jj}}$ negatively on the full-active region; its strict minimizer satisfies $C^{-1} B^\top x < 0$ there. The preceding pair construction recovers $\bar{D}$, with diagonal one. Set $C_* = \bar{D}^{-1}$, $H_* = (h_j)$, $B_* = H_* C_*$, and $A_* = S + H_* C_* H_*^\top$. This is the normalization with $\text{diag } C_*^{-1} = \mathbf{1}$.

All the selections just made are intrinsic. Multiple maximal or minimal vertices identify exactly the missing extreme. Otherwise the number of rank-one covers of A distinguishes a missing singleton from the first case. Ranges identify the required pairs without active-set labels. The same construction must therefore give the same normalized matrix for every minimal competitor. Its remaining scales are exactly the e positive endpoint scales; permutations are discrete. Minimality rules out coordinate deletion. Independence already proved noncoalescence. $\square$

Example H.2 (A codimension-one cell disappearance). For $e = 3$, $k = 5$, put $v = (1, 1, -1)$, $p = (1, -1, 0)$, $q = (1, 0, 1)$ and take the rows of B_ϵ to be

$$p, \quad -p - \epsilon v, \quad q, \quad -q, \quad v.$$

Let $C = I_3$ and $A = B_\epsilon B_\epsilon^\top + I_5$. If $\epsilon > 0$, the image $B_\epsilon^\top X^\circ$ is all of $\mathbb{R}^3$: the coefficients of p, q can be arbitrary and the coefficient of v can have either sign while all five source coefficients are positive. At $\epsilon = 0$ that image is the half-space $v \cdot y > 0$. It meets every open sign orthant except $y_1, y_2 < 0 < y_3$. Thus exactly the active set $\{1, 2\}$ disappears, while rank stays three. For small negative ϵ the same seven cells persist by strict witnesses, and the eighth remains excluded. Theorem H.1 reconstructs the quotient on the transition and on this neighbouring nonvisible stratum. The wall is the single parameter equation $\epsilon = 0$.

There is a genuine exception in endpoint dimension two: losing one cell releases a scalar modulus. We give its entire minimal fibre. Write $s_+ = \max(s, 0)$ and $s_- = \max(-s, 0)$.

Theorem H.3 (Two-endpoint three-cell fibres). *Suppose B has rank two and exactly three active sets have open cells. The minimal endpoint dimension is two. Up to exchanging the two endpoints, the intrinsic function has one of the following forms. The linear forms $u^\top x, v^\top x$ are independent, and every displayed cell is nonempty in X° .*

Chain. For an observed largest matrix A ,

$$G(x) = x^\top A x - (u^\top x)_-^2 - (v^\top x)_-^2, \quad v^\top x < 0 \implies u^\top x < 0. \quad (\text{H.1})$$

All normalized two-endpoint representations are

$$B(a) = [u, au + v], \quad C(a) = \begin{pmatrix} 1 & a \\ a & 1 + a^2 \end{pmatrix}, \quad a_- \leq a \leq a_+, \quad (\text{H.2})$$

where

$$a_- = \sup_{x \in X, u^\top x > 0} \frac{-v^\top x}{u^\top x}, \quad a_+ = \inf_{x \in X, v^\top x < 0} \frac{u^\top x}{v^\top x}.$$

Here the missing set is $\{2\}$; $A - uu^\top - vv^\top > 0$.

Missing empty cell. For the observed smallest matrix S ,

$$G(x) = x^\top S x + (u^\top x)_+^2 + (v^\top x)_+^2, \quad \text{not both } u^\top x, v^\top x > 0. \quad (\text{H.3})$$

Put

$$a_0 = \max \left\{ \sup_{u^\top x > 0} \frac{v^\top x}{u^\top x}, \sup_{v^\top x > 0} \frac{u^\top x}{v^\top x} \right\}, \quad x \in X.$$

All normalized representations, for $a \in (-1, 1)$ with $a \geq a_0$, are

$$D(a) = \begin{pmatrix} 1 & a \\ a & 1 \end{pmatrix}, \quad C = D(a)^{-1}, \quad B = [u, v]C, \quad A = S + [u, v]C[u, v]^\top. \quad (\text{H.4})$$

Missing full cell. The dual form is $G = x^\top A x - (u^\top x)_-^2 - (v^\top x)_-^2$, with the two negative regions disjoint. All normalized representations have $B = [u, v]$, $C = D(a)$, where $|a| < 1$,

$$a \geq \max \left\{ \sup_{u^\top x < 0} \frac{v^\top x}{u^\top x}, \sup_{v^\top x < 0} \frac{u^\top x}{v^\top x} \right\}, \quad A - [u, v]D(a)^{-1}[u, v]^\top > 0.$$

In each case one then applies arbitrary positive endpoint scales and permutations. On the interior of each nontrivial parameter interval the continuous fibre dimension is three: two scales and the displayed modulus. No endpoint can be deleted. All ratio bounds are finite polyhedral tests and include the boundary of X by continuity.

Proof. The three local matrices have either a chain order or a fork order. Their rank-one differences span a two-dimensional range, so any representation has at least two endpoints. A minimal competitor has independent cross columns and three of the four active polynomials; it has the same order type. We can therefore work with its two-dimensional KKT fan.

In the chain case normalize the first endpoint pivot and the second Schur pivot to one. This gives exactly $C(a)$ and $B(a)$ in (H.2). Write $U = u^\top x$, $V = v^\top x$. The change $w_1 = z_1 + az_2$, $w_2 = z_2$ completes the quadratic terms to $w_1^2 + w_2^2 + 2Uw_1 + 2Vw_2$. The missing cell $\{2\}$ has the strict inequalities $aU + V < 0$ and $U - aV > 0$. The image of X° in the (U, V) -plane is convex. Since it meets the empty, first-only, and full cells, absence of the fourth forces it to avoid $U > 0, V < 0$. Indeed, if $a \geq 0$, such a point is either in the fourth cell or in the empty cell with $V < 0$; joining the latter to a strict full-cell point crosses the fourth. If $a < 0$, it is either in the fourth cell or in the full cell with $U > 0$; joining the latter to a strict empty-cell point has the same consequence. This proves the nesting in (H.1). The three optimizers in w are respectively $(0, 0)$, $(-U, 0)$ and $(-U, -V)$. Their feasibility and inactive-coordinate inequalities are exactly $aU + V \geq 0$ when $U > 0$, and $-U + aV \geq 0$ when $V < 0$. These are precisely $a \geq a_-$ and $a \leq a_+$. Hence they are both necessary and sufficient, including by continuity all cell boundaries. The two successive rank-one deficits determine u, v , with signs fixed by their zero and active regions. The normalization thus gives every minimal competitor. Its Schur complement is the observed smallest matrix, independent of a , so positivity introduces no further restriction.

For the missing-empty case complete the square and use the dual problem:

$$G(x) = x^\top Sx + \max_{w \geq 0} \{2(U, V)w - w^\top Dw\}, \quad D = C^{-1}, \quad [u, v] = BC^{-1}$$

after normalizing $\text{diag } D = \mathbf{1}$. Only the zero, first-only and second-only dual cells occur. Their convex planar image cannot contain $U, V > 0$: for $a \leq 0$ that quadrant is entirely in the both-active cell; for $a > 0$, a point outside its central wedge is in one singleton cell, and joining it to a strict point of the other singleton cell crosses the both-active wedge. Thus the positive regions are disjoint. The maximizers are $(0, 0)$, $(U, 0)$, $(0, V)$. Their inactive inequalities are $V - aU \leq 0$ on $U > 0$ and $U - aV \leq 0$ on $V > 0$, precisely the bound $a \geq a_0$. The two rank-one increments above S , with signs fixed by their active regions, recover u, v . The reconstruction (H.4) follows, and its Schur complement is $S > 0$.

For the missing-full case apply the same three-sector argument to the primal minimization with $\text{diag } C = \mathbf{1}$ and replace (U, V) by $(-U, -V)$. Its inactive inequalities give the stated lower bound. The unobserved full Schur complement must additionally remain positive, which is exactly the last displayed condition. These constructions exhaust the minimal competitors because their rank-one labels and their planar order type are intrinsic. The ratio inequalities are homogeneous linear inequalities after multiplication by the indicated positive denominator (or reversal for a negative denominator), and hence are tested on a finite polyhedral cone. The free positive scalings and the nonconstant normalized scalar give the fibre count. Minimality excludes deletion. $\square$

Proposition H.4 (Rank-one cross-block stratum). *Let $e = 2$, $Q > 0$, $B = ud^\top$ with $u \neq 0$, $d \neq 0$, and $C > 0$. Define*

$$\kappa(d, C) = \max_{z \geq 0} \{2d^\top z - z^\top Cz\}.$$

It is the maximum of the feasible active-face values

$$0, \quad (d_1)_+^2/C_{11}, \quad (d_2)_+^2/C_{22}, \quad d^\top C^{-1}d \text{ when } C^{-1}d \geq 0.$$

Writing $\kappa_- = \kappa(d, C)$ and $\kappa_+ = \kappa(-d, C)$ gives

$$G_Q(x) = x^\top Ax - \kappa_-(u^\top x)_-^2 - \kappa_+(u^\top x)_+^2. \quad (\text{H.5})$$

If u has both signs on X° , the minimal endpoint dimension is one when $\kappa_- \neq \kappa_+$ and zero when they agree. If u has only one sign on X , the function is quadratic and its minimal dimension is zero.

For a fixed nonquadratic function normalized as $G = x^\top Sx - (u^\top x)_-^2$, its entire fibre within the rank-one cross-block stratum is

$$B = ud^\top, \quad \kappa(d, C) - \kappa(-d, C) = 1, \quad A = S + \kappa(-d, C)uu^\top, \quad C > 0,$$

subject to $A - BC^{-1}B^\top > 0$. On a fixed strict active-face stratum this fibre has dimension four, including the two endpoint scales.

Proof. For $u^\top x < 0$ scale z by $-(u^\top x)$; for positive sign scale it by $u^\top x$. Completing the resulting maximum proves (H.5). Maximizing on the two axes or in the interior gives exactly the listed feasible-face values. If the two coefficients are unequal, absorb the smaller multiple of $uu^\top$ into A and orient and scale u so that the remaining hinge has coefficient one. This is a one-endpoint representation. Its positive definiteness follows from $\kappa(\pm d, C) \leq d^\top C^{-1}d$ and the original Schur inequality. A nonquadratic function cannot have zero endpoints. In the other cases a single positive quadratic form represents the function.

For the fibre statement the intrinsic rank-one Hessian difference fixes the line and normalized magnitude of u . Comparing the two half-space polynomials gives exactly the displayed equations. Conversely those equations reproduce the function. On each strict face, κ is smooth and homogeneous of degree two in d . The difference has Euler derivative two on its level one, so that level is regular. The two coordinates of d and three entries of C , minus one equation, give dimension four. $\square$

Example H.5 (Coalescence without coordinate deletion). Take $C = I_2$, $d = (1, -1)$, a mixed-sign u , and $A > 2uu^\top$. Then $\kappa_- = \kappa_+ = 1$ and $G_Q(x) = x^\top(A - uu^\top)x$. Its minimal endpoint dimension is zero, yet deleting either original endpoint changes the value on one open half-space. Thus two Hessians can coalesce while no individual coordinate is redundant. For $k = 2$, the rank-one condition on a nonzero 2×2 cross block is a codimension-one rank-loss locus. It is different from the one-cell walls of Theorems H.1 and H.3.

APPENDIX I. ENDPOINT FANS, RECONSTRUCTION LEMMAS, AND GENERAL FIBRES

The forward active-set formulas are classical quadratic programming. The issue here is the exhaustion of inverse representations when the active labels are not given. We expose the two-dimensional fan and then separate the reconstruction arguments used on the first nonvisible strata.

Lemma I.1 (Planar complementary fans). *Let $Y \subset \mathbb{R}^2$ be an open convex cone obtained as the image of X° under two independent linear forms U, V . In the chain normalization*

$$B = [u, au + v], \quad C = \begin{pmatrix} 1 & a \\ a & 1 + a^2 \end{pmatrix},$$

the four strict primal active sectors are exactly

I	strict inequalities	(z_1, z_2)
$\emptyset$	$U > 0, aU + V > 0$	$(0, 0)$
$\{1\}$	$U < 0, V > 0$	$(-U, 0)$
$\{2\}$	$aU + V < 0, U - aV > 0$	$(0, -(aU + V)/(1 + a^2))$
$\{1, 2\}$	$V < 0, U - aV < 0$	$(-U + aV, -V)$

If the empty, first-only, and full sectors meet Y and the second-only sector does not, then Y contains no point with $U > 0, V < 0$. On the retained cone the value function is exactly the chain form (H.1); feasibility is equivalent to the two ratio bounds in (H.2).

For the dual maximization $\max_{w \geq 0} \{2(U, V)w - w^\top D(a)w\}$, where $D(a) = \begin{pmatrix} 1 & a \\ a & 1 \end{pmatrix}$ and $|a| < 1$, the strict sectors are

J	strict inequalities	(w_1, w_2)
$\emptyset$	$U < 0, V < 0$	$(0, 0)$
$\{1\}$	$U > 0, V - aU < 0$	$(U, 0)$
$\{2\}$	$V > 0, U - aV < 0$	$(0, V)$
$\{1, 2\}$	$U - aV > 0, V - aU > 0$	$D(a)^{-1}(U, V)^\top$

If the first three sectors meet Y and the last does not, the regions $U > 0$ and $V > 0$ are disjoint in Y . The value is then $U_+^2 + V_+^2$, with exactly the bound $a \geq a_0$ in (H.4). Replacing (U, V) by $(-U, -V)$ gives the missing-full primal case.

Proof. Each row follows by solving stationarity on the indicated positive coordinates and requiring the other multipliers to be positive. These equations prove both necessity and sufficiency of every row, including uniqueness by strict convexity. Weak versions cover the sector boundaries.

For the chain assertion, suppose first that $a \geq 0$ and Y contains $U > 0, V < 0$. At such a point $U - aV > 0$. Avoidance of the missing sector forces $aU + V \geq 0$. Equality is impossible in the open set Y , since a small perturbation enters the missing sector. When $a = 0$ the strict inequality is already impossible. When $a > 0$, join this point to a strict full-sector point. The second coordinate remains negative along the segment; at the full-sector endpoint both U and $aU + V$ are negative. At the crossing $aU + V = 0$ one has $U = -V/a > 0$, so immediately beyond it the segment lies in the missing sector, a contradiction.

If $a < 0$, a point with $U > 0, V < 0$ has $aU + V < 0$. Avoidance of the missing sector forces $U - aV < 0$, again strictly by openness. Join it to a strict empty-sector point. The first coordinate stays positive. At the crossing $U - aV = 0$ one has $V = U/a < 0$ and $aU + V < 0$; the segment therefore crosses the missing sector. This is again impossible.

Thus $V < 0$ implies $U < 0$ on Y . The three candidate optimizers in the completed-square coordinates $w_1 = z_1 + az_2, w_2 = z_2$ are $(0, 0)$, $(-U, 0)$, and $(-U, -V)$. For $U > 0$ the remaining inactive inequality is $aU + V \geq 0$, equivalent to $a \geq -V/U$. For $V < 0$ the remaining feasibility inequality is $-U + aV \geq 0$, equivalent to $a \leq U/V$. For $U < 0 < V$, stationarity and the inactive multiplier are already satisfied. These are all possibilities off the two separating lines. They prove necessity and sufficiency of the displayed supremum and infimum bounds. Continuity extends the equality to the boundary of X .

For the dual assertion, if $a \leq 0$ the positive quadrant lies in the both-active sector. Suppose $0 < a < 1$ and a point of Y has $U, V > 0$ but lies outside that sector. It is in one singleton sector, say $V < aU$. Join it to a strict point of the other singleton sector, which has $V > 0$ and $U < aV$. The second coordinate stays positive and the continuous ratio U/V crosses the interval $(a, 1/a)$. On this interval both-active inequalities hold. This contradicts the missing-sector hypothesis. Exchanging coordinates treats the other possibility. The only optimizers on Y are therefore $(0, 0), (U, 0), (0, V)$. Their remaining multiplier inequalities are precisely $V \leq aU$ on $U > 0$ and $U \leq aV$ on $V > 0$. These give the stated ratio bound and show its sufficiency. Sign reversal gives the primal missing-full assertion. $\square$

Proposition I.2 (Exhaustion of the three-cell inverse fibre). *Under the hypotheses of Theorem H.3, every minimal competitor is one of the three families displayed there. The cell labels determine the signs of the two rank-one forms, the chain or fork order type, and the corresponding global gluing on X . There is no further gluing choice with the same labelled function.*

Proof. Column independence gives four distinct algebraic active-set matrices, ordered as the Boolean lattice on two vertices by Lemma G.2. Removing a singleton leaves a three-element chain; removing the empty or full vertex leaves one of the two forks. In each case the ranges

of two rank-one adjacent differences span a two-dimensional space. Every representation of the same function has a cross block containing those ranges, so it needs at least two endpoints. A two-endpoint competitor therefore has independent cross columns. It has exactly three visible matrices, by the strict-point argument of Lemma N.2; hence it has the same chain or fork order type.

For a chain, normalize its first diagonal pivot and its second Schur pivot to one by positive endpoint scaling. Completing the square then forces $C = \begin{pmatrix} 1 & a \\ a & 1+a^2 \end{pmatrix}$ and $B = [u, au + v]$. The two successive rank-one deficits determine $uu^\top$ and $vv^\top$. Their orientations are fixed by the side on which the respective decrement occurs; reversing either sign would put a different polynomial on a labelled open set. Lemma I.1 proves exactly the global nesting and interval bounds. The full Schur complement is the observed smallest matrix, so positivity adds no hidden condition.

For a missing-empty fork use $S = S_{\{1,2\}}$, $D = C^{-1}$ and $H = BC^{-1}$. Completing the square and dualizing gives

$$G(x) = x^\top Sx + \max_{w \geq 0} \{2(H^\top x)^\top w - w^\top Dw\}.$$

Normalize the diagonal of D to one. The two observed increments above S determine u, v , positively oriented on their increment regions. The dual table forces exactly $D(a)$ and its ratio bound, and reversing the completion of the square gives (H.4). Its Schur complement is $S > 0$. The missing-full case is the sign-reversed primal table; its unobserved full Schur complement must be positive, which is exactly the additional inequality in Theorem H.3.

All sector conditions were derived from the unique optimizer and were shown sufficient on the whole cone, not only near a witness point. Every minimal competitor must therefore have the same oriented rank-one labels and one of the stated scalar parameters. This proves exhaustion and excludes alternative gluing. On an admissible interval reduced to a single point, only the two endpoint scales remain; on a nontrivial open interval they coexist with the scalar modulus, giving dimension three. Endpoint permutations remain a discrete ambiguity. $\square$

Lemma I.3 (Reconstruction of a missing singleton). *Let $e \geq 3$, $\text{rank } B = e$, and suppose the only missing active set is a singleton. The observed data determine the normalized missing cross column and then the entire normalized representation.*

Proof. The largest and smallest matrices A, S are present. Exactly $e - 1$ rank-one covers of A remain; factor their deficits as $u_i u_i^\top$ and orient u_i positively in the empty-active region. A rank-two deficit whose range contains exactly one of the known column lines must correspond to the pair consisting of that column and the missing column. Indeed the range of a deficit is the span of exactly its active columns, and independent columns have no additional known column line in their span. Two such pair ranges intersect in the missing line, since their three original columns are independent. Such two pairs exist because $e \geq 3$.

Choose a unit vector v on that line with positive orientation on the empty region. For a recovered pair express its deficit uniquely as $[u_i, v]K[u_i, v]^\top$. If the missing normalized column is dv , its normalized two-endpoint correlation is some a with $|a| < 1$, and

$$K = \frac{1}{1-a^2} \begin{pmatrix} 1 & -ad \\ -ad & d^2 \end{pmatrix}.$$

Thus $d = \sqrt{K_{22}/K_{11}}$ is intrinsic and positive. All normalized columns are now known; $A - S$ and formula (G.3) determine the coupling. This construction uses only observed labels and ranges, so it is forced for every minimal competitor. $\square$

Lemma I.4 (Reconstruction of a missing extreme). *Let $e \geq 3$ and $\text{rank } B = e$. If exactly the full active cell is missing, the observed singleton and pair deficits below A determine the full normalized*

representation. If exactly the empty cell is missing, the dual singleton and pair increments above $S_{[e]}$ do so.

Proof. For the first assertion, singleton deficits determine the positively oriented normalized columns $u_j = B_j/\sqrt{C_{jj}}$. Every pair is observed since $e \geq 3$. In the pair basis $U_{ij} = [u_i, u_j]$, its deficit has coefficient matrix $\bar{C}_{\{i,j\},\{i,j\}}^{-1}$. A left inverse of U_{ij} recovers this coefficient, and inversion gives the corresponding normalized principal block. All pairs determine every off-diagonal entry and the diagonal is one. Together with the observed A this gives the whole matrix.

For the second assertion write $S = S_{[e]}$, $D = C^{-1}$, and $H = BC^{-1}$. Block elimination gives

$$S_I - S = H_J D_{JJ}^{-1} H_J^\top, \quad J = I^c.$$

Every dual singleton and pair is present. Normalize $h_j = H_j/\sqrt{D_{jj}}$ and orient it negatively on the full-active region, where $C^{-1}B^\top x < 0$. The preceding pair argument recovers $\bar{D}$ with diagonal one. Then

$$C_* = \bar{D}^{-1}, \quad B_* = (h_j)C_*, \quad A_* = S + (h_j)C_*(h_j)^\top$$

is the unique normalized representative, with $\text{diag } C_*^{-1} = \mathbf{1}$. Multiple maximal or minimal observed vertices distinguish the missing extremes intrinsically, so the construction does not require supplied active-set names. $\square$

Together with the unchanged-extremes reconstruction when the missing vertex is neither an extreme nor a singleton, these lemmas cover every case of Theorem H.1. The rank of the span of observed Hessian differences is e , even when an extreme is absent, because the ranges adjacent to that extreme span all cross columns. This supplies the common minimality and competitor argument before any reconstruction is performed.

I.1. A finite presentation beyond the first strata. An explicit scalar normal form is more informative than a quantifier-elimination output. Nevertheless arbitrary simultaneous cell disappearance and rank loss admit the following finite inverse description. It treats all active cones, including those with empty interior.

Theorem I.5 (Semialgebraic presentation of all minimal endpoint fibres). *This is a reference-presentation equality theorem: the matrix Q is supplied. The function-only minimal-fibre theorem is Theorem 7.2. Fix retained dimension k and a positive reference representation Q with e endpoints. For each $0 \leq e' \leq e$, the set of positive e' -endpoint representations of the same labelled function G_Q is semialgebraic and has an explicit finite equality test. For a candidate Q' , form its active data K'_J, S'_J and the reference data K_I, S_I . For each pair I, J , let $v_1, \dots, v_r$ generate the polyhedral cone $K_I \cap K'_J$. Then equality of the two functions is equivalent to*

$$v_a^\top (S_I - S'_J) v_b = 0 \quad \text{for every } I, J \text{ and } 1 \leq a, b \leq r. \quad (\text{I.1})$$

The zero cone imposes no condition. The least e' for which these conditions and positive definiteness are feasible is the minimal endpoint dimension. The normalized minimal fibre is obtained by additionally setting $\text{diag } C' = \mathbf{1}$; all unnormalized representations follow by positive endpoint scales, with permutations left as a finite ambiguity.

As the reference matrix varies in a semialgebraic family with fixed k, e , there is a finite semialgebraic partition on which the minimal endpoint dimension and the semialgebraic homeomorphism type of each normalized candidate fibre are constant. This includes arbitrary losses of visibility, changes of cross-block rank, and coalescence of active-set matrices.

Proof. The active formulas remain valid without full rank or strict visibility, because each principal endpoint block is positive definite. On $K_I \cap K'_J$ the two values are $x^\top S_I x$ and $x^\top S'_J x$. These intersection cones cover X . If their difference vanishes on a cone, it vanishes at every generator and

at every sum of two generators. Polarization gives all equalities in (I.1). Conversely, expanding a nonnegative linear combination of the generators shows that those equalities make the quadratic difference vanish on the whole cone. This proves the exact finite test, including lower-dimensional intersections.

For semialgebraicity one may write the same test without choosing generators:

$$\bigwedge_{I,J} \forall x (x \in K_I \cap K'_J \implies x^\top (S_I - S'_J)x = 0).$$

Every coefficient is rational in the entries of Q, Q' , with denominators products of positive principal minors. Clearing these positive denominators and applying real quantifier elimination [5] gives a finite Boolean combination of polynomial equalities and inequalities. Alternatively, extreme rays of each pointed intersection cone are obtained from finitely many choices of active linear equalities; partition by the relevant minors and their signs to obtain a finite generator implementation. No generic-rank assumption is used.

There is always a feasible candidate at $e' = e$, namely Q . Testing the finitely many smaller dimensions therefore determines the minimum. Positive diagonal scaling normalizes every positive endpoint block, so it loses no representation. Now regard each normalized equality set as a semialgebraic set over the reference parameter space. Semialgebraic triviality [16] applied to each of these finitely many projections gives finite partitions with constant fibre homeomorphism type, including the empty fibre. A common refinement fixes the least nonempty dimension and proves the final assertion. $\square$

The last theorem is an effective presentation and a finite-stratification result, not an assertion that every higher-dimensional fibre is a product of the scalar intervals of Theorem H.3. Its semialgebraic triviality step is classical. The value-function equality test specifies the inverse object to which that theorem is applied; the first-stratum theorems supply explicit reconstruction where a much sharper description is possible.

APPENDIX J. AN EXACTLY CLASSIFIED NATIVE STOCHASTIC FAMILY

Fix $z_1, \dots, z_m \in \mathbb{R}^k$ spanning $\mathbb{R}^k$. Consider a single categorical observation with $2m + 1$ cells and probabilities

$$p_{j,+}(a, \tau) = a_j/2 + z_j^\top \tau, \quad p_{j,-}(a, \tau) = a_j/2 - z_j^\top \tau, \quad p_0(a, \tau) = 1 - \sum_{j=1}^m a_j. \quad (\text{J.1})$$

The centre is $\tau = 0$, where $a_j > 0$ and $\sum_j a_j < 1$. For all small τ in a fixed cone $\mathcal{K}$ with nonempty interior these probabilities are positive. The masses a are free nuisance coordinates and τ are retained coordinates. The observation metric is the ordinary, unaltered Hellinger metric. Thus the construction specifies the probability model and not an independently adjustable quadratic metric.

Theorem J.1 (Native image, secants, and exact ray complexity). *For (J.1), profiling the free mass coordinates at the centre gives*

$$Q(a) = 4 \sum_{j=1}^m \frac{z_j z_j^\top}{a_j} > 0. \quad (\text{J.2})$$

Put $\mathcal{V} = \text{span}\{z_j z_j^\top : 1 \leq j \leq m\}$ and $s = \dim \mathcal{V}$. The native image $\{Q(a) : a_j > 0, \sum a_j < 1\}$ is relatively open in $\mathcal{V}$, has dimension s , and its normalized nonzero secants are exactly the unit sphere in $\mathcal{V}$ for the Frobenius norm. In particular that secant set has dimension $s - 1$.

Exactly s legal ray values $v^\top Q v$, $v \in \mathcal{K}$, are necessary in the worst case to determine the unknown native quotient, for both fixed and deterministic adaptive exact queries. There is a fixed

family of s rays attaining the bound. The same count determines the quadratic cone ball, and hence any fixed root decoder from which that ball's radial gauge is recovered.

Taking the contrasts e_i , $1 \leq i \leq k$, and $e_i + e_j$, $i < j$, gives $s = k(k+1)/2$. Thus full-dimensional native families with sharp polarization complexity exist in every retained dimension, without varying the observation metric or using a numerical rank witness.

Proof. At $\tau = 0$, the two entries in pair j have probability $a_j/2$. Their retained probability derivatives are z_j and $-z_j$, while the mass derivative in that pair has entries $1/2, 1/2$. Their Fisher cross product cancels. The retained information is therefore $4z_j z_j^\top / a_j$, summed over pairs. The remaining cell has zero retained derivative. The nuisance information is

$$I_{aa} = \text{diag}(a_1^{-1}, \dots, a_m^{-1}) + \frac{\mathbf{1}\mathbf{1}^\top}{1 - \sum_j a_j} > 0.$$

Consequently its Schur complement is exactly (J.2). The spanning condition proves positive definiteness.

The coordinate change $a_j \mapsto c_j = 4/a_j$ maps the mass domain diffeomorphically onto an open subset of $\mathbb{R}^m$. The linear map $(c_j) \mapsto \sum_j c_j z_j z_j^\top$ is onto $\mathcal{V}$, and every surjective linear map is open. Hence the image is relatively open with dimension s . All secants lie in $\mathcal{V}$. Fix an interior image point and a sufficiently small relative ball around it. Its differences contain a relative ball about zero, so every nonzero direction in $\mathcal{V}$ is the direction of a secant. This proves the precise sphere assertion, including $s = 1$.

Evaluation by the rank-one forms $H \mapsto v^\top H v$ with v in the interior of $\mathcal{K}$ separates $\mathcal{V}$. Otherwise a nonzero quadratic polynomial would vanish on an open set. Finite-dimensional linear algebra therefore selects s such evaluations forming a basis of $\mathcal{V}^*$. Their values determine Q .

For the adaptive lower bound, fix Q_0 in a relative open ball of the native image and run a proposed algorithm making fewer than s queries. There is a nonzero $H \in \mathcal{V}$ annihilated by every recorded evaluation. Both $Q_0 + \epsilon H$ and $Q_0 - \epsilon H$ remain in the native image for sufficiently small $\epsilon > 0$. They produce the same first answer as Q_0 , hence the same second query and answer, and inductively the same transcript and stopping decision. They are distinct, so the algorithm cannot recover both. No continuity of the query rule is assumed. A cone ball with nonempty interior determines its positive quadratic form by its radial gauge, proving the fibre assertion.

Finally the listed contrast tensors form a basis of Sym_k : the $e_i e_i^\top$ give the diagonal entries and subtracting two such diagonal tensors from $(e_i + e_j)(e_i + e_j)^\top$ gives the (i, j) off-diagonal tensor. This proves the all-dimensional statement. $\square$

Corollary J.2 (Native realization at every positive quotient). *For every $Q_0 \in \text{Sym}_n^{++}$ there is a fixed paired-contrast experiment whose native quotient image contains a neighbourhood of Q_0 in Sym_n . Its design and contrasts are fixed, and only its centre masses vary. Consequently the fully visible endpoint class of Section G occurs in a native polynomial probability model with the ordinary Hellinger metric.*

Likewise, every finite-map reduction in Theorem C.2 can be embedded, on a sufficiently small feasible neighbourhood of its central fibre, in a strictly positive polynomial probability observation. Given any $H > 0$ on its residual space, that embedding may be chosen so that its minimum Hellinger distance has the two-sided relative comparison

$$\min_x h(P_{\text{rem}}(x), P_{\text{base}}(t)) = \frac{1}{2} \min_b q_b(H, t)(1 + O(\varepsilon(t) + \omega(t))),$$

where $\omega(t) \rightarrow 0$ and equality here denotes the corresponding two-sided bounds, including exact zeros. Thus the extracted finite sheets and the positive-metric envelope are native stochastic objects for this class.

Proof. Choose vectors $v_1, \dots, v_N$ whose rank-one tensors form a basis of Sym_n , for example the vectors used in Theorem J.1. Choose $\eta > 0$ so small that $Q_0 - \eta \sum_j v_j v_j^\top > 0$, and factor this difference as $WW^\top$ with columns $w_1, \dots, w_n$. Put $m = N + n$ and choose all centre masses $a_j = 1/(2m)$. Use contrasts

$$z_j = \sqrt{a_j \eta / 4} v_j \quad (j \leq N), \quad z_{N+i} = \sqrt{a_{N+i} / 4} w_i \quad (i \leq n).$$

Formula (J.2) gives $Q(a) = Q_0$. The contrast tensors span Sym_n , so the image is open there. Apply this construction with $n = k + e$ to any fully visible matrix Q_0 , marking its last e retained coordinates as nonnegative endpoints. Profile the free masses first and then those endpoints. This gives exactly G_{Q_0} , and all nearby matrices remain in the same native experiment.

For the reduction, write $O(x) = (F(x), G(x))$, $o(t) = (a(t), b(t))$, and $o_0 = (a_0, b_0)$. Apply the preceding construction to H and set the paired model's retained coordinate to $O(x) - o_0$ or to $o(t) - o_0$, with the same fixed masses. Every probability is polynomial in x on the remote side, and the marked base is semialgebraic. Positivity holds after restricting to sufficiently small neighbourhoods of the finitely many central inverse points. This restriction does not change the small-time minimizing germ, since the residual is bounded away from zero on the discarded compact part.

On a compact positive probability neighbourhood the square-root identity gives a uniform norm equivalence and, when both observations tend to the centre, the stronger relative comparison

$$h(P', P)^2 = \frac{1}{4} R^\top H R (1 + O(\omega)), \quad R = O(x) - o(t),$$

where ω bounds the two displacements from o_0 . Trials at the inverse origins have residual $O(c(t))$. Norm equivalence therefore localizes Hellinger minimizers and quadratic trials to the same fast tube and bounds their residuals by $O(c(t))$. At those points one may take $\omega(t) = K(|o(t) - o_0| + c(t)) \rightarrow 0$. Combining with (C.6) proves the asserted relative bounds, without dividing by a residual that may vanish. The contrasts span the observation space, so equality of the probability vectors is exactly equality of the observations. Exact cancellations are preserved. $\square$

Corollary J.3 (Rank loci and fixed root patterns). *For a varying contrast matrix Z , the native dimension is exactly the rank of the matrix with columns $\text{vec}_{\text{sym}}(z_j z_j^\top)$. Its rank-drop loci are given by the corresponding minors. Positive mass changes cause no additional dimension drop. Under a prescribed retained-coordinate map $\tau = B\xi$, the formula is the same with contrasts $B^\top z_j$.*

Every fixed combination of simple root shifts, inward endpoint sums, and repeated interior variances can be used as the coordinates τ in (J.1). This gives polynomial probability observations with the same quotient classification on that pattern, and uniform conclusions on compact sets with positive cell margins and fixed contrast-rank bounds.

Proof. The dimension assertion is Theorem J.1; polynomial minors give exactly the rank loci. Substitution gives the pullback contrasts. A simple root displacement and an endpoint sum are linear polynomial functions of root coordinates. For a repeated interior cluster the function

$$X = \frac{1}{2} \sum_{i=1}^d \left(u_i - \frac{1}{d} \sum_h u_h \right)^2$$

is polynomial, nonnegative, and realizes every small nonnegative value with zero mean. These choices independently realize the indicated product cone near separated clusters; the means of repeated interior clusters are held fixed, and bounded mass changes remain feasible. Substituting these polynomials into (J.1) proves the model assertion. Compact strict margins give the stated uniformity. $\square$

For two roots at $r \pm h$, the variance is $X = h^2$. In the coordinate $h \neq 0$ the pulled-back information is $4h^2 Q_X$, which vanishes as $h \rightarrow 0$. At the collision the nonzero variance coordinate is second order instead, and the local root scale is $n^{-1/4}$ rather than $n^{-1/2}$; this follows precisely from the local experiment below. Thus the rank formula distinguishes a coordinate transition from loss of the quadratic variance experiment. We do not extend compact fixed-pattern uniform constants across this degeneracy.

The class (J.1) is distinct from the two-component binary polynomial model. The latter retains its earlier rational-moment description and its proved sharp two-variance native family. The present theorem supplies a general classified native class; it does not assert that every higher-degree binary pattern has full dimension. Also, the unknown a is not supplied as part of an exact-query problem. If the full numerical centre law is supplied, the masses and (J.2) are directly known and no oracle reconstruction is needed.

APPENDIX K. THE LOCAL EXPERIMENT REPRESENTED BY A RAY COST

We use squared Hellinger distance $h^2(p, q) = \sum_e (\sqrt{p_e} - \sqrt{q_e})^2$, with no factor of one half. Let p be a strictly positive finite probability vector. Suppose the feasible local alternatives, indexed by $a \in \mathbb{R}^r$ and v in a cone $\mathcal{K} \subset \mathbb{R}^k$ with nonempty interior, have

$$p_{n,a,v} = p + n^{-1/2}(La + Dv) + o(n^{-1/2}), \quad (\text{K.1})$$

uniformly on bounded index sets. The columns have zero sums and (L, D) is injective. The remainder is in ordinary Euclidean norm on the finite cell space. Define

$$W = \text{diag}(p_e^{-1}), \quad I = (L, D)^\top W (L, D), \quad Q = I_{DD} - I_{DL} I_{LL}^{-1} I_{LD}.$$

The empty free block has its natural convention. Fixed clock designs with limiting proportions $w_j > 0$ give the same formula with $W = \text{diag}(w_j/p_{je})$.

Theorem K.1 (Canonical efficient discrimination). *For independent observations from (K.1), the local log likelihood has the expansion, for each bounded local index and uniformly on compact index sets when the remainders are uniform,*

$$\log \frac{dP_{n,a,v}^{\otimes n}}{dP_{n,0,0}^{\otimes n}} = (a, v)^\top Z_n - \frac{1}{2}(a, v)^\top I (a, v) + o_P(1), \quad Z_n \implies N(0, I), \quad (\text{K.2})$$

provided the centre is taken to be exactly $p_{n,0,0} = p$. After removal of the unrestricted nuisance translation, the Gaussian limit experiment is

$$Y \sim N(Qv, Q), \quad v \in \mathcal{K}. \quad (\text{K.3})$$

Its scalar discrimination cost is

$$c(v) = v^\top Q v, \quad \text{Aff}(N(0, Q), N(Qv, Q)) = e^{-c(v)/8}, \quad \text{Power}_\alpha(v) = \Phi(\Phi^{-1}(\alpha) + \sqrt{c(v)}). \quad (\text{K.4})$$

Here the last expression is the Neyman–Pearson power for testing the null against the specified efficient local alternative at level α .

Two such labelled quotient experiments with the same retained coordinates are statistically equivalent by Markov randomizations in both directions if and only if their matrices Q agree. Thus the ray costs determine an intrinsic discrimination object, not just a chosen matrix representation. Exact-ray complexity is nevertheless not sample complexity.

Proof. Write $s = La + Dv$ and $\delta_n = p_{n,a,v} - p$. Positivity gives a uniform bound on $|\delta_{ne}/p_e|$. Taylor expansion of $\log(1 + u)$, with the third-order remainder bounded by a constant times $|u|^3$, yields

$$\sum_{i=1}^n \log \frac{p_{n,a,v}(X_i)}{p(X_i)} = \sum_e (N_e - np_e) \frac{\delta_{ne}}{p_e} - \frac{n}{2} \sum_e \frac{\delta_{ne}^2}{p_e} + o_P(1).$$

The first-order expectation vanishes because $\sum_e \delta_{ne} = 0$. The replacement of the quadratic empirical counts by their expectations has order $o_P(1)$, and $n \sum_e |\delta_{ne}|^3 = O(n^{-1/2})$. The finite-dimensional multinomial central limit theorem and (K.1) now give (K.2). Uniformity follows from bounding the remainder against the norm of the same finite-dimensional count vector. The zero-sum score columns remove the $pp^\top$ covariance term. For a deterministic clock allocation, apply the same argument independently to each group and use $n_j/n \rightarrow w_j$.

The Gaussian score $Z = (Z_L, Z_D)$ has mean $I(a, v)$ and covariance I under the local alternative. The efficient score

$$Y = Z_D - I_{DL} I_{LL}^{-1} Z_L$$

has mean Qv , covariance Q , and zero covariance with Z_L . The two blocks are therefore independent. The mean of the nuisance block is $I_{LL}a + I_{LD}v$ and can take every value as a ranges freely. Equivalently, after the nuisance reparametrization $a \mapsto a + I_{LL}^{-1} I_{LD}v$, the Gaussian experiment is the product of the efficient experiment and an unrelated full translation family. Discarding that nuisance translation yields precisely (K.3). This is the quotient operation; it does not claim that data concerning an unknown nuisance parameter can be simulated without a parameter.

Whitening by $Q^{-1/2}$ gives normal distributions with common identity covariance and mean difference $Q^{1/2}v$. Integrating the product of square-root densities by completing the square gives the affinity in (K.4). The log likelihood has variance $c(v)$ and means $-c(v)/2$ and $c(v)/2$ under the null and alternative; its threshold test gives the displayed power, also at $c(v) = 0$ by continuity.

Identical Q give identical quotient experiments. Conversely Hellinger affinity cannot decrease under a Markov kernel: apply the Cauchy–Schwarz inequality conditionally on each output. Kernels in both directions therefore force equality of affinities of each labelled parameter pair, in particular of 0 and v . Formula (K.4) implies equality of $v^\top Qv$ on the cone. Since that cone has nonempty interior, the two quadratic polynomials and their matrices are equal. This proves the converse. $\square$

Corollary K.2 (Variational costs and singular root scaling). *Suppose the feasible local chart has the stronger score expansion $p_{t,a,v} = p + t(La + Dv) + O(t^{1+\epsilon})$, $\epsilon > 0$, with bounded free coordinates realizing each bounded feasible v , and every almost-minimizing local law has bounded scaled coordinates. Then*

$$\lim_{t \downarrow 0} \frac{4}{t^2} \inf_a h^2(p_{t,a,v}, p) = v^\top Qv.$$

With $t = n^{-1/2}$ the same matrix is the efficient information in Theorem K.1. In a repeated interior cluster, variance tv is realized by root displacements of order $n^{-1/4}$; inward endpoint sums and simple root shifts have order $n^{-1/2}$.

Proof. The square-root identity gives $4t^{-2}h^2(p_{t,a,v}, p) = \|La + Dv\|_W^2 + o(1)$ uniformly on bounded sets. Its minimum over a is $v^\top Qv$, obtained by $a_* = -I_{LL}^{-1} I_{LD}v$. Testing its feasible realization proves the upper bound; the bounded-coordinate assumption and the same expansion prove the lower bound for every almost-minimizing sequence. Setting $t = n^{-1/2}$ gives (K.1). The cluster identity $X = \frac{1}{2} \sum u_i^2$ for zero-mean root displacements gives the square-root scale $\sqrt{t}$; sums are linear. These are realized score paths, including the cone boundary. $\square$

The binary whole-model coefficient inverse and cluster estimates retained in the supporting volume verify these hypotheses on every compact full-rank fixed-pattern stratum. The paired-contrast model verifies them directly in its probability coordinates. Thus the two settings share the efficient discrimination interpretation. This identifies the intrinsic meaning of a variational ray while leaving separate the number of independent observations needed to estimate it.

APPENDIX L. NATURAL COMPATIBILITY OF THE THREE QUOTIENTS

The preceding binary theorem determines a metric from a pre-existing observation mechanism. The next statement identifies the associated root sheets and endpoint profile without an embedding into a designed contrast model. The distinction between labelled roots and their symmetric observation is important here.

Theorem L.1 (Calibrated binary structural compatibility). *Fix a compact separated configuration in Theorem 2.1, with positive cell and exposure margins. Near its repeated-root centre, the polynomial map from labelled roots to cluster sums, variances and inward endpoint shifts is finite. Over strictly positive variances its degree is 2^k , corresponding exactly to exchange of the two roots in each interior cluster.*

For a fixed retained direction $w = (x, z) \in \mathbb{R}_+^{k+e}$, the complete leading observation residual at scale t , with unrestricted free sum and mixing directions $a \in \mathbb{R}^{k+1}$, is the affine sheet

$$E_w = \{La + Dw : a \in \mathbb{R}^{k+1}\}. \quad (\text{L.1})$$

Here the probability score blocks are obtained from (2.3) by multiplying row j by the four-vector $\pi_j(1 - \pi_j)(M_1 - M_2)$. The root labels give identical observation sheets, so their multiplicity is removed in the intrinsic residual set. Its metric tensor envelope is the affine-sheet envelope of Theorem F.1 whenever $w \neq 0$; for $w = 0$ the minimum is zero.

The following three operations therefore give compatible objects derived from the same model: the finite root quotient and its leading residual, profiling the free coordinates to obtain Q with the native Hankel inverse (2.6), and profiling the inward endpoints to obtain $G_Q(x)$. More precisely, locally at the fixed centre,

$$\lim_{t \downarrow 0} \frac{4}{t^2} \inf_{\text{free sums, mixing, endpoints}} \sum_j \lambda_j h^2(p_j(t), p_j(0)) = G_Q(x), \quad (\text{L.2})$$

where the interior variances are fixed to tx , and the other variables range over their feasible local neighbourhood. The endpoint-free version with all of w fixed equals $w^\top Qw$. These limits and the score expansion are uniform for bounded retained directions and on compact separated centre sets. In the one-colour subclass, the endpoint dimensions and fibres are exactly those in Theorem 2.3.

Proof. For a labelled interior pair with displacements u_1, u_2 , write $s = u_1 + u_2$ and $v = (u_1 - u_2)^2/4$. Its factor is exactly $(T - r - s/2)^2 - v$. The inverse is $u_1 = s/2 + \sqrt{v}$, $u_2 = s/2 - \sqrt{v}$, with exchange as the only other label. Distinct separated clusters cannot mix in a sufficiently small neighbourhood. Endpoint coordinates are their inward displacements. This proves finiteness, the stated degree, and feasibility at $v = 0$ by continuity; all small nonnegative variances are realized.

In the symmetric coordinates, the map to the vector of clock log odds is analytic and has derivative V , which is invertible by Theorem 2.1. It is consequently a local analytic diffeomorphism on an open extension of those coordinates. Each calibrated probability table determines its scalar mixing probability by any cell where $M_1 - M_2$ is nonzero. Strict probability margins make these coordinate changes and their derivatives uniformly bounded. The observation map is therefore locally bi-Lipschitz in the symmetric coordinates. This statement concerns the chosen local germ, not a uniform inverse through changes of cluster pattern or channel rank.

Set the symmetric coordinates to (ta, tw) . Taylor expansion gives $p(t) = p(0) + t(La + Dw) + O(t^2)$ uniformly on bounded sets. Conversely, a bounded observation residual divided by t forces all symmetric coordinate increments to be $O(t)$ by the local inverse. Taking subsequential limits, and realizing arbitrary bounded a , proves both inclusions in (L.1). The full score is injective, so for nonzero w the sheet does not contain zero and its augmented affine matrix has full rank, as required in Theorem F.1.

For the minimization in (L.2), zero free increments and zero endpoint increments give a trial of cost $O(t^2)$ for bounded x . Norm equivalence between Hellinger distance and probability increments, followed by the local inverse, forces every minimizing sequence to have all symmetric coordinate increments $O(t)$. Positive definiteness of the score Gram matrix also bounds the scaled minimizers uniformly. On these bounded sets,

$$\frac{4}{t^2} \sum_j \lambda_j h^2(p_j(t), p_j(0)) = \|La + D(x, z)\|_W^2 + o(1), \quad W = \text{diag}(\lambda_j/p_{jc}).$$

The minimum over a is the nuisance Schur complement, and the remaining minimum over $z \geq 0$ is exactly $G_Q(x)$. The unrestricted free minimizer and the nonnegative endpoint minimizer are feasible in the local chart for small time. Thus testing them gives the upper bound; localization and the uniform expansion give the lower bound. The same argument with w fixed gives its quadratic cost. Compact separation and positivity bounds supply all the asserted uniform constants. $\square$

This compatibility theorem does not claim that distinct labelled root sheets must yield distinct residual sheets. In this natural symmetric model they do not. Nor does calibration of the channels identify the unknown-channel experiment with the calibrated one. The general finite-map results, the paired benchmark, and the preserved unknown-channel binary inverse retain their separate hypotheses.

Preparation and verification. The development and drafting of the residual-scale and divisorial criteria in Section E, the one-cell and two-endpoint fibre arguments in Section H, and the interpolation-based Hankel classification in Sections 2–L used language-model assistance. Their proofs are supplied in the article. The accompanying exact algebra checks test specified identities and finite instances; they do not certify the universal theorems. This revision is prepared for independent mathematical and author review.

APPENDIX M. REAL FACTORIZATION AND POSITIVE-TIME ACCESSIBILITY

We isolate two points at which complex or formal order calculations would not suffice. All analytic functions in this appendix are real analytic.

Lemma M.1 (Factors of a normal-crossings product). *In the local analytic ring at a point with coordinates $(r_1, \dots, r_s, z)$, suppose*

$$fg = u \prod_{i=1}^s r_i^{a_i}, \quad u(0) \neq 0.$$

Then $f = u_f r^b$ and $g = u_g r^c$ with analytic units u_f, u_g and $b + c = a$. If instead

$$\sum_{j=1}^M R_j^2 = u r^a,$$

then every a_i is even and $R_j = r^{a/2} V_j$ with analytic V_j and $\sum_j V_j^2 = u$. In particular the V_j have no common zero near the point.

Proof. Each coordinate ideal (r_i) is prime: its quotient is the integral domain of analytic germs in the remaining coordinates. An analytic germ divisible by r_i can be divided analytically, as seen by its convergent power series. Let b_i and c_i be the finite maximal orders of divisibility of f and g by r_i . The primality just noted gives $b_i + c_i = a_i$. Divide by all these coordinate powers. The remaining factors have product u , so their values at the point are nonzero and each is a unit. This proves the first assertion without an unstated factorization assumption.

For the second, let b_i be the smallest r_i -divisibility order among the R_j . After dividing by $r_i^{b_i}$, at least one restricted germ on $r_i = 0$ is nonzero. A sum of squares of real analytic germs is identically zero only if every germ is zero: evaluate on a real neighbourhood. Thus the restricted sum of squares is nonzero and its r_i -order is exactly $2b_i$. It follows that $a_i = 2b_i$. Every R_j is divisible by every $r_i^{b_i}$, hence by their product, by successive coordinate division. Dividing the identity proves the remaining assertions. Since u is nonvanishing and equals a real sum of squares, it is positive on a sufficiently small neighbourhood. $\square$

Lemma M.2 (Proper pruning and accessible faces). *Let Γ be compact subanalytic and equal to the closure of $\Gamma \cap \{t > 0\}$, where $t \geq 0$ is analytic on an ambient neighbourhood. Let $\psi : M \rightarrow \Gamma$ be a proper analytic surjection from a nonsingular real analytic manifold. Retain the closure in M of $\{t \circ \psi > 0\}$. Its image is still all of Γ . The same statement holds after a further proper analytic surjection onto M .*

Suppose after normal crossings a retained local orthant chart has

$$t \circ \psi = u(r, z)r^a, \quad r_i \geq 0, \quad u > 0,$$

and its full interior has positive time. For every coordinate face with $a_i > 0$, every choice of sufficiently small fixed positive values of the other r_j and an interior value of z gives a feasible positive-time analytic arc transverse to that face, by setting $r_i = s \downarrow 0$. Its time order is a_i . If $R \circ \psi = r^b V$ with V nowhere zero, its residual norm has order b_i on this arc.

Proof. For $y \in \Gamma$ choose a sequence $y_n \in \Gamma$ with $t(y_n) > 0$ and $y_n \rightarrow y$. Choose any lifts $x_n \in M$. Properness over the compact set Γ supplies a convergent subsequence with limit x , and continuity gives $\psi(x) = y$. Its limit is in the retained closure. This proves surjectivity after pruning; the same argument proves the assertion after any further proper surjection.

On a connected analytic component on which $t \circ \psi$ is not identically zero, its nonnegative analytic function has a zero set with empty interior, so the positive-time part is dense. Components on which it vanishes identically contribute no retained interior and may be discarded. On a local normal-crossings chart choose orthants whose interiors have positive time, replace their signed coordinates by nonnegative r_i , and take smaller closed boxes. The transverse arcs specified in the statement belong to those boxes and their images lie in Γ . Holding the other coordinates positive leaves a nonzero constant times s^{a_i} for time and a nonzero vector times s^{b_i} for the residual, proving both orders. This is an explicit feasible arc, not an appeal to a complex divisor. $\square$

Theorem M.3 (Proper real order presentation). *Let Γ be as in Lemma M.2, and let $R_1, \dots, R_M$ be analytic on an ambient neighbourhood. Suppose there is no common residual zero on $\Gamma \cap \{0 < t \leq t_0\}$. On a sufficiently small compact time neighbourhood of the zero-time fibre there is a finite proper-surjective system of real corner charts with*

$$t = u(r, z)r^a, \quad R = r^b V(r, z), \quad r_i \geq 0, \quad a_i = 0 \implies b_i = 0, \quad (\text{M.1})$$

where u and $|V|$ have positive lower and finite upper bounds. Every face used in a ratio b_i/a_i is reached by a feasible positive-time transverse arc. Assuming the zero-time fibre is nonempty, put

$$\rho = \max\{b_i/a_i : \text{retained chart faces with } a_i > 0\}.$$

Then, on the feasible germ, $|R| \geq ct^\rho$ for a constant $c > 0$, and an accessible arc attains order ratio ρ . If the positive-time slices are all nonempty for small time, their minimum residual norm has order exactly ρ .

Proof. The uniformization theorem of [3, Theorem 0.1] gives a proper analytic surjection from a nonsingular manifold onto the compact subanalytic set. Pull back t and R . Prune as in Lemma M.2. On each remaining component the analytic function $t \sum_j R_j^2$ is not identically zero, since the residual is nonzero at positive time. Apply the proper normal-crossings theorem [3,

Corollary 4.9] to that function on a neighbourhood of the compact inverse image of the zero-time fibre. The resulting manifold and proper surjection preserve the feasible image. Prune once more.

In every resulting coordinate neighbourhood the product is a coordinate monomial times a unit. The first part of Lemma M.1 factors the time function and the sum of squares separately; its second part factors the vector residual and proves that its remaining vector factor is nowhere zero. Shrink to finitely many closed corner boxes covering the compact inverse zero-time fibre. The positive-time orthants have $u > 0$, and the units have uniform bounds on these boxes. Their images cover the feasible germ: otherwise positive-time points with times tending to zero outside the image would have lifts converging, by properness, to the inverse zero-time fibre covered by the boxes. Choose the boxes with interiors covering that fibre to obtain the contradiction.

If $a_i = 0$ but $b_i > 0$, take $r_i = 0$ and all other divisor coordinates positive and sufficiently small. The time is positive whereas every residual coordinate vanishes, a contradiction. Thus (M.1) holds. Lemma M.2 supplies all the stated transverse arcs with the correct orders, including the face realizing the finite maximum.

Rescale the boxes so that $0 \leq r_i \leq 1$. The inequalities $b_i \leq \rho a_i$ imply $r^b \geq (r^a)^\rho$. Combining this with the unit bounds yields $|R| \geq ct^\rho$ on every chart and hence on the germ. On the maximizing transverse arc, t is a positive unit times s^{a_i} and $|R|$ is a positive unit times s^{b_i} . This gives an upper bound by a constant times t^ρ on all sufficiently small time slices, since the arc's time is eventually strictly monotone. The matching lower bound proves the final statement. $\square$

The theorem computes orders for an analytic residual on a subanalytic feasible graph. A continuous semialgebraic observation can be treated by taking its compact graph and using the analytic coordinate projections for time and residual. The polar-gap suprema in Proposition C.4 likewise have semialgebraic graphs. Extracting their orders by these real charts and then testing the strict inequalities in (C.3) connects the resolution presentation to the finite-sheet theorem. The extra residues and limiting derivative matrices in (C.7) are indispensable; monomial orders alone are not asserted to recover them.

APPENDIX N. COVERING AND MINIMAL-COMPETITOR LEMMAS

The two arguments below expand the topological and counting transitions used in the finite-sheet and reconstruction theorems.

Lemma N.1 (A parameterized definable covering). *Use the geometric hypotheses of Section C, with $a(t)$ and $r(t)$ only continuous semialgebraic on $(0, t_0)$. Define*

$$\mathcal{B} = \{(t, U) : 0 < t < t_0, |U| < 2r(t)\}, \quad \mathcal{X} = \{(t, x) : x \in \text{int } N, (t, F(x) - a(t)) \in \mathcal{B}\}.$$

The map $\Psi(t, x) = (t, F(x) - a(t))$ is a finite covering of $\mathcal{B}$, proper over every compact subset of that base. Its fibre cardinality is constant. Every covering component is the graph of a continuous semialgebraic inverse sheet over $\mathcal{B}$, and those sheets exhaust all inverse points. After shortening the time interval the sheets are analytic for positive time. The centre arcs have convergent Puiseux representatives after a finite power substitution in time. Every sheet is defined on the closed smaller tube $|U| \leq r(t)$.

Proof. There are no critical or boundary inverse points over $\mathcal{B}$. The inverse function theorem for F , followed by the continuous shift $a(t)$, makes Ψ a local homeomorphism. This does not require differentiating a or r . If $K \subset \mathcal{B}$ is compact, its time coordinates stay away from zero and t_0 . A sequence in $\Psi^{-1}(K)$ has a subsequence converging in that compact time interval times N . Its image stays in K ; the limit cannot be on ∂N . Hence the inverse image is compact. Each fibre is a compact discrete set and is finite. A proper local homeomorphism is a covering: take disjoint local inverse neighbourhoods of the finite fibre, and use compactness to rule out additional inverse points over a shrinking target neighbourhood. Its image is both open and closed in the connected

base and is nonempty, since the centre fibres are nonempty. It is therefore onto, with constant finite cardinality.

The map $(t, U) \mapsto (t, U/r(t))$ identifies the base with an interval times a ball; it is simply connected. Every covering component consequently maps homeomorphically onto the base. The components are semialgebraic, so their inverse graphs give definable labels. Continuous semialgebraic one-variable arcs are analytic off a finite set. Shrinking the interval removes that set, and the analytic inverse function theorem then makes the sheets analytic at every positive-time point. The bounded centre arcs have convergent Puiseux expansions by one-variable semialgebraic preparation. Finally the closed smaller tube is contained in the open larger tube, proving the last assertion. $\square$

Lemma N.2 (Visibility of every minimal competitor). *Suppose a labelled profile G has a fully visible e -endpoint representation. Every representation of G with e endpoints is fully visible. More generally, if B' has independent columns, an active cone with nonempty interior in X° contains a strict KKT point.*

Proof. The intrinsic matrix difference between the largest and smallest local quadratics has rank e . Every local difference for a competing representation has range contained in $\text{ran } B'$. Thus B' has rank e . There are at most 2^e active polynomials, while G has exactly 2^e distinct polynomials on open sets. Every one must therefore occur on a full-dimensional active cone; a finite union of lower-dimensional polyhedral cones cannot cover such an open set.

For a fixed active set I , each active coordinate is a row of $-C_{II}^{-1}B_I^\top x$. No such row is identically zero, because B_I has independent columns. Each inactive multiplier is $(B_j^\top - C_{jI}C_{II}^{-1}B_I^\top)x$, a nonzero functional because B_j is not in the span of B_I . The finitely many zero hyperplanes of these functionals cannot cover a full-dimensional active cone in X° . A point off them satisfies all its feasible inequalities strictly. This proves both statements. $\square$

Lemma N.3 (The ambient factor convention). *For a positive block matrix with e endpoints, take an invertible $e \times e$ factor L with $C = L^\top L$ and set $T = -L^{-\top}B^\top \in \mathbb{R}^{e \times k}$. Then*

$$G_Q(x) = x^\top (A - BC^{-1}B^\top)x + \text{dist}(Tx, L\mathbb{R}_+^e)^2.$$

Two such invertible factors of the same C differ by one left orthogonal matrix, and that same matrix transports T . Positive endpoint scales do not change the cone generated by the columns of L .

Proof. Expand $|Lz - Tx|^2$ and use $T^\top L = -B$. If $L_1^\top L_1 = L_2^\top L_2$, then $O = L_2 L_1^{-1}$ satisfies $O^\top O = I_e$ and $T_2 = OT_1$. Replacing L by LD , for positive diagonal D , changes the lengths but not the generating rays of its simplicial cone. These are statements in the minimal ambient space $\mathbb{R}^e$, not assertions about unused orthogonal directions in a larger factorization. $\square$

APPENDIX O. THEOREM HIERARCHY, PRESERVATION, AND EVIDENCE

The principal theorem is Theorem 5.1. Its inputs are the native inverse-information identity and the synchronized contact limit. Its mechanism is Lemma 5.3 combined with the native three-site relations. The explicit five-curvature Corollary 6.1, the observable budget image, and contact stability are consequences of that mechanism. The function-input endpoint theorem is a companion inverse theorem, not a hypothesis needed to reconstruct the metric in the endpoint-free principal theorem.

Statement family	Role	Source/proof status
Native Hankel inverse information	Structural input	Complete interpolation proof retained.
Synchronized contact	Statistical input	Complete proof retained; uniform class made explicit.
Conormal rigidity; three-site criterion	Principal theorem	New, all shared dimensions $d \geq 2$.
Five directional curvatures	Explicit sharp corollary	New reconstruction and exceptional example.
Reciprocal fibres	General geometric input	Abstract proof separated from native specialization.
Observable budget region and fibres	Consequence	New transport through contact tomography.
Function-input minimal fibre	Companion inverse theorem	New exact graph input and terminating promised search.
Finite maps, weighted jets, divisorial tests	Companion geometry	All inherited proofs retained in appendices.
Endpoint visibility and planar fibres	Companion explicit inverse theory	All inherited proofs and sector tables retained.
Native benchmarks, local experiments, binary inverse	Companion statistical theory	All inherited proofs retained.

The mathematical baseline is A2 revision 107. Its source commit and the independent report commit are, respectively,

a17d3cad6aeebe75f29102ecb2d1ef6784927f4e
b3d0c5ce18a5f6e4491ed70f50a6991ea846dcd8.

All earlier source files remain unchanged in the repository. The complete mathematical parts imported by revision 107 are included here, with their original theorem labels, except for replacing its introductory positioning and dependency prose by the present hierarchy. The previous positioning text is also retained as historical background below. Prepared copies expose hypotheses or correct the scope of descriptive sentences; they do not remove theorem proofs. The deterministic preparation script and preservation audit record these changes.

The exact checks concern specified polynomial identities, rational matrices, normal spaces, KKT cases, and finite examples. They do not prove the universal theorems. The native build receipt binds the input manifest, final log, checks, and PDF to one exact source commit; it is archival evidence, not mathematical validation. Language-model assistance was used in developing the conormal arguments, drafting this revision, and preparing the source checks. The resulting proofs are submitted to author and independent referee scrutiny. Neither successful compilation nor assistance constitutes independent theorem certification or an acceptance decision.

APPENDIX P. BACKGROUND COMPARISONS FOR THE COMPANION RESULTS

P.1. Theorem-level relation to earlier work. The following background comparisons concern the retained companion results. The principal conormal interaction and its inverse-optimization and design-equivalence comparisons are stated separately in the introduction and Theorem 5.1. None of the classical ingredients listed below is claimed as a new construction. The distinctions are part of the statements, not qualifications to be supplied after applying them.

Quadratic measurements and singular statistical models. The map $\eta \mapsto ((a_i^\top \eta)^2)_i$ is the real quadratic measurement map familiar from phase retrieval. Its sign ambiguity and identifiability are studied by Balan, Casazza, and Edidin [12]. The representation as linear measurements of $\eta\eta^\top$ is also the starting point of PhaseLift [13]; dropping the rank constraint produces a different

feasible set. We use the elementary sufficient hypothesis that the measurement tensors span Sym_d , rather than claiming a sharp phase-retrieval measurement count. Drton [15] explains why tangent cones, including nonconvex ones at singularities, govern likelihood-ratio limits in semialgebraic statistical models. Theorem 4.1 is not a new general tangent-cone principle. It proves that synchronized polynomial roots give this particular cone in a calibrated binary experiment, computes its deterministic Hellinger contact in the experiment's native Hankel metric, and proves that a reconstructed endpoint profile determines that contact for every loading family and target. The finite-root-label quotient of Theorem L.1 is a regular companion result; it is not used to disguise this cone as an affine sheet.

Moment spaces and the location of the inverse. Moment spaces and their statistical design role are classical; see Dette and Studden [9]. At fixed clocks, $\sum_j \omega_j w(T_j) w(T_j)^\top$ is simply the linear image of a positive weight orthant. It is a finite polyhedral moment cone, not the full truncated moment cone with arbitrary support. Vandermonde rank explains its span and dimension. Lemma B.1 likewise uses classical polynomial interpolation and extraction of principal-part coefficients. The model-specific content of Theorem 2.1 is that the calibrated polynomial log-odds scores force those coefficients and therefore force the *inverse nuisance-efficient information* into the stated fixed congruence class. A generic information matrix written as a weighted sum of regression tensors does not establish this inverse-information statement. Proposition B.5 identifies its square-design boundary: the same subdesign congruence can fail once an extra clock has positive weight.

Design geometry with a fixed budget. The equivalence of design extremum problems in Kiefer and Wolfowitz [10], and the information-matrix viewpoint developed systematically by Pukelsheim [17], concern a much broader optimization theory. Our scalar allocation Corollary B.4 is a direct Cauchy–Schwarz calculation within that tradition. Theorem B.3 instead describes the entire image of the positive exposure simplex in moment coordinates and every exposure vector over each image point. The reciprocal budget function is minimized on an affine moment fibre. Its explicit Hessian and the dimension $\dim \ker T = 2 - e$ then distinguish a hypersurface with unique designs from a convex region with two-point or circular design fibres. This is the precise fixed-budget conclusion used in the nonlinear contact theorem; no unnormalized-cone secant assertion is transferred to the budget image.

Total positivity and endpoint visibility. The Cauchy determinant formula, Cauchy–Binet formula, complementary minors, and total positivity are classical; Pinkus [11] provides their systematic matrix setting. We use a single ordered-site sign matrix $\Sigma_{aa} = (-1)^a$ after forming $Q = R^{-1}$. Theorem 2.3 translates those matrix signs into positive-kernel and active-region assertions for the polynomial binary family. The novelty claimed is not strict total positivity of a Cauchy Gram matrix, but the explicit consequence for which native endpoint cells occur and which inverse fibre theorem applies. These conclusions depend on the one-colour and endpoint-count hypotheses as stated.

Forward complementarity and inverse value functions. Positive definite quadratic minimization over an orthant is a monotone linear complementarity problem, treated in Cottle, Pang, and Stone [14]. Its piecewise linear optimizer and piecewise quadratic value are the familiar forward objects of parametric quadratic programming, including Bemporad, Morari, Dua, and Pistikopoulos [1, 2]. Schur complementation and convex duality are also standard [6]. The inverse object here is a labelled value function with neither endpoint count nor active-set names supplied. Theorems G.3, H.1, and H.3 recover or classify all minimal positive representations under their respective visibility hypotheses. The planar table in Lemma I.1 explicitly supplies the gluing and exhaustion needed for the last assertion. Theorem I.5 gives a general equality test and then applies classical quantifier elimination and semialgebraic triviality [5, 16]. Its finite stratification is not presented as a new triviality theorem or as an explicit normal-form list for every higher-rank degeneration.

Residual geometry and resolution. Proper real uniformization and normal crossings are taken from Bierstone and Milman [3]. The scalar and sum-of-squares factorization arguments are included to expose which real faces are accessible. The new use of these tools is an exact marked-tube derivative certificate preserving the polar numerators' cancellations, together with the residual-scale rather than radius-scale comparison. Theorem D.1 is an explicit proper weighted-initial-map statement with a localization proof and coefficient-uniform constants. It does not classify every singularity by its order vector. The transformed residual image and its metric are intrinsic to the feasible observation germ, whereas the fast derivative bounds also depend on the selected transverse presentation. Proposition D.2 and Example D.3 make this distinction testable.

Efficient information, identifiability, and exact queries. Nuisance-efficient information is obtained by a Schur complement; local asymptotic normality and comparison of labelled experiments use the classical frameworks of van der Vaart and Blackwell [7, 4]. Those operations are not new information-theoretic principles. Allman, Matias, and Rhodes [8] establish broad latent-structure identifiability results by algebraic and tensor methods. Our calibrated scalar-clock experiment has a different, explicitly interpolated score inverse, and it is kept distinct from the inherited unknown-channel inverse in Appendix Q. Finally, the exact recovery count for a deterministic noiseless dual-value oracle is a linear-algebra consequence of the native inverse-information span. It is secondary to the geometric classification and is neither a statistical sample complexity nor a guarantee for the different Q -value oracle.

These comparisons delimit the theorem-level contribution. They do not assert an exhaustive negative priority result over every formulation of inverse complementarity or singular design. The proofs, rather than a claim that no related theorem exists, specify the advance submitted for independent review.

APPENDIX Q. THE BINARY INVERSE: PRECISE QUANTIFIERS AND SCORE COLUMNS

We state explicitly the correction to the earlier coefficient-inverse argument. Let

$$K(z) = \alpha_1 f_1(z) u_1 v_1^\top + \alpha_2 f_2(z) u_2 v_2^\top,$$

where f_1, f_2 are coprime monic degree- d polynomials, $d \geq 1$, with roots in $[0, D]$, $\alpha_a > 0$, $\alpha_1 + \alpha_2 = 1$, and $U = (u_1, u_2), V = (v_1, v_2)$ are invertible strictly positive column-stochastic matrices. Let $T_1, \dots, T_\ell > D$ be distinct, with $\ell \geq 2d + 1$. Put $q(z) = \mathbf{1}^\top K(z) \mathbf{1}$ and $P_j = K(T_j)/q(T_j)$.

Lemma Q.1 (Existence of a separating clock pair). *For each fixed clock index i there exists an index j such that*

$$\frac{f_1(T_j)}{f_1(T_i)} \neq \frac{f_2(T_j)}{f_2(T_i)}.$$

No assertion is made for an arbitrary prescribed pair. The matrix $K(T_j)K(T_i)^{-1}$ for such a pair has distinct eigenvalues and its eigenvectors recover the columns of U after normalization by their entry sums. On a compact full-rank fixed-pattern family a finite cover supplies locally chosen pairs with uniform eigenvalue gaps.

Proof. If the displayed equality held for every j , the degree-at-most- d polynomial $f_1 - f_1(T_i)f_2/f_2(T_i)$ would vanish at every clock, hence identically. Monicity would make $f_1 = f_2$, contradicting coprimality. A separating pair therefore exists. Matrix multiplication gives

$$K(T_j)K(T_i)^{-1} = U \operatorname{diag} \left(\frac{f_1(T_j)}{f_1(T_i)}, \frac{f_2(T_j)}{f_2(T_i)} \right) U^{-1}.$$

The two simple eigenlines and their stochastic normalizations are analytic locally. A chosen strict gap persists in a neighbourhood. Compactness supplies a finite cover by such neighbourhoods and the minimum of their finitely many positive gap bounds. This does not select one universal pair on the entire family. $\square$

Lemma Q.2 (Displayed partial-fraction independence). *For distinct cluster locations r_j in one component, the first-order polynomial columns corresponding to cluster sums and repeated interior variances are, respectively,*

$$-\frac{f_a(z)}{z - r_j}, \quad -\frac{f_a(z)}{(z - r_j)^2},$$

with the second column retained only when the multiplicity is at least two and the cluster is interior. The inward sum at the upper endpoint reverses the sign of the first column. All these columns are linearly independent. Combined with the analytic coefficient/stochastic inverse, they imply injectivity of the full free-and-retained score matrix.

Proof. A linear dependence, divided by the nonzero polynomial f_a , is a rational identity

$$\sum_j \frac{a_j}{z - r_j} + \sum_{\substack{j \text{ repeated} \\ \text{and interior}}} \frac{b_j}{(z - r_j)^2} = 0.$$

At each repeated pole, multiplying by $(z - r_j)^2$ and evaluating there gives $b_j = 0$. Multiplying the remaining identity by $z - r_j$ and evaluating gives $a_j = 0$ for every cluster. A sign reversal does not affect independence. Different components have separately recovered polynomial coefficients, so their columns are independent in that product coordinate space.

For completeness, coefficient recovery from the normalized clock laws uses the monic normalizer $q = \mathbf{1}^\top K \mathbf{1}$. A homogeneous ambiguity has $\deg e < d$, $\deg H \leq d$, and $H(T_j) = e(T_j)P_j$. Thus the polynomial $qH - eK$ of degree at most $2d$ vanishes identically. Taking a marginal and multiplying by the inverse channel gives $q \mid ef_1$ and $q \mid ef_2$. A Bezout identity for the coprime components implies $q \mid e$, hence $e = 0$ and $H = 0$. A nonsingular minor gives the analytic coefficient inverse. Lemma Q.1 recovers U . The row sums of $U^{-1}K(z)$ are $\alpha_a f_a(z)$; their leading coefficients give α_a , and division gives f_a . Normalizing these rows gives $v_a^\top$. Thus both channels, weights, and polynomial coefficients are recovered locally. Differentiating this inverse gives a left inverse in coefficient and stochastic coordinates. Applying it to a zero combined score reduces to the polynomial independence just proved and to zero stochastic coordinate increments. This proves the final assertion. $\square$

All compact-uniform statements here hold within the full-rank fixed-pattern set with the required gaps and probability margins. They do not assert uniformity through channel-rank loss, cluster merging, a vanishing design weight, or a discriminant not satisfying the polar-gap bounds.

APPENDIX R. RELATION TO THE PRESERVED MATHEMATICAL RECORD

The complete preceding principal article is retained as the supporting volume. Its whole-model local spectral theorem, binary rational-moment bounds, fixed-experiment two-variance theorem, finite affine realization, product construction, vector weighted cancellation, one-endpoint classification, and exact diagnostic examples are not replaced or removed. The present endpoint theorem extends the mixed-sign one-endpoint phenomenon to an open class in every endpoint dimension; it does not delete the earlier quadratic one-endpoint fibres.

The archive retains the full multiplicity and boundary atlas, the exact rank-deficient cubic fibre and identifying inequalities, the inverse through channel-rank loss, both cubic walls, signed second-order and critical equality-shell calculations, real weighted atlas, finite-jet and Puiseux-residue comparisons, and fixed-format entrance results. In particular, for the earlier cubic family $f = z(z - r)^2$, $g = (z - s)^3$, the previously established identifying locus remains, under its original hypotheses,

$$\{\det U \neq 0\} \cup \{(3D - r)s - 2rD < 0\} \cup \{(1 - \alpha)(r + 3s)(2r - 3s)^2 - 4\pi r^3 < 0\}.$$

The finite-map theorem is not substituted for this different model. The correction in Lemma Q.1 is the operative clock-pair quantifier for the preserved inverse proof.

The assertions in the present article are proved here. Finite symbolic tests accompanying the revision check displayed algebra and representative active cones; they are not formal verification of the universal theorems. Likewise, source preservation and native compilation are evidence about the manuscript and its build, not substitutes for independent mathematical review.

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